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Analyzing Trip Chaining Behavior in New York State Using 2009 and 2017 National Household Travel Survey



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October 2025

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October 2025

Prepared for
New York State Department of Transportation

Prepared by
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Oak Ridge, TN 37831
managed by
UT-BATTELLE LLC
for the
US DEPARTMENT OF ENERGY
under contract DE-AC05-00OR22725

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ACKNOWLEDGMENTS

The authors would like to express their sincere appreciation to Richard Batchelder of the New York State Department of Transportation for his guidance and to the department for project support that allowed the authors to conduct this study.

EXECUTIVE SUMMARY

Trip chaining, defined as the sequential linking of trips by individuals throughout a given day, provides critical insights into daily mobility patterns and activity sequencing. Understanding these patterns has significant implications for transportation demand forecasting, congestion management, and local economic activity. This analysis examines trip chaining behaviors in New York State (NYS) for the years 2009 and 2017 and compares the Middle Atlantic Census Division with other U.S. regions in 2022, utilizing data from the National Household Travel Survey (NHTS). Through demographic, geographic, and temporal analysis, this study characterizes how populations organize travel for work, personal errands, and social activities, providing empirical evidence of evolving trip chaining behaviors to inform transportation planning strategies.

Workers employed in manufacturing, construction, maintenance, and farming demonstrate significantly higher rates of direct home-work commuting patterns without intermediate trip stops. Analysis of both 2009 and 2017 data (Figure ES.1) reveals that direct return-to-home travel after work represents the predominant pattern across all occupational categories. This pattern is particularly pronounced among manufacturing, construction, maintenance, and farming workers (77% in 2009 and 79% in 2017 in NYS), rates that are statistically significantly higher than those observed for clerical or professional workers (63% in 2009 and 62% in 2017). These findings suggest reduced schedule flexibility among blue-collar occupational categories. Comparable direct travel patterns are observed among workers unable to telecommute (Figure ES.2) and those with dependent children, although the latter demographic exhibits marginally elevated rates of pickup and drop-off trip activities (Figure ES.3).

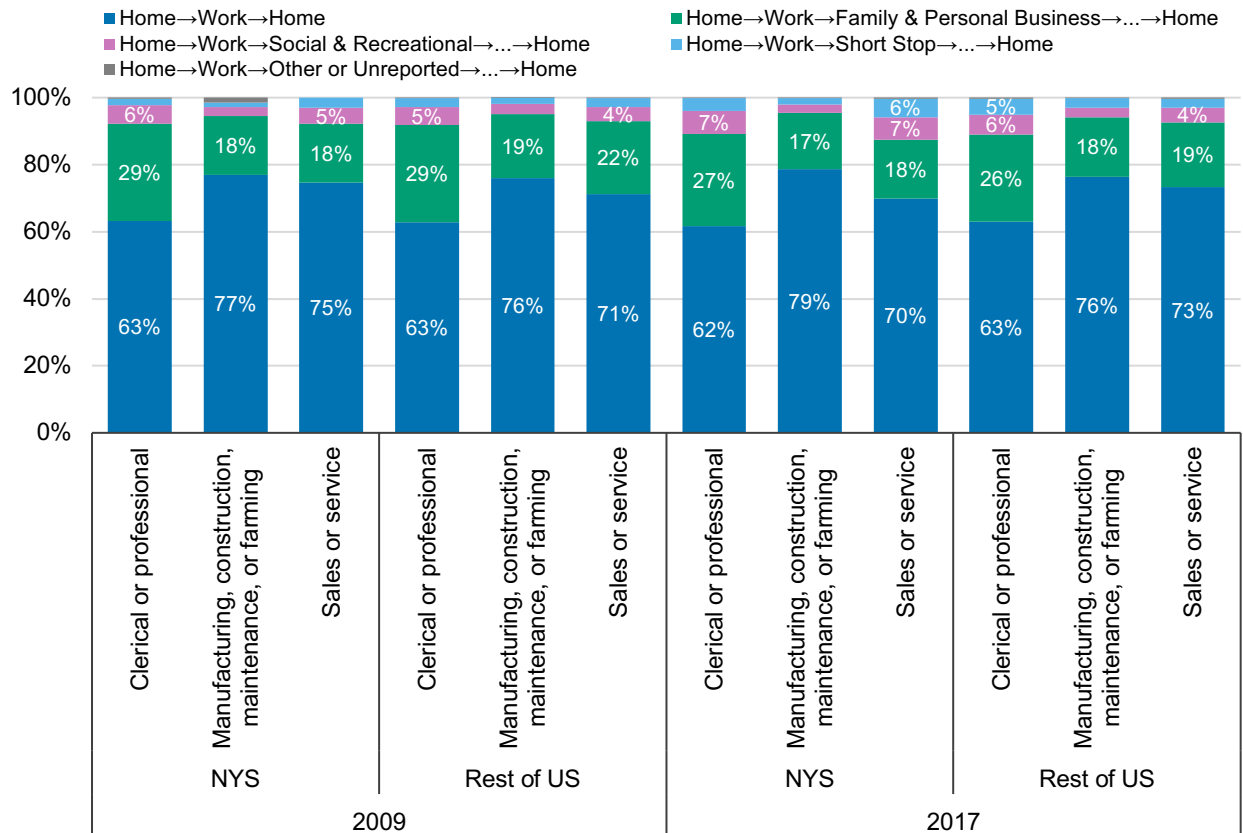


Figure ES.1 Post-Work Activity Patterns by Occupation in NYS and Rest of the United States: Home→Work Trip Chains

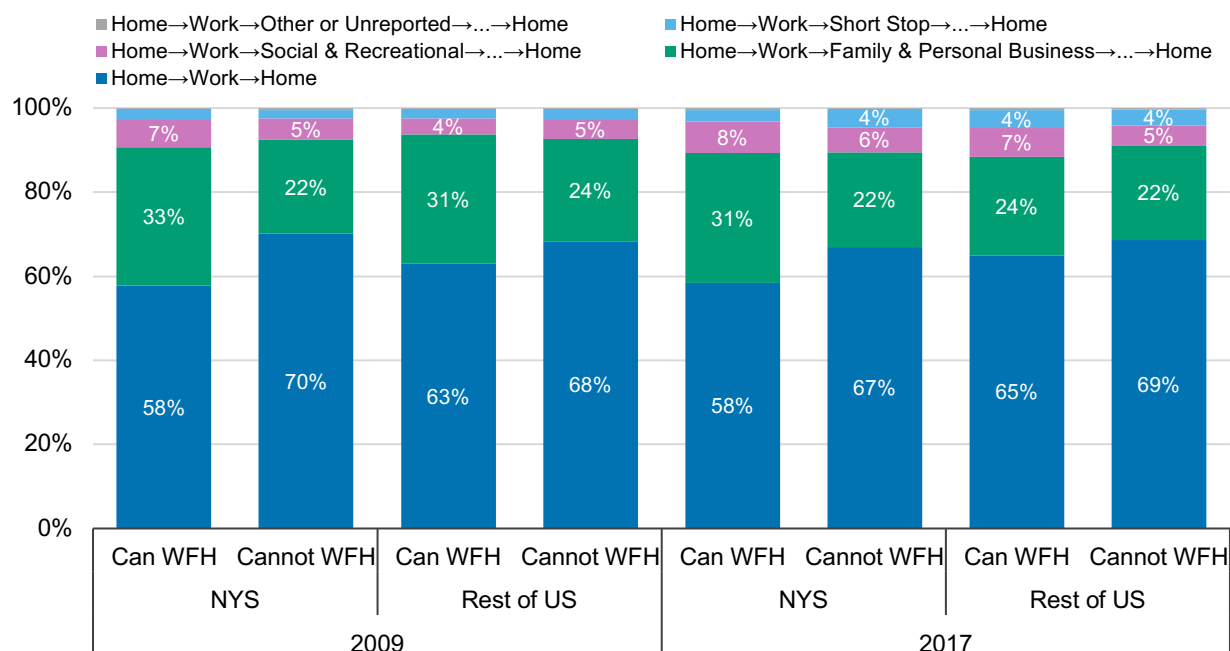


Figure ES.2 Post-Work Activity Patterns by Option of Work from Home (WFH) in NYS and Rest of the United States: Home→Work Trip Chains

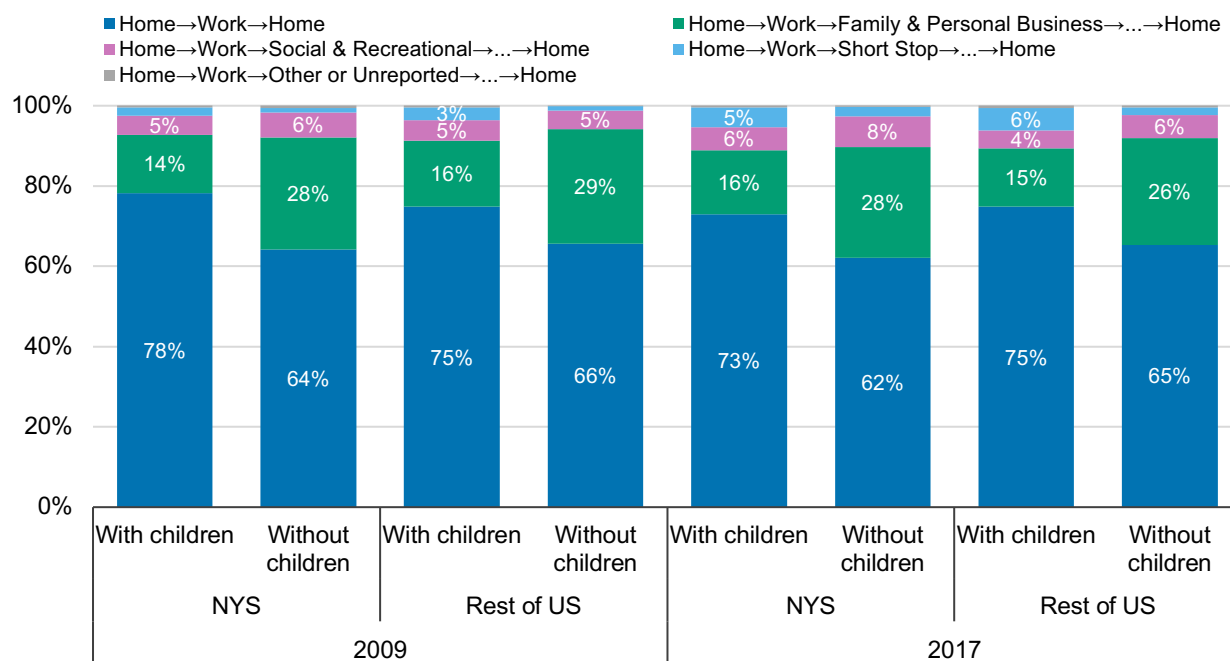


Figure ES.3 Post-Work Activity Patterns by Lifecycle in NYS and Rest of the United States: Home→Work Trip Chains

Reduced occupational flexibility may correspondingly limit public transportation utilization patterns. For example, manufacturing, construction, maintenance, and farming workers exhibited the lowest rates of modal transfer from private vehicles to public transit systems, though this observed difference did not achieve statistical significance when examined across the rest of the United States (Figure ES.4).

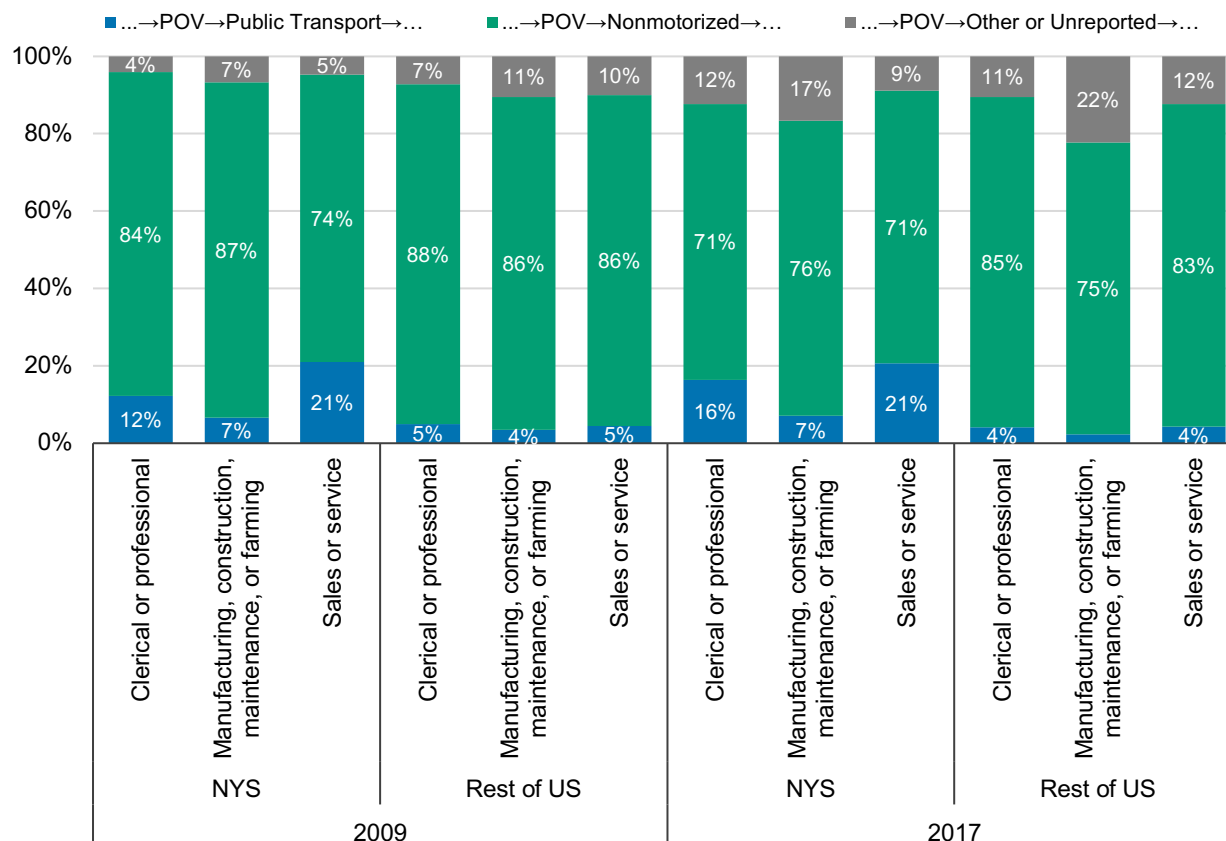


Figure ES.4 Transportation Mode Transition Following a Privately Operated Vehicle (POV) Trip by Occupation in NYS and Rest of the United States

Trip-chaining behaviors exhibit significant spatial variation between New York City (NYC) and the remainder of NYS (Figure ES.5). Within NYC, workers without personal vehicles demonstrate reduced propensity for direct home-bound travel following work and increased likelihood of integrating errands into their commuting patterns, leveraging the city’s comprehensive transit infrastructure to facilitate multi-purpose trips. Conversely, in non-NYC areas of NYS, zero-vehicle workers exhibit higher rates of direct post-work home travel, with 2017 NHTS data indicating 73% of vehicle-owning workers versus 75% of non-vehicle-owning workers engaging in this behavior. This inverse pattern observed outside NYC likely reflects constrained transportation alternatives and extended travel times, which serve as barriers to post-work activity engagement. Analogous spatial differences emerge in social and recreational trip patterns: zero-vehicle households in NYC demonstrated significantly elevated rates of post-work social and recreational activities across both study years, with 11% of zero-vehicle workers participating in such activities in 2017 compared to 6% of vehicle-owning workers.

Analysis by population density reveals an inverse relationship between residential density and direct post-work home travel patterns. In 2017, workers with telecommuting capabilities residing in high-density areas (>4,000 persons per square mile) exhibited significantly lower rates of direct home-bound travel (56%) compared to their counterparts in rural localities (65%), representing a statistically significant difference (Figure ES.6).

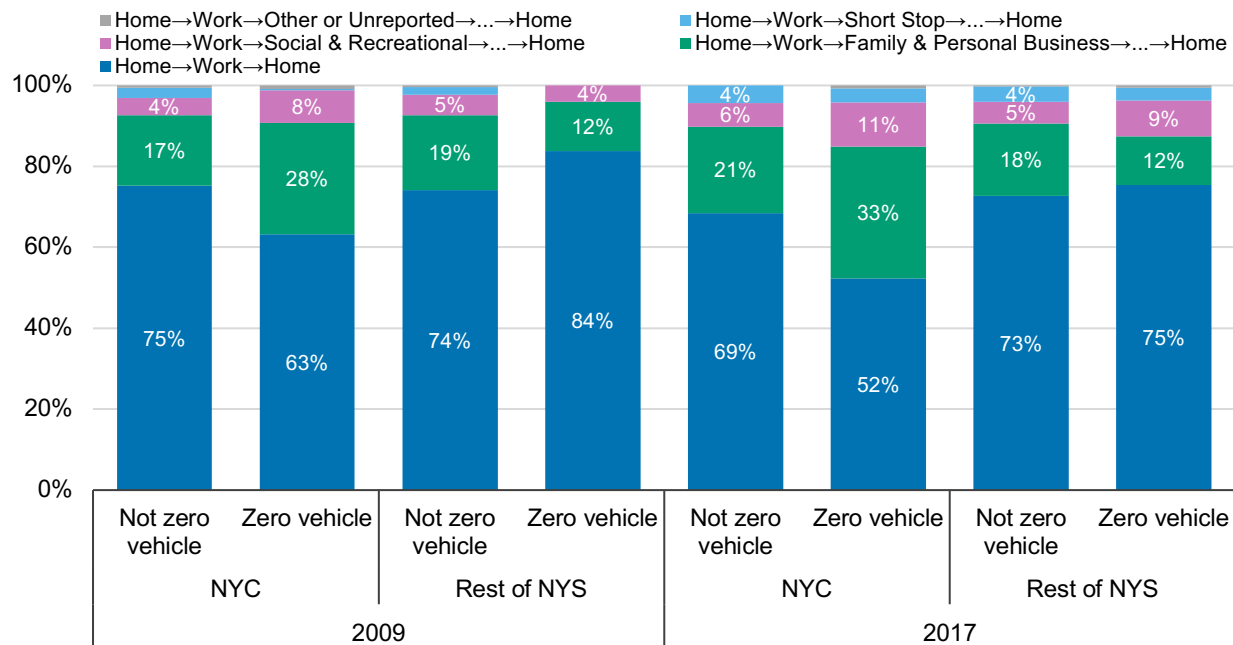


Figure ES.5. Post-Work Activity Patterns by Vehicle Ownership in NYC and Rest of NYS: Home→Work Trip Chains

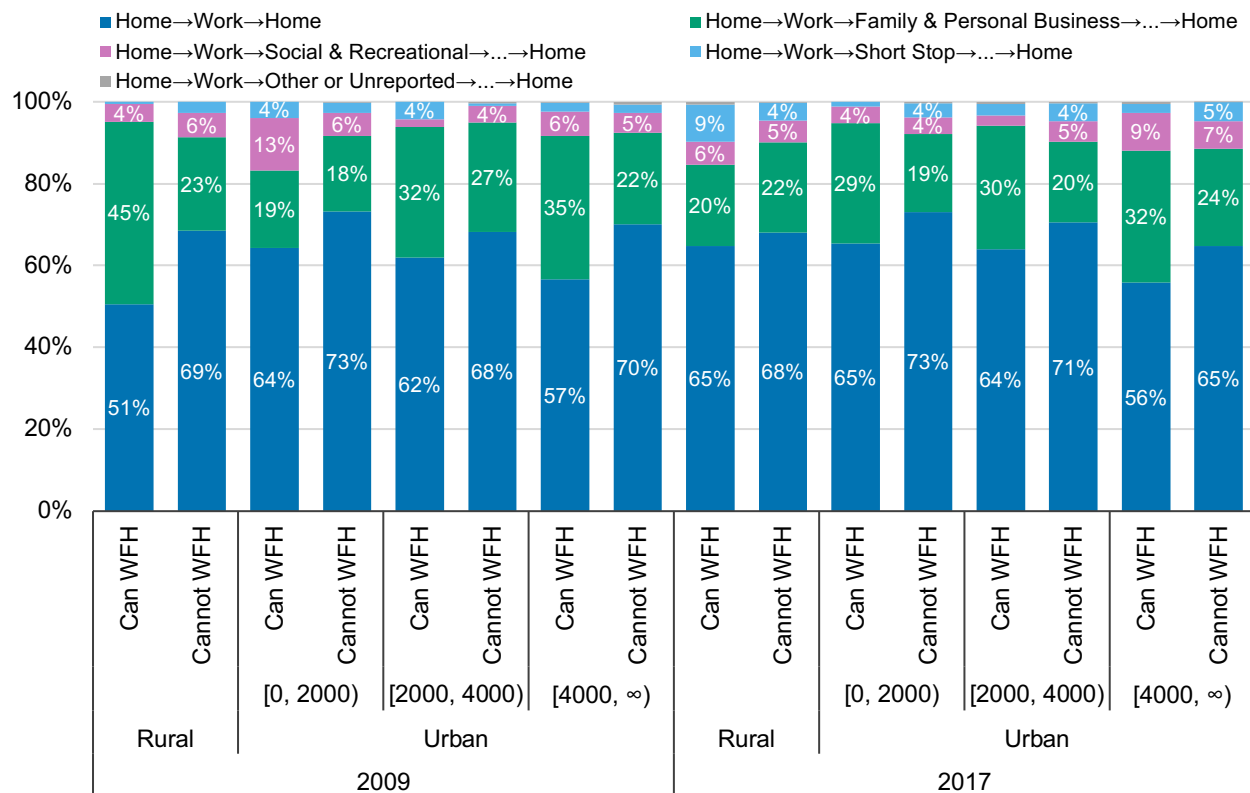


Figure ES.6 Post-Work Activity Patterns by Option of WFH and Population Density: Home→Work Trip Chains

Sex-based analysis of NYC commuting patterns reveals significant temporal shifts in post-work travel behavior (Figure ES.7). Female workers in NYC exhibited a statistically significant decline in direct home-bound travel, decreasing from 67% in 2009 to 56% in 2017. Concurrently, the proportion of female workers stopping for family or personal business activities increased substantially from 24% to 32%, indicating an expanding role in household management and errand completion during post-work periods. Male workers in NYC demonstrated a similar declining trend in direct home travel (74% to 66%), though this change did not achieve statistical significance. However, male workers showed a statistically significant increase in social and recreational stops from 6% to 9% over the study period. Regional comparative analysis in 2017 reveals that NYC female workers exhibited significantly lower rates of direct post-work home travel and significantly higher rates of family or personal business stops relative to all other demographic groups examined.

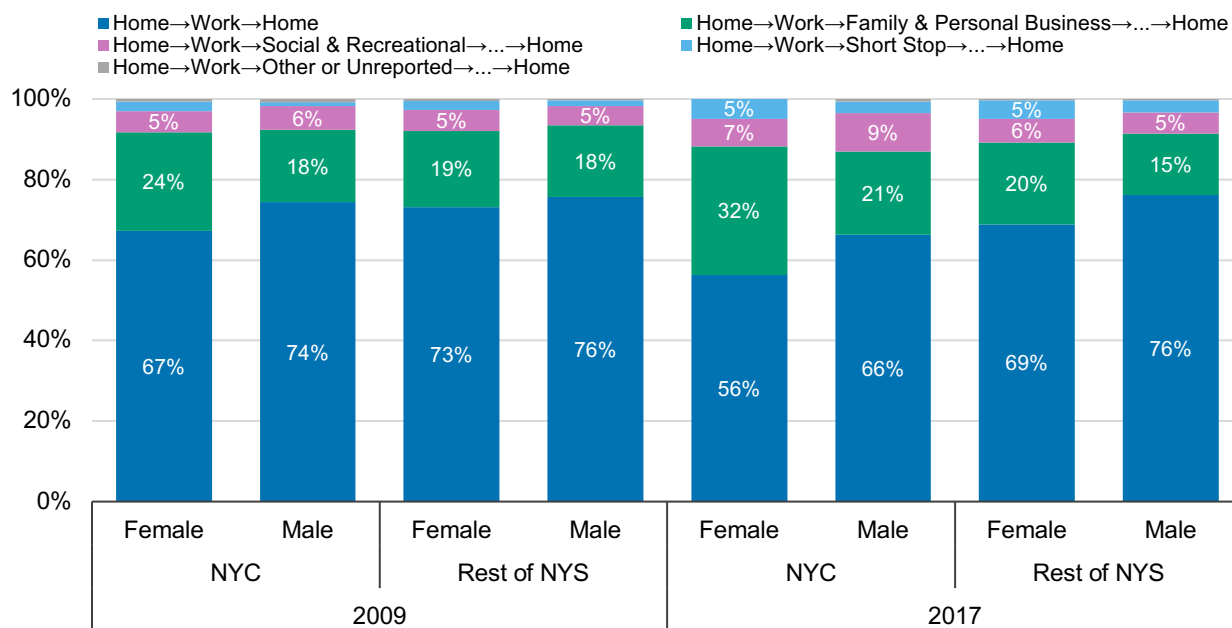


Figure ES.7. Post-Work Activity Patterns by Sex in NYC and Rest of NYS: Home→Work Trip Chains

Analysis of 2022 NHTS data for the Middle Atlantic Census Division demonstrates a pronounced shift toward direct home-to-work commuting patterns. In 2022, 90% of workers in both college-educated and non-college-educated cohorts engaged in direct work travel, representing a statistically significant increase relative to previous survey years for both demographic groups (Figure ES.8). Concurrently, intermediate stops for family and personal business activities declined to minimal levels (3–4%), while short stops remained similarly limited (3–5%). By 2022, previously observed differences between educational attainment groups no longer achieved statistical significance. Comparable patterns emerge across the rest of United States. These temporal changes may be attributable to methodological factors, specifically the reduced sample size in the 2022 survey iteration. Among survey respondents, the proportion reporting work-related trips on the designated survey day was further diminished, potentially introducing selection bias that may have influenced the observed travel behavior patterns.

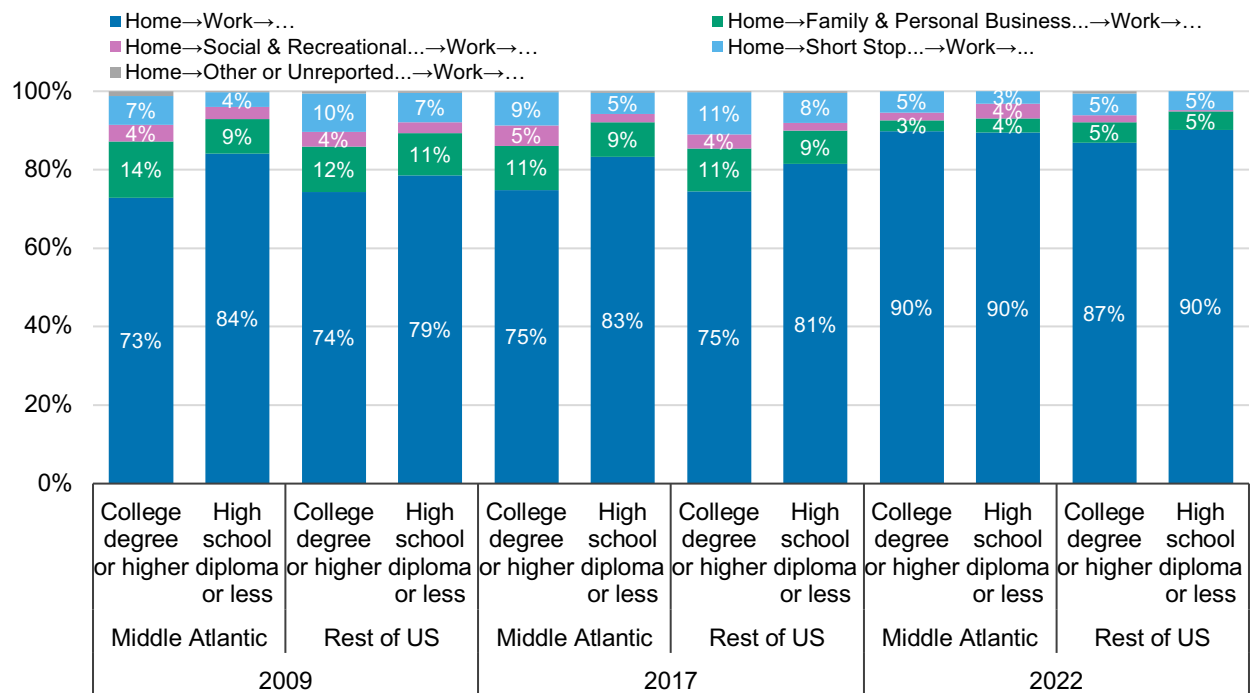


Figure ES.8 Pre-Work Activity Patterns by Education Level in the Middle Atlantic Census Division and Rest of the United States

1. INTRODUCTION

1.1 BACKGROUND

Trip chaining, defined as the sequential linking of multiple trips within individual travel patterns, provides critical insights into daily mobility behaviors and activity organization. This study examines individual trip chains using data from the National Household Travel Survey (NHTS), where each chain represents a sequence of spatially and temporally connected trip records. Trip chaining behavior reflects the systematic organization of activities across geographic and temporal dimensions, particularly the sequential ordering of destinations throughout daily activity patterns. Analysis of post-work trip destinations, for instance, reveals prevalent behavioral routines and underlying mobility requirements. These travel patterns have significant implications for transportation demand forecasting, traffic congestion management, and commercial location planning, as they directly influence the spatial and temporal distribution of travel flows and economic activity.

Most of the literature has focused on individual trip characteristics (Pan et al. 2025; 2024). However, research on multi-destination journey patterns and trip-chaining methodologies has been established since the 1980s (Thill and Thomas 1987). Subsequent investigations have examined trip chaining behaviors across demographic segments (McGuckin and Murakami 1999), the influence of temporal and environmental factors (Vande Walle and Steenberghen 2006; Yang et al. 2016), and interactions with mode choice decisions (Hensher and Reyes 2000). The proliferation of comprehensive travel diary datasets, particularly through large-scale surveys such as the NHTS, has enabled researchers to develop more nuanced understanding of population-specific trip chaining adoption patterns and their underlying determinants.

This analysis examines trip chaining behaviors in New York State (NYS) utilizing 2009 and 2017 NHTS datasets. Additionally, 2022 NHTS data enables comparative analysis between the Middle Atlantic Census Division and the rest of the United States. Through temporal comparative analysis, this study characterizes evolving trip chaining patterns across demographic and geographic segments. The findings provide empirical support for transportation planning applications and contextualize the heterogeneity of travel requirements across population subgroups and temporal periods. While this research does not evaluate specific policy interventions or propose design modifications, it presents a comprehensive empirical characterization of intrastate trip chaining patterns and their evolution over time.

1.2 DATASET

This study used data from the NHTS, a recurring survey sponsored by the Federal Highway Administration (FHWA) that is widely recognized as a key source for understanding travel behavior in the United States (Federal Highway Administration 2017). The NHTS collects detailed information on individuals' daily travel, linked to personal, household, and vehicle characteristics, and includes variables such as trip frequency, distance, duration, mode, and purpose.

The 2009 NHTS included 151,147 households nationwide, with 25,510 from the core national sample and 124,637 from add-on participants. Among these, 14,102 households were from NYS, which—combined with the national sample—produced 16,165 usable NYS households. The 2017 NHTS surveyed 129,696 households nationwide, representing more than 264,000 individuals, and NYS's add-on sample contained 17,209 households.

In 2020, FHWA launched the Next Generation NHTS (NextGen NHTS) to provide a continuous travel monitoring program with more frequent national and regional data products. Each cycle targets at least 7,500 households to capture emerging trends. The 2022 NHTS collected responses from 7,893

households, 16,997 individuals, and 31,074 trips. Due to the smaller sample size, the 2022 survey is representative at the national and Census Division levels rather than at the state or local level (Federal Highway Administration 2023). Nine Census Divisions are available in the data, where New York State falls within the Middle Atlantic Division.

1.3 TRIP CHAIN DEFINITION

In this report, we define an individual's trip chain as the sequence of trips undertaken throughout a single day, where each trip corresponds to a record in the NHTS trip file. The primary focus of this analysis is on activity transitions (inferred from trip purposes) and transportation mode transitions, with attention to the following aspects:

- **Diversity of Activities.** We quantify the number of distinct activity types (i.e., trip purpose categories) each individual engages in over the course of a day. Home is not considered as an activity.
- **Pre-Work Transitions.** For individuals with a work trip, we investigate the activity after leaving home and before arriving at work, as well as the number of intermediate stops or distinct destinations visited.
- **Post-Work Transitions.** For individuals who made a work trip during the day, we examine the next activity following work before returning home. We also analyze how many distinct destinations (places) are visited after work and before home.
- **Non-Workday Activity Transitions.** For individuals who did not make a work trip on the assigned travel (survey) day, we identify the types of activities they engaged in and describe the overall structure of their daily trip chain.
- **Transportation Mode Transitions**
 - We analyze the mode used after a trip made with a Privately Operated Vehicle (POV), excluding cases where the following trip is also via POV.
 - Similarly, we analyze the mode used after a trip made using public transit, excluding cases where the following trip also uses public transit.

The activity transition sequence is derived from the trip purpose field in the NHTS dataset (Table 1-1). The mode of transportation is derived from the trip mode field in the NHTS dataset (Table 1-2). These categories were used to define transition patterns between activities and to analyze modal transfers associated with these transitions.

Table 1-1. Activity Type and Corresponding Trip Purposes (variable: WHYTO)

Type of activity	Trip purpose defined in 2022 NHTS	Trip purpose defined in 2017 NHTS	Trip purpose defined in 2009 NHTS
Home	Regular home activities (chores, sleep), Work from home (paid)	Regular home activities (chores, sleep), Work from home (paid)	Home
Work	Work, Work-related meeting / trip, Attend school as a student	Work, Work-related meeting / trip, Attend school as a student	Work, Go to work, Return to work, Attend business meeting/trip, Other work related, School/religious activity, Go to school as student
Family & Personal Business	Attend child care, Attend adult care, Buy goods (groceries, clothes, appliances, gas), Buy services (dry cleaners,	Attend child care, Attend adult care, Buy goods (groceries, clothes, appliances, gas), Buy services (dry cleaners,	Medical/dental services, Shopping/errands, Buy goods: groceries/clothing/hardware store, Buy services: video

	banking, service a car, pet care), Buy meals (go out for a meal, snack, carry-out), Other general errands (post office, library), Health care visit (medical, dental, therapy), Religious or other community activities	banking, service a car, pet care), Buy meals (go out for a meal, snack, carry-out), Other general errands (post office, library), Health care visit (medical, dental, therapy), Religious or other community activities	rentals/dry cleaner/post office/car service/bank, Buy gas, Family personal business/obligations, Use professional services: attorney/accountant, Attend funeral/wedding, Use personal services: grooming/haircut/nails, Pet care: walk the dog/vet visits, Attend meeting: PTA/home owners association/local government
Social & Recreational	Recreational activities (visit parks, movies, bars, museums), Exercise (go for a jog, walk, walk the dog, go to the gym), Visit friends or relatives	Recreational activities (visit parks, movies, bars, museums), Exercise (go for a jog, walk, walk the dog, go to the gym), Visit friends or relatives	Social/recreational, Go to gym/exercise/play sports, Rest or relaxation/vacation, Visit friends/relatives, Go out/hang out: entertainment/theater/sports event/go to bar, Visit public place: historical site/museum/park/library
Short Stop	Drop off /pick up someone (personal), Change type of transportation	Drop off /pick up someone, Change type of transportation	Transport someone, Pick up someone, Take and wait, Drop someone off
Other activities	Something else (specify)	Something else, Not ascertained, I don't know, I prefer not to answer	Other reason, Appropriate skip, Refused, Don't know, Not ascertained

Table 1-2. Mode of Transportation Type and Its Corresponding Modes

Mode of transportation	Mode defined in 2022 NHTS	Mode defined in 2017 NHTS	Mode defined in 2009 NHTS
Nonmotorized	Walk, Bicycle (including bikeshare, ebike, etc.), E-scooter	Walk, Bike	Walk, Bicycle
POV	Car, SUV, Van, Pickup truck, Motorcycle, Recreational Vehicle	Car, SUV, Van, Pickup truck, Motorcycle / Moped, RV (motor home, ATV, snowmobile), Rental car (Including Zipcar / Car2Go)	Car, SUV, Van, Pickup truck, Motorcycle, RV
Public transport	Public or commuter bus, Amtrak, Commuter rail, Subway or elevated rail	Public or commuter bus, Amtrak / Commuter rail, Subway / elevated / light rail / street car, City-to-city bus (Greyhound, Megabus)	Local public bus, Commuter bus, City to city bus, Amtrak/inter city train, Commuter train, Subway/elevated train, Street car/trolley
Other modes	Airplane, Taxicab or limo service, Other ride-sharing service, Paratransit/ Dial a ride, Other (specify)	Airplane, Golf cart / Segway, Boat / ferry / water taxi, Taxi / limo (including Uber / Lyft), School bus, Something Else	Light electric veh (golf cart), Shuttle bus, School bus, Taxicab, Ferry, Special transit-people w/disabilities, Other

In this report, we focused on key transition points. Specifically, the first trip following or preceding key activities such as "home" or "work."

- **Pre-Work Trip Chaining Patterns.** We analyzed the share of patterns reflecting how individuals travel before arriving at work:
 - Home → Work → Home: The individual traveled directly from home to work without making any stops.
 - Home → Next activity → ... → Work → ...: The individual stopped for another activity before going to work. The "next activity" refers to the first activity after leaving home.
- **Post-Work Trip Chaining Patterns.** We analyzed the share of patterns reflecting how individuals travel after leaving work:
 - Home → Work → Home: The individual returned directly home after work.
 - Home → Work → Next activity → ... → Home: The individual conducted other activities after work. The "next activity" refers to the first activity after leaving work.
- **Non-Workday Activity Patterns.** This includes all weekends, holidays, and vacation or off days for both employed and unemployed individuals. While it is possible to analyze weekends and other non-workdays separately, doing so often limits sample size. Therefore, we group all non-workdays together, defined as days when individuals made no work trips. We analyzed:
 - Home → Next activity... → ...: The "next activity" refers to the first activity after leaving home, helping us understand typical behavior on non-workdays.
- **Transportation Mode Transition.** We examined how individuals transition between modes during their travel:
 - ... → Target mode → Next mode → ...: For a given target mode (e.g., public transport or personal vehicle), we identified the next mode used in the sequence when people switch to a different mode.

We also analyzed the number of different types of activities people engaged in on a given day. As defined in Table 1-1, activity types include work, family and personal business, social and recreational activities, and short stops. We grouped individuals trip pattern into the following categories:

- **No trip.** Did not make any trips on the survey day.
- **One type of activity.** Made one or more trips, but all for the same type of activity.
- **Two types of activities.** Made trips for two distinct types of activities.
- **Three or more types of activities.** Made trips for three or more different activity types.

2. TRIP CHAINING BEHAVIOR BETWEEN NYS AND REST OF US

2.1 TRIP CHAINING BY OCCUPATION

Table 2-1 shows the number of distinct activity types (based on trip purposes) individuals engaged in over a day. In both years in NYS, the majority of people reported two or more activity types. Around 40% of clerical or professional workers falling into the two-activity category. Clerical or professional workers show slightly higher shares in the 3+ activity group than the other two groups, indicating more varied daily routines, with 16% in both NYS and the rest of the United States in 2017. Between 2009 and 2017, there were no statistically significant differences among any groups or the two regions.

Table 2-1. Share of Daily Number of Activity Types by Occupation in NYS and Rest of the United States

NHTS year	Region	Occupation	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	Clerical or professional	9%	32%	43%	17%
		Manufacturing, construction, maintenance, or farming	10%	42%	36%	11%
		Sales or service	7%	39%	38%	16%
	Rest of US	Clerical or professional	6%	32%	42%	20%
		Manufacturing, construction, maintenance, or farming	9%	41%	37%	13%
		Sales or service	7%	37%	38%	17%
2017	NYS	Clerical or professional	10%	35%	39%	16%
		Manufacturing, construction, maintenance, or farming	17%	39%	33%	12%
		Sales or service	14%	40%	34%	12%
	Rest of US	Clerical or professional	10%	35%	38%	16%
		Manufacturing, construction, maintenance, or farming	13%	44%	33%	11%
		Sales or service	13%	41%	33%	13%

2.1.1 Pre-Work Trip Chaining

Figure 2-1 shows the distribution of activity patterns involving activities before arriving at work. Direct home-to-work travel is the most common pattern across all occupations and years, with over 70% of individuals following it across all regions and groups. This is particularly pronounced among manufacturing and construction workers, reaching 82% in NYS in 2017, reflecting their structured, time-constrained routines.

Family and personal business was the most common pre-work activity among those not traveling directly to work, with 10–16% of individuals across occupations making such stops across regions and groups. The highest share in 2009 appeared among manufacturing workers in the rest of the United States. Social and recreational stops remained consistently low (around 2–4%) across all groups and years, indicating these are generally not typical pre-work activities.

Clerical or professional workers show a greater tendency to make short stops before work. In NYS, this rose slightly from 8% in 2009 to 10% in 2017 (though not statistically significant), while in the rest of the United States, the increase from 9% to 12% was statistically significant, possibly reflecting more flexible schedules or added morning tasks.

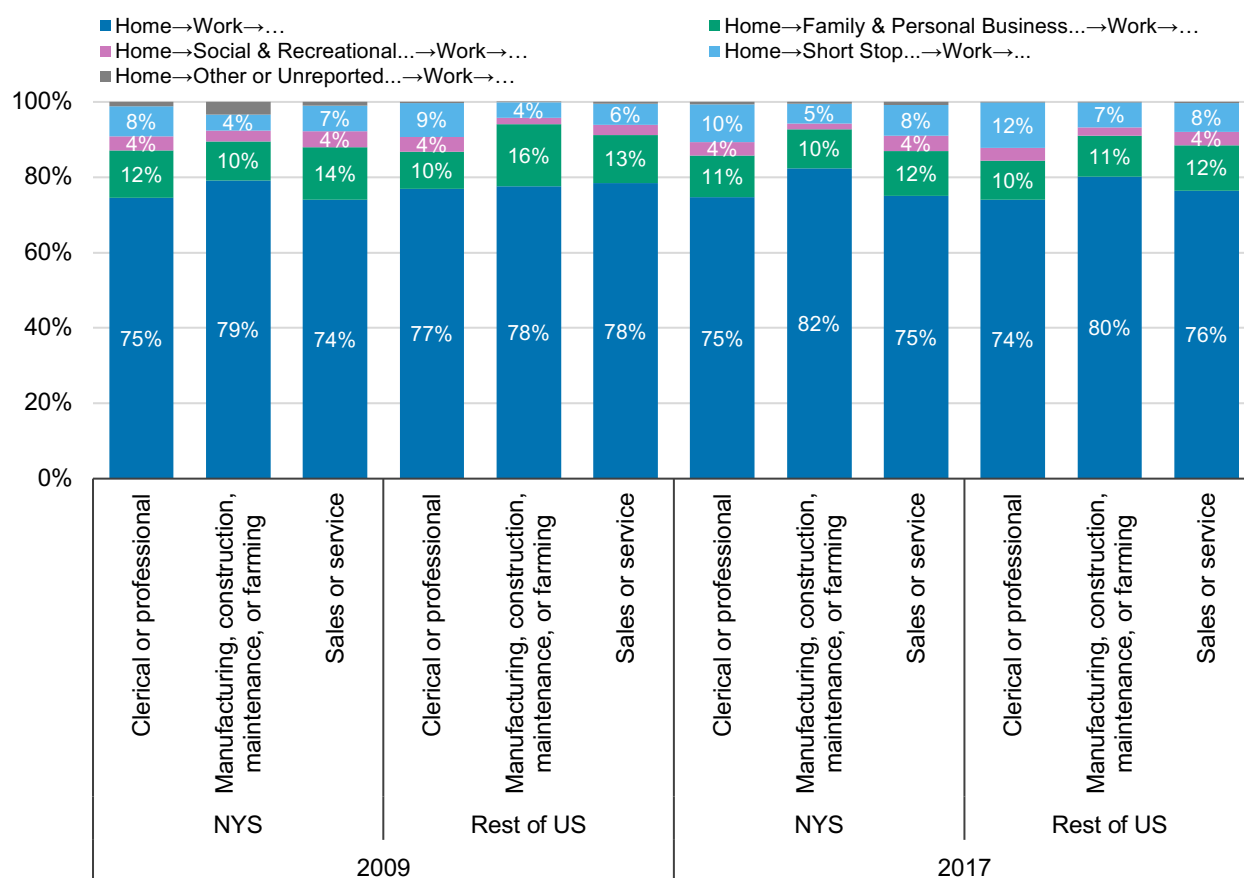


Figure 2-1. Pre-Work Activity Patterns by Occupation in NYS and Rest of the United States

While Figure 2-1 shows the first trip individuals made after leaving home, Table 2-2 provides a breakdown of the exact number of trips made between leaving home and arriving at work. Individuals who went directly to work without making any intermediate stops are categorized as “No trip.”

The table shows that direct commute to work (no trip) remains the dominant pattern, statistically significantly higher than other trip chaining behaviors across all occupations and years. This is especially common among manufacturing and construction workers, with 75–85% commuting directly in 2009 and 2017. One stop before work (one trip) is the most common form of activity chaining beyond a direct commute. This is most prevalent among clerical or professional workers, particularly in the rest of the United States (16% in 2017 and 14% in 2009), possibly reflecting more flexibility or responsibilities before work. Between 2009 and 2017, direct commuting increased slightly for sales or service workers in both regions.

2.1.2 Post-Work Trip Chaining

Across all occupations and both years, most individuals follow a direct home→work→home travel pattern which is statistically significantly higher than other trip chaining patterns, with the highest shares among manufacturing, construction, maintenance, or farming workers: 77% in 2009 and 79% in 2017 in NYS (Figure 2-2). These rates are statistically significantly higher than those for clerical or professional workers in the same region (63% in 2009 and 62% in 2017), suggesting the latter group may have more schedule flexibility or reasons for intermediate stops.

Table 2-2. Distribution of Number of Trips Before Work by Occupation in NYS and Rest of the United States

NHTS year	Region	Occupation	No trip	One trip	Two trips	3+ trips
2009	NYS	Clerical or professional	75%	12%	8%	6%
		Manufacturing, construction, maintenance, or farming	79%	11%	7%	2%
		Sales or service	74%	10%	8%	8%
	Rest of US	Clerical or professional	75%	14%	7%	4%
		Manufacturing, construction, maintenance, or farming	82%	9%	5%	3%
		Sales or service	75%	11%	8%	6%
2017	NYS	Clerical or professional	77%	14%	6%	3%
		Manufacturing, construction, maintenance, or farming	78%	15%	4%	3%
		Sales or service	78%	10%	7%	5%
	Rest of US	Clerical or professional	74%	16%	7%	4%
		Manufacturing, construction, maintenance, or farming	80%	11%	5%	4%
		Sales or service	76%	11%	7%	6%

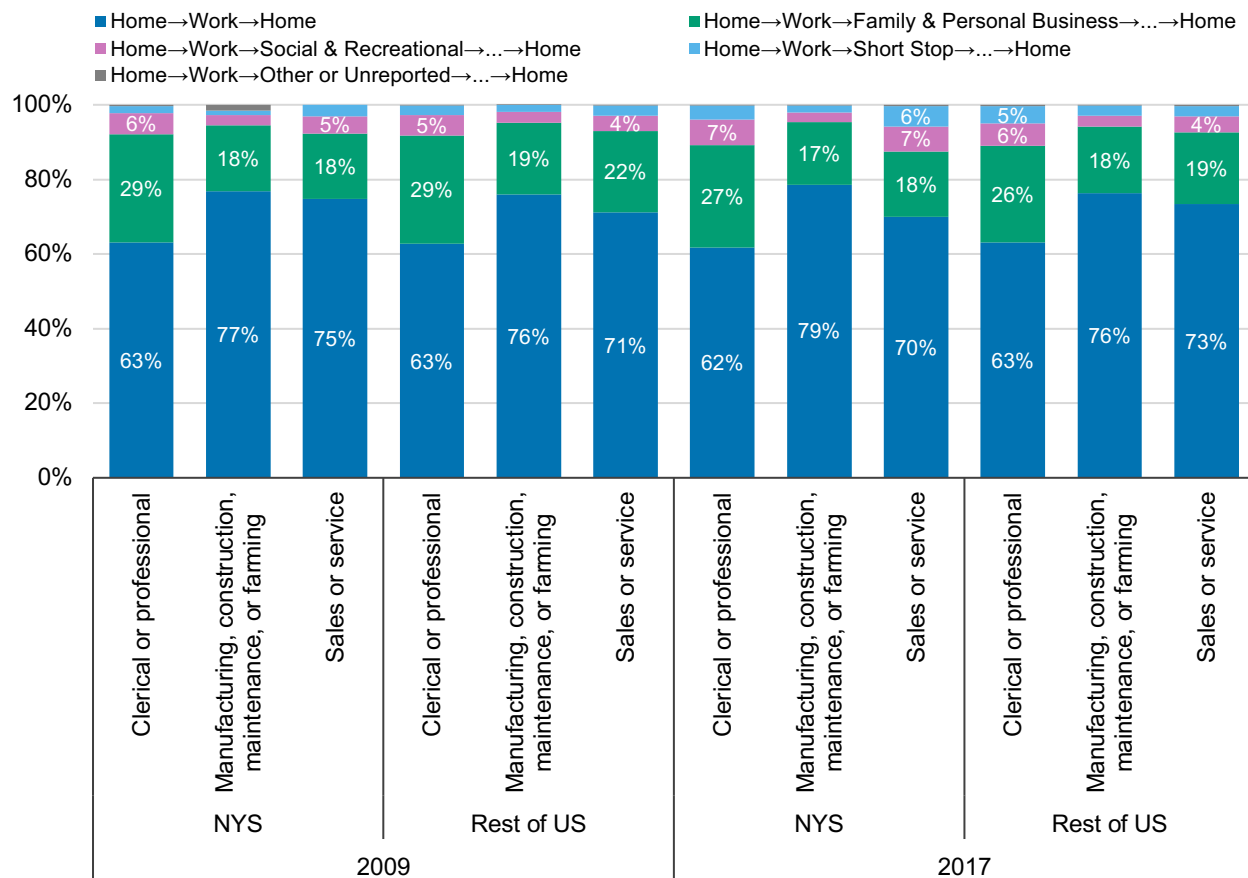


Figure 2-2. Post-Work Activity Patterns by Occupation in NYS and rest of the United States: Home→Work Trip Chains

The second most common pattern is stopping for family or personal business before heading home. This is more prevalent among clerical or professional workers, reaching 29% in 2009 across both NYS and the rest of the United States, with a modest, non-significant decline to 26–27% in 2017. Regional patterns are largely consistent. However, in the rest of the United States, clerical or professional workers show a statistically significant increase in short stops after work, rising from 3% in 2009 to 5% in 2017.

While Figure 2-2 shows the first trip individuals made after leaving work, Table 2-3 provides a breakdown of the exact number of trips made between leaving work and arriving home. Individuals who went directly back home without making any intermediate stops are categorized as “No trip.” Clerical or professional workers were more likely to make multiple post-work trips, particularly in 2009. For example, in NYS, 26% of clerical or professional workers made two trips after work, statistically significantly higher than 19% among manufacturing workers. By 2017, clerical or professional workers still had the highest share of making two more trips after work. Manufacturing, construction, maintenance, or farming workers had the highest rates of going straight home across all years and regions, with the rest of the United States showing a statistically significant increase from 40% in 2009 to 50% in 2017, likely tied to more fixed schedules or job structure. Sales or service workers in NYS also saw a statistically significant decline in the share making three or more trips, from 30% in 2009 to 20% in 2017.

Table 2-3. Distribution of Number of Trips After Work by Occupation in NYS and Rest of the United States

NHTS year	Region	Occupation	No trip	One trip	Two trips	3+ trips
2009	NYS	Clerical or professional	30%	10%	26%	34%
		Manufacturing, construction, maintenance, or farming	48%	10%	19%	23%
		Sales or service	39%	9%	22%	30%
	Rest of US	Clerical or professional	30%	10%	23%	37%
		Manufacturing, construction, maintenance, or farming	40%	10%	23%	28%
		Sales or service	37%	11%	20%	32%
2017	NYS	Clerical or professional	40%	16%	19%	25%
		Manufacturing, construction, maintenance, or farming	48%	12%	17%	23%
		Sales or service	48%	15%	17%	20%
	Rest of US	Clerical or professional	37%	16%	21%	26%
		Manufacturing, construction, maintenance, or farming	50%	15%	15%	20%
		Sales or service	46%	13%	19%	21%

2.1.3 Non-Work Days Trip Chaining

Figure 2-3 presents the distribution of daily trip chains for individuals who did not report any work-related travel on the survey day. Family and personal business is the most common first activity after leaving home, statistically significantly higher than other activities across all occupations and years, accounting for over half of non-work trip chains. The highest shares are consistently observed among manufacturing workers.

While most occupational differences are not statistically significant, there are two notable exceptions in the rest of the United States in 2017. First, 68% of manufacturing workers began their day with family or personal business, statistically significantly higher than the 61% observed among clerical or service workers. Second, among manufacturing workers, this share also increased statistically significantly from

62% in 2009 to 68% in 2017, suggesting a growing reliance on these types of trips on non-work days, possibly reflecting practical household responsibilities or changes in daily routines.

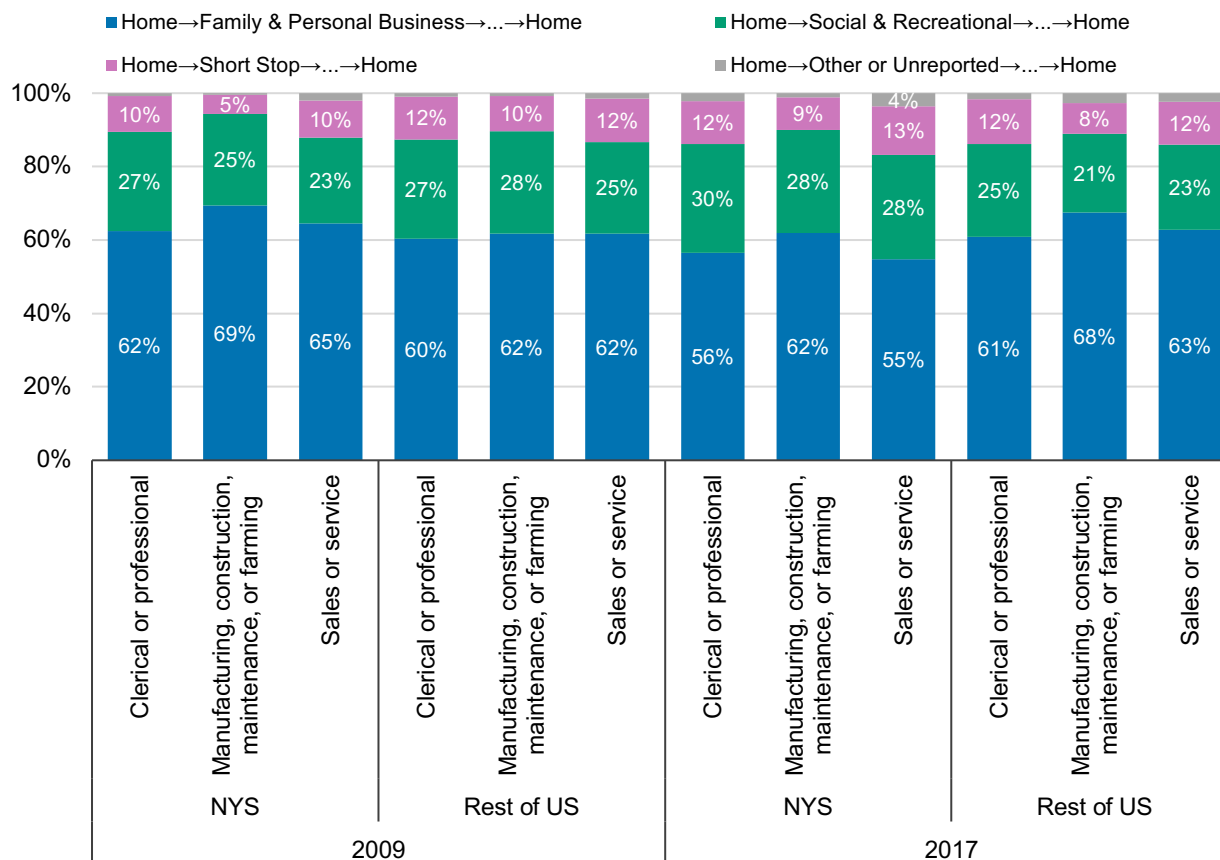


Figure 2-3. Activity Patterns on Non-Work Days by Occupation in NYS and Rest of the United States

2.1.4 Transportation Mode Transition

Figure 2-4 summarizes the mode of transportation used immediately after a trip made by personal vehicle. Across all occupations and years, nonmotorized modes (e.g., walking, biking) were the most common transfers, statistically significantly higher than other modes of transportation. In 2009, nonmotorized modes exceeded 85% for most groups. By 2017, this share declined slightly in NYS, particularly among clerical or professional workers (from 84% to 71%), though the change was not statistically significant.

Public transport use after Privately Operated Vehicle (POV) trips was relatively rare overall but more common in NYS, especially among sales or service workers. In NYS, 21% of sales or service workers transferred to public transport in both 2009 and 2017, compared to only 4–5% in the rest of the United States. This difference between the two regions was statistically significant in 2009. In the rest of the United States, the share of manufacturing workers who switched to other or unreported modes of transportation increased significantly, from 11% to 22%. This suggests that a growing number of these workers may be using alternative or less commonly reported transportation modes after initially traveling by personal vehicle, potentially reflecting increased flexibility or changes in last-mile travel behavior.

Figure 2-5 shows the mode of transportation used after a public transport trip, excluding cases where the next trip was also by public transit. Nonmotorized modes (walking, biking) were the most common

follow-up across most occupations and years, especially in NYS. In 2009, for example, 70–80% of NYS workers continued their trip by foot or bike, statistically significantly higher than those switching to POV across all occupations. In contrast, POV as a follow-up mode was much more common in the rest of the United States, particularly in 2009. Among sales or service workers, 45% used a POV next—comparable to the share using nonmotorized modes—while 37% of clerical workers also transferred to POV, though this was still statistically significantly lower than nonmotorized use.

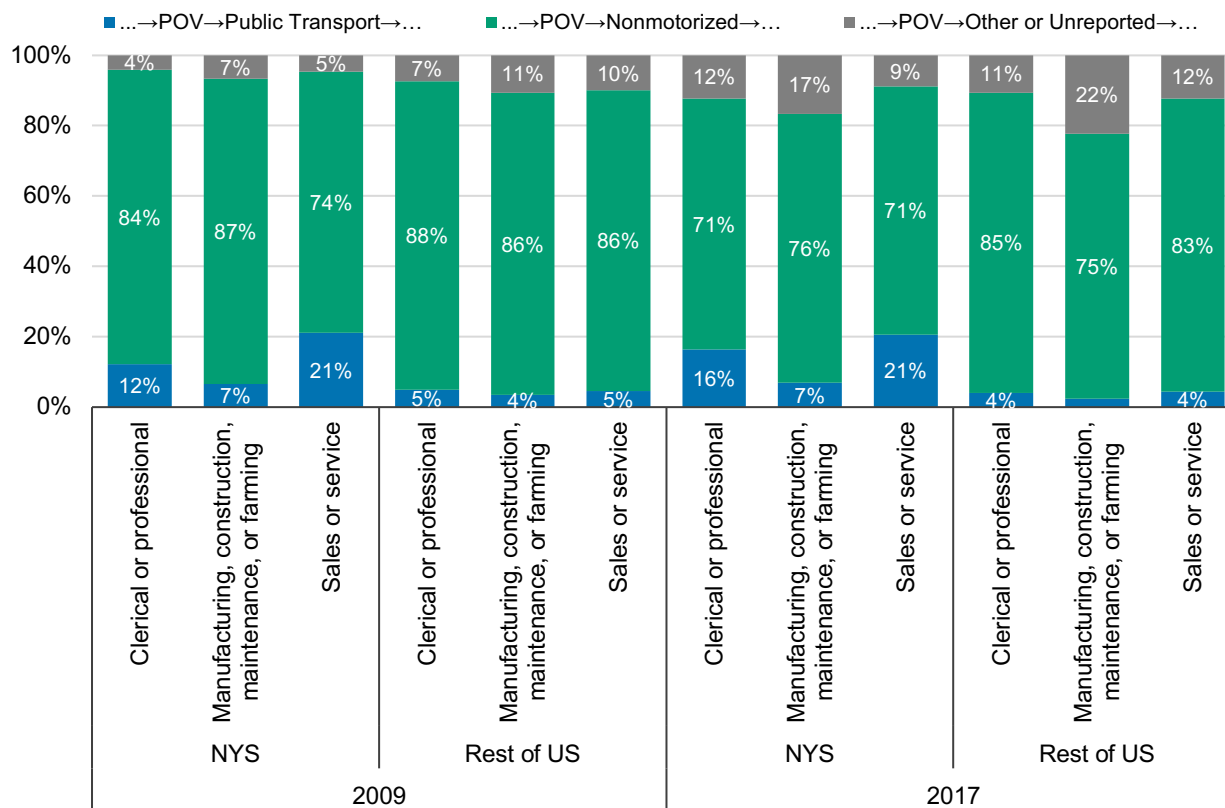


Figure 2-4. Transportation Mode Transition Following a POV Trip by Occupation in NYS and Rest of the United States

2.2 TRIP CHAINING BY OPTION OF WFH

In NYS, workers who could work from home (WFH) were statistically significantly more likely to engage in 3 or more activity types per day (24%) compared to those who could not WFH (15%) in 2009 (Table 2-4). Conversely, those who cannot WFH were more concentrated in the 1-activity category (37% vs. 30%), a statistically significant difference. This pattern suggests that WFH-capable individuals may have had more flexibility to engage in multiple types of activities during the day. In the rest of the United States, the pattern was similar. At the same time, those who cannot WFH were also more likely to have zero trips than their counterpart. In 2017, for example, 12% of non-WFH individuals in NYS reported no trips, statistically significantly higher than only 8% of individuals who could WFH.

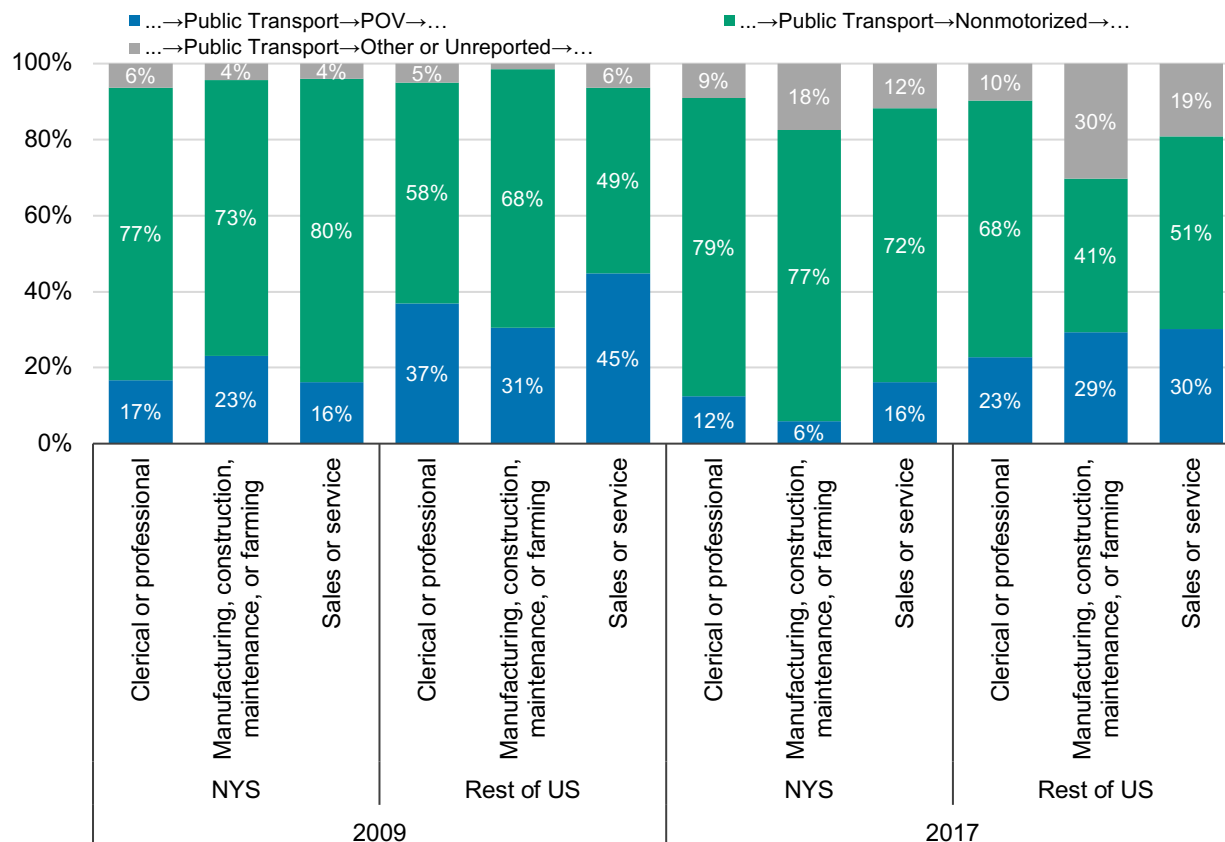


Figure 2-5. Transportation Mode Transition Following a Public Transport Trip by Occupation in NYS and Rest of the United States

Table 2-4. Daily Number of Activity Types by Option of WFH in NYS and Rest of the United States

NHTS year	Region	Option of WFH	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	Can WFH	7%	30%	39%	24%
		Cannot WFH	8%	37%	40%	15%
	Rest of US	Can WFH	6%	29%	42%	23%
		Cannot WFH	6%	36%	40%	18%
2017	NYS	Can WFH	8%	32%	42%	18%
		Cannot WFH	12%	39%	36%	14%
	Rest of US	Can WFH	8%	33%	39%	20%
		Cannot WFH	10%	40%	36%	14%

2.2.1 Pre-Work Trip Chaining

In both 2009 and 2017, workers who could not WFH were statistically significantly more likely to travel directly from home to work without making any stops compared to those who could WFH (Figure 2-6). In 2009 in NYS, 77% of non-WFH workers made a direct trip, compared to 68% of WFH-capable workers. A similar pattern was observed in 2017 (79% vs. 74%). This difference may reflect the greater flexibility of WFH-capable workers in planning their schedules and integrating additional activities into

their morning routine. In contrast, WFH-capable workers were more likely to make pre-work stops, particularly for short stops or family and personal business. For example, in 2009 in NYS, 12% of WFH workers made a short stop before work, statistically significantly higher than the 7% of non-WFH workers. The 2017 data in NYS further supports this pattern: 11% of WFH workers made short stops compared to 7% of non-WFH workers. The share of making stops for family and personal business was also slightly higher among WFH workers, although the difference was not always statistically significant.

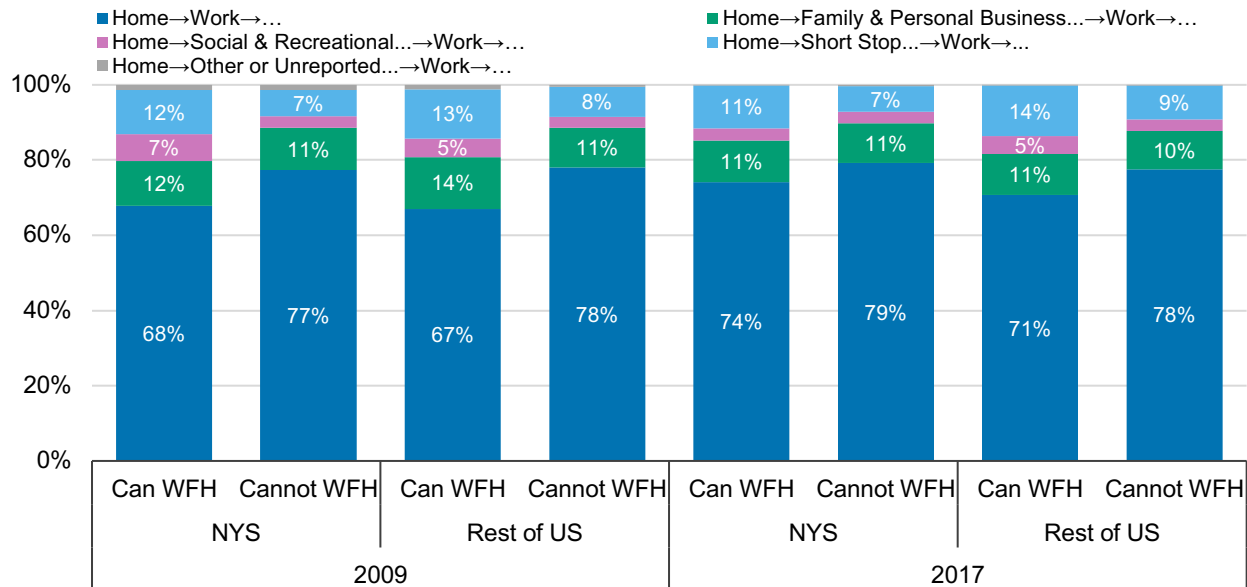


Figure 2-6. Pre-Work Activity Patterns by Option of WFH in NYS and Rest of the United States

2.2.2 Post-Work Trip Chaining

In NYS, workers who could not WFH were statistically significantly more likely to return directly home after work (70%) compared to those who could WFH (58%) in 2009 (Figure 2-7). Conversely, those who could WFH were statistically significantly more likely to engage in family & personal business trips after work (33% vs. 22%). In the rest of the United States, similar trends were observed. The 2017 data show that patterns in NYS and the rest of the United States remained consistent. This difference may reflect greater schedule flexibility or autonomy among WFH-capable workers, who are often in professional or managerial roles with more control over their time. These individuals may combine errands or responsibilities into their commute. On the other hand, workers unable to WFH may have more rigid schedules, limited discretionary time, or higher job-related stress and fatigue, making them more likely to return home directly after work.

In 2017, within NYS, we further analyzed the departure time patterns of individuals whose first trip after work was for family or personal business. Those who could work from home showed a more varied departure time pattern than their counterparts (Figure 2-8), highlighting their greater flexibility.

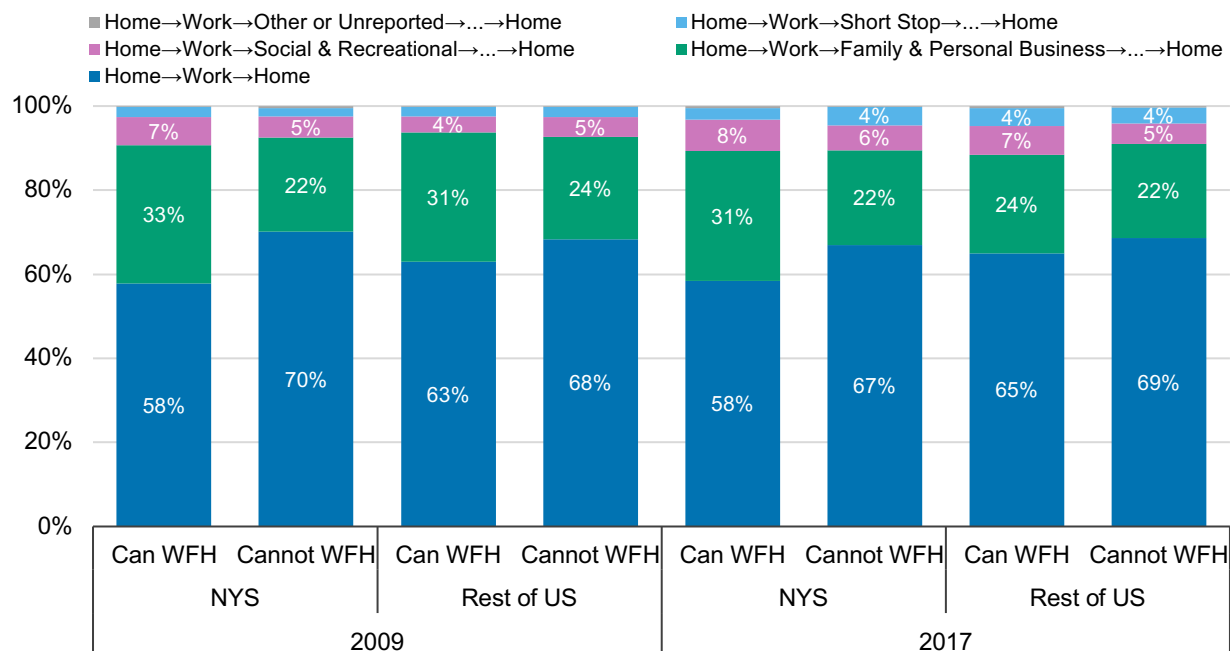
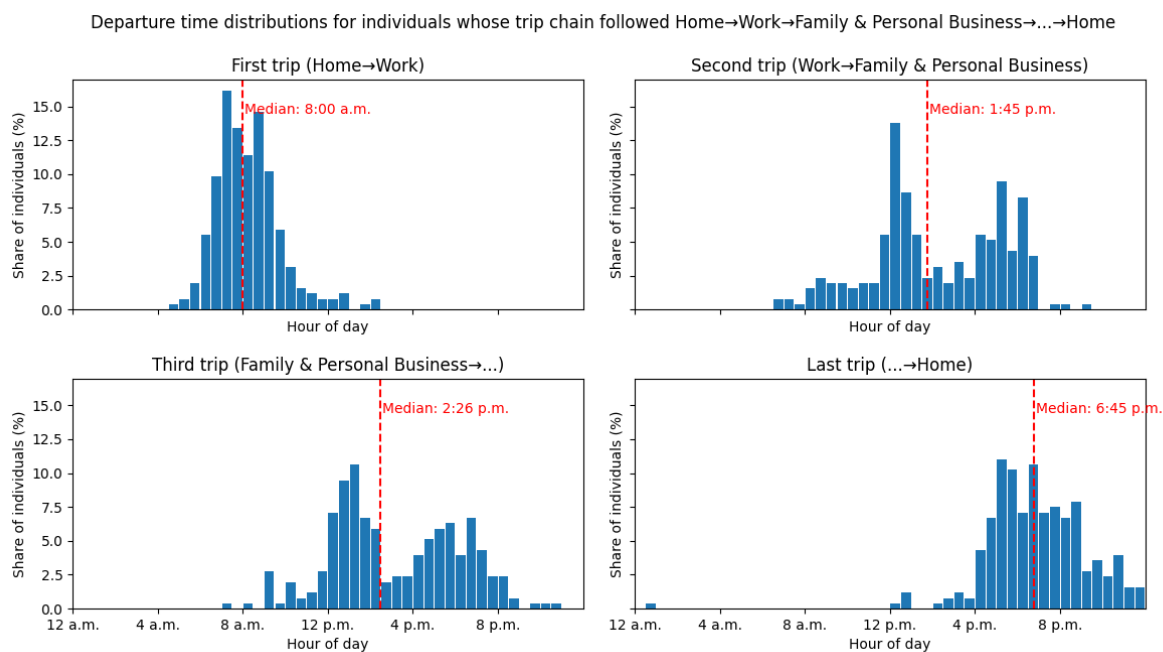
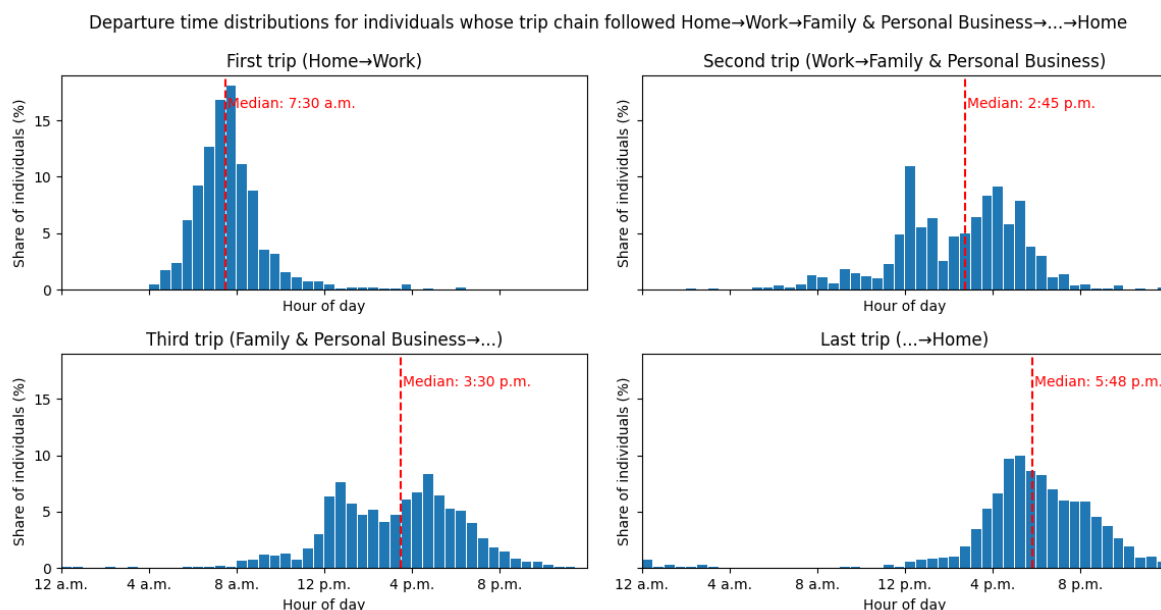


Figure 2-7. Post-Work Activity Patterns by Option of WFH in NYS and Rest of the United States: Home→Work Trip Chains



(a) Individuals who can WFH



(b) Individuals who cannot WFH

Figure 2-8. Departure Time Distribution of Individuals Whose First Trip After Work is Family & Personal Business Activity

2.2.3 Non-Work Trip Chaining

In both 2009 and 2017, both groups exhibited similar activity patterns on non-work days, with family and personal business being the most common first activity after leaving home (Figure 2-9). In 2009, in NYS, the share of individuals making family and personal business trips was identical for both groups (65%). In the rest of the United States in 2009, those who could WFH were statistically significantly less likely to start their day with family and personal business trips compared with their counterpart (57% vs. 63%). This group was more likely to engage in social and recreational activities (29% vs. 26%) and short stops (12% vs. 10%), although the differences were not statistically significant.

By 2017, in NYS, the percentage of those who could WFH whose day began with family and personal business trips decreased significantly to 52%, compared to 58% for non-WFH individuals. Meanwhile, the share of those who could WFH engaging in social and recreational activities rose statistically significantly to 32%, though this was not statistically different from the 29% for non-WFH individuals. In the rest of the United States in 2017, patterns were similar to 2009.

2.2.4 Transportation Mode Transition

In NYS in 2009, WFH-capable individuals were more likely to switch to a nonmotorized mode (79%) after a public transport trip, compared with 76% among those who could not WFH (Table 2-5). The difference is not statistically significant. However, those who could not WFH were statistically significantly more likely to switch to a POV afterward (18%) than their counterpart (12%). In the rest of the United States, there were no significant differences between WFH and non-WFH individuals in the choice of follow-up modes.

By 2017, the transfer patterns in NYS remained largely similar. In the rest of the United States, non-WFH individuals were statistically significantly more likely to switch to a POV (29%) than those who could WFH (17%). Meanwhile, individuals who could WFH were more likely to switch to nonmotorized modes

(69%) compared with 59% for non-WFH people, though this difference is not statistically significant. These shifts suggest that flexibility in work location may be associated with more use of nonmotorized travel after using public transport.

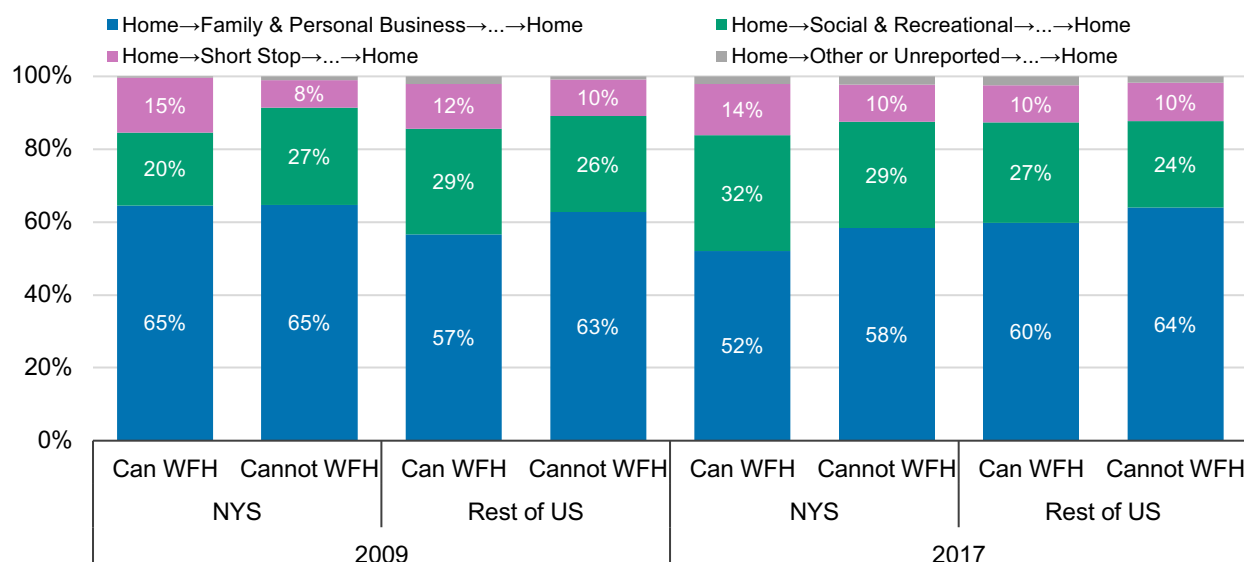


Figure 2-9. Activity Patterns on Non-Work Days by Option of WFH in NYS and Rest of the United States

Table 2-5. Transportation Mode Transition Following a Public Transport Trip by Option of WFH in NYS and Rest of the United States

NHTS year	Region	Option to WFH	...→Public Transport→POV→...	...→Public Transport→Nonmotorized→...	...→Public Transport→ Other or Unreported→...
2009	NYS	Can WFH	12%	79%	9%
		Cannot WFH	18%	76%	6%
	Rest of US	Can WFH	40%	58%	3%
		Cannot WFH	41%	55%	4%
2017	NYS	Can WFH	10%	79%	11%
		Cannot WFH	14%	76%	10%
	Rest of US	Can WFH	17%	69%	14%
		Cannot WFH	29%	59%	12%

2.3 TRIP CHAINING BY AGE

The number of distinct activity types individuals engaged in per day varied by age group but remained similar across regions and survey years (Table 2-6). In 2009, the majority of individuals across all groups reported either one or two activity types per day. Among those between 25 and 65, 36–37% in both NYS and the rest of the United States engaged in one activity type. Younger individuals (age 5 to 25 as NHTS does not report any information for age under 5) were statistically significantly more likely to report only one activity (46% in NYS and 41% in the rest of the United States) than their counterpart. By 2017, there was an increase in the share of individuals reporting no trips, rising statistically significantly from 11% to 16% among those between 25 and 65 and also statistically significantly from 14% to 21% among younger individuals in NYS. A corresponding decrease was seen in those reporting two or more activity types per day.

Table 2-6. Share of Daily Number of Activity Types by Age in NYS and Rest of the United States

NHTS year	Region	Age	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	Younger than 25	14%	46%	30%	9%
		25 to 65	11%	37%	38%	14%
		Older than 65	30%	41%	24%	5%
	Rest of US	Younger than 25	12%	41%	34%	13%
		25 to 65	11%	36%	38%	16%
		Older than 65	26%	41%	26%	6%
2017	NYS	Younger than 25	21%	43%	27%	9%
		25 to 65	16%	37%	34%	13%
		Older than 65	28%	41%	26%	6%
	Rest of US	Younger than 25	21%	42%	28%	9%
		25 to 65	16%	38%	33%	13%
		Older than 65	28%	41%	25%	6%

2.3.1 Pre-Work Trip Chaining

In both years, younger workers (under 25) were consistently the most likely to go directly from home to work in both NYS and the rest of the United States (Figure 2-10). In 2009, 90% of younger workers in NYS and 86% in the rest of the United States went straight to work, with the difference statistically significant. This pattern persisted in 2017, with 88% in NYS and 84% in the rest of the United States. For workers aged 25 to 65, direct home-to-work trips were slightly more common in NYS than in the rest of the United States. Family and personal business stops before work showed a clear age gradient across both regions. Older workers were statistically significantly more likely to make these stops than younger workers. In NYS in 2017, 23% of workers over 65 made family/personal business stops, statistically significantly higher than the 11% of workers aged 25–65 and the 5% of those under 25. A similar pattern existed in the rest of the United States, though the shares were slightly lower.

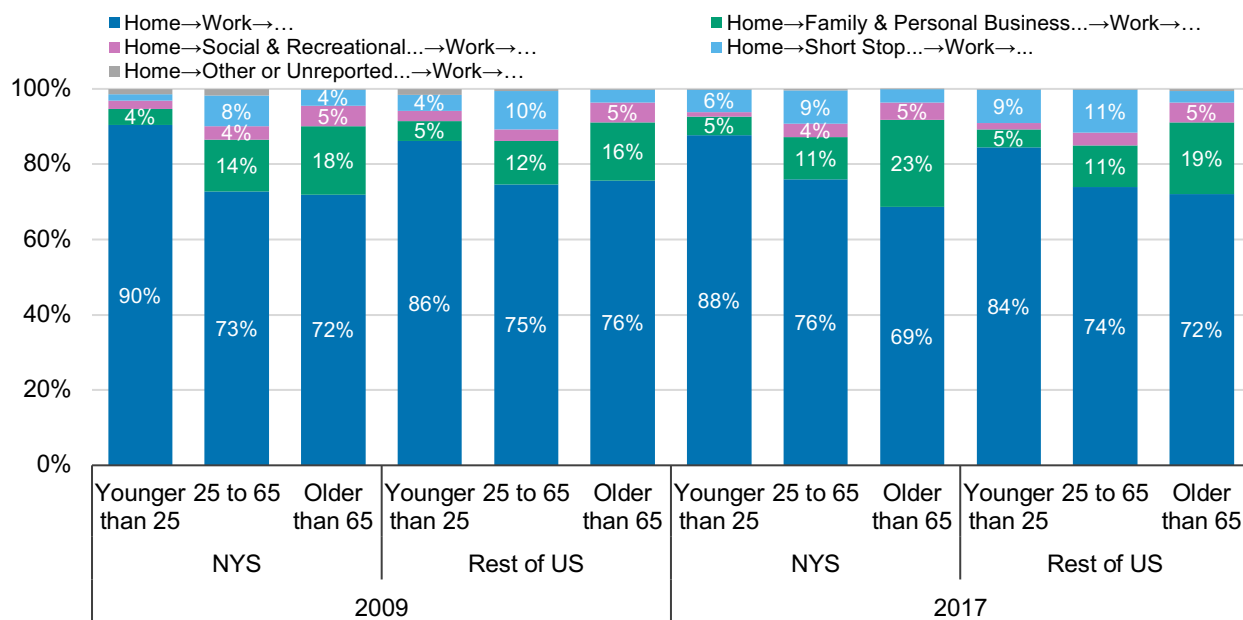


Figure 2-10. Pre-Work Activity Patterns by Age in NYS and Rest of the United States

2.3.2 Post-Work Trip Chaining

In both years, across all regions, younger workers (under 25) were statistically significantly more likely to go directly home after work compared to workers aged 25 to 65 and those over 65 (Figure 2-11). For younger workers (under 25), both in 2009 and 2017, the share going directly home after work was similar between NYS and the rest of the United States, with no statistically significant differences. In 2017, the share was 76% in NYS and 79% in the rest of the United States. The consistently high rates suggest limited regional variation in post-work behavior for this age group. Among workers aged 25 to 65, NYS workers were slightly less likely to go directly home after work compared to their counterparts in the rest of the United States. For older workers (over 65), the differences were more pronounced. In both years, NYS older workers were less likely to go directly home after work compared to their counterparts in the rest of the United States. Family and personal business stops were consistently more common in NYS across all age groups, especially for older workers. In 2017, 40% of older workers in NYS made family/personal business stops compared to 31% in the rest of the United States, a statistically significant difference. This pattern held for the working-age group as well, with 25% in NYS versus 24% in the rest of the United States. Though smaller in magnitude, these differences suggest a slightly greater tendency among NYS workers to engage in family-related activities after work.

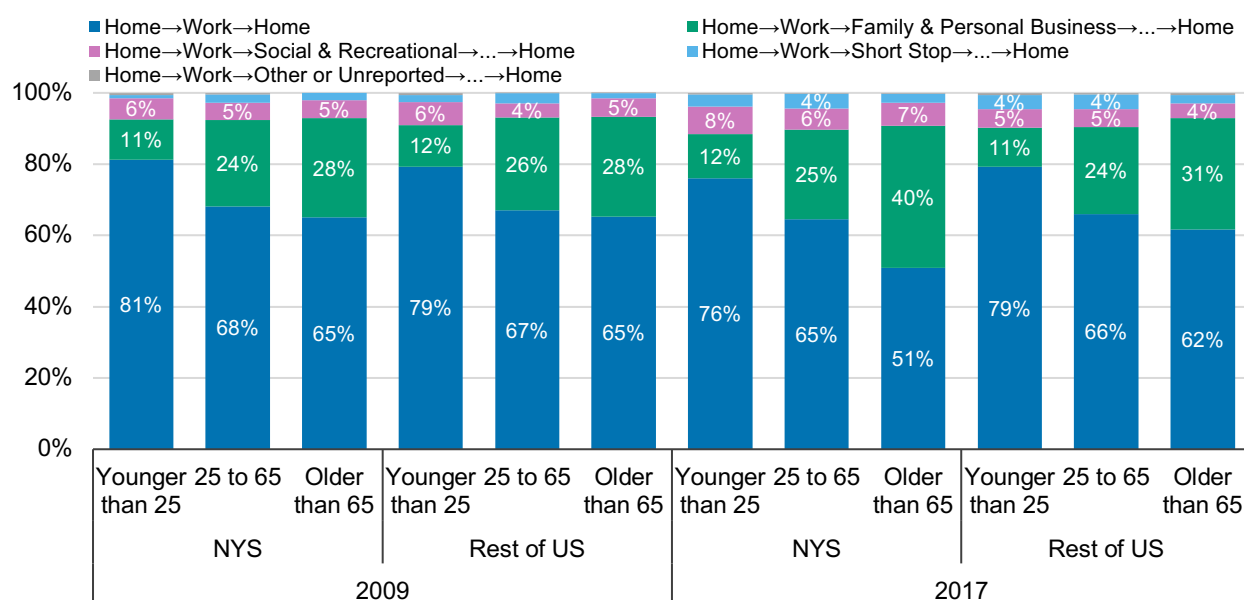


Figure 2-11. Post-Work Activity Patterns by Age in NYS and Rest of the United States: Home→Work Trip Chains

2.3.3 Non-Work Days Trip Chaining

On non-work days, younger individuals (under 25) had similar shares of social and recreational trips between NYS and the rest of the United States in both years. For example, 37% in NYS vs. 35% in the United States in 2017, with no statistically significant change over time. Among working-age adults (25 to 65), the share making family and personal business trips in NYS decreased statistically significantly from 65% in 2009 to 59% in 2017, while the United States showed no significant change (61% in 2009 and 62% in 2017). In 2009, NYS had a statistically significantly higher share than the United States, but this difference was no longer significant by 2017. For older adults (65+), the share making family/personal business trips remained stable around 70% in both NYS and the United States across both years, with no statistically significant differences by region or year. The share of working-age adults in

NYS making short stops increased statistically significantly from 10% in 2009 to 14% in 2017, while the share in the United States remained at 14%, with no significant change.

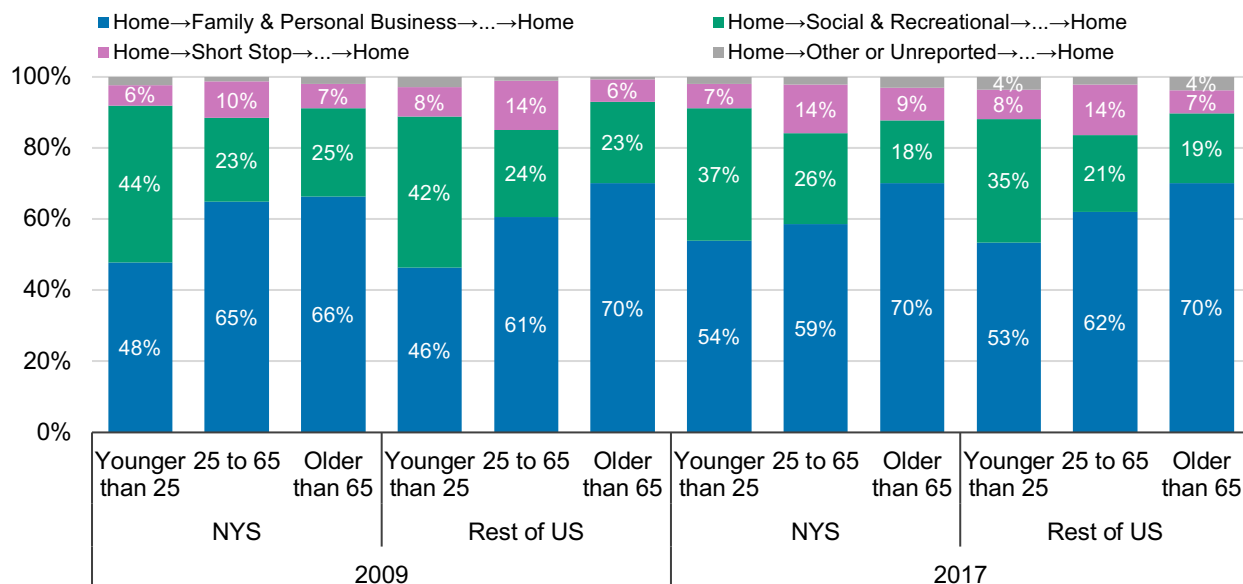


Figure 2-12. Activity Patterns on Non-Work Days by Age in NYS and Rest of the United States

2.3.4 Transportation Mode Transition

Figure 2-13 shows that, in both years, NYS workers across all age groups were statistically significantly more likely to transfer from POV to public transport compared to the United States, especially among younger individuals and working-age adults. In 2009, 13% of younger workers in NYS transferred from POV to public transport compared to 4% in the United States, a statistically significant difference. In 2017, this pattern persisted, with 11% in NYS vs. 3% in the United States. Among working-age adults (25 to 65) in NYS, the share transferring from POV to public transport increased statistically significantly from 13% in 2009 to 15% in 2017, while remaining unchanged at 3% in the United States. For older adults (65+), NYS also consistently showed a higher share of transfers to public transport, though with smaller differences. Nonmotorized transitions (e.g., walking, biking) were common across all age groups and regions, but consistently higher in the United States than NYS, especially among working-age and older adults. In 2017, 86% of working-age adults in the United States transferred from POV to nonmotorized modes compared to 76% in NYS, a statistically significant difference. For both years and regions, the share of individuals who transferred from POV to other modes was statistically significantly higher than the other two groups.

2.4 TRIP CHAINING BY LIFECYCLE

In 2009, NYS individuals with children were statistically significantly less likely to report no trips (12%) compared to those without children (17%). This pattern also similar in the rest of the United States (11% vs. 16%), with statistically significant differences (Table 2-7). However, by 2017, these differences disappeared: both groups in NYS reported 19%, and the rest of the United States showed similar rates (18% with children vs. 20% without). In 2009, around one-third of all groups recorded two activity types per day, with no statistically significant differences between lifecycle stages. By 2017, however, NYS individuals without children (32%) had a slightly higher share than those with children (30%). The proportion of individuals engaging in 3 or more activity types per day declined slightly from 2009 to 2017.

across all groups. The rest of the United States saw a statistically significant drop for those with children from 16% to 12%.

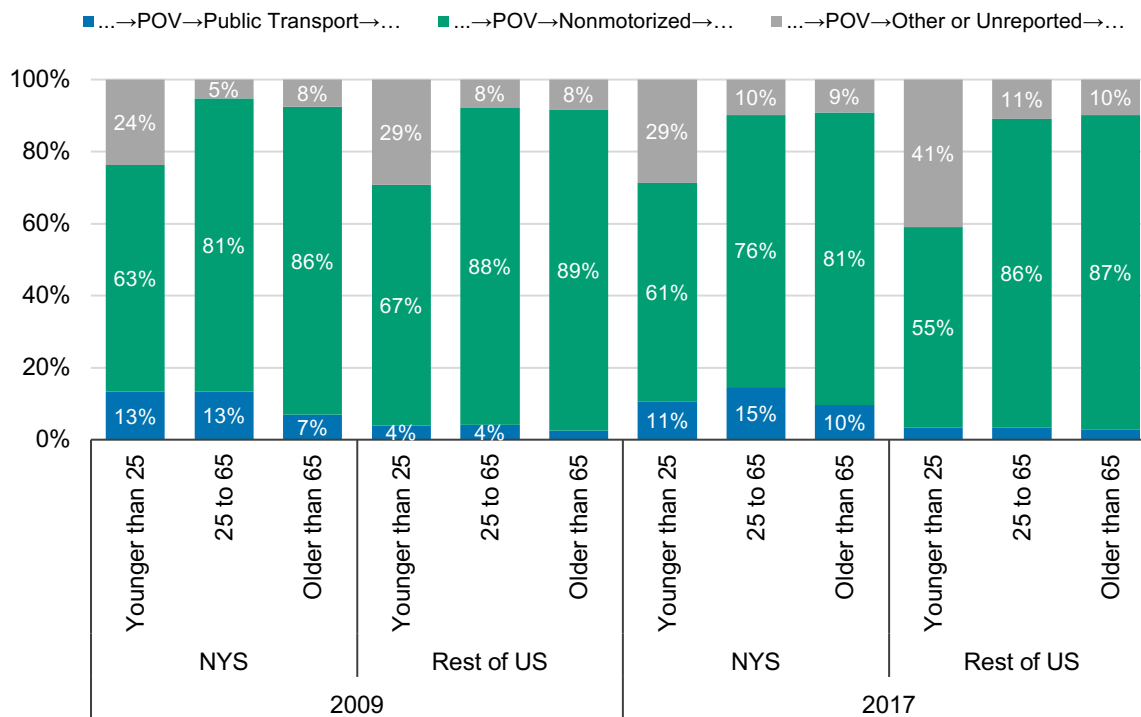


Figure 2-13. Transportation Mode Transition Following a POV Trip by Age in NYS and Rest of the United States

Table 2-7. Share of Daily Number of Activity Types by Lifecycle in NYS and Rest of the United States

NHTS year	Region	Lifecycle category	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	With children	12%	41%	34%	13%
		Without children	17%	38%	35%	10%
	Rest of US	With children	11%	37%	36%	16%
		Without children	16%	39%	34%	11%
2017	NYS	With children	19%	39%	30%	12%
		Without children	19%	39%	32%	10%
	Rest of US	With children	18%	39%	31%	12%
		Without children	20%	40%	31%	9%

2.4.1 Pre-Work Trip Chaining

A statistically significant higher share of NYS workers with children (81%) went directly from home to work compared to those without children (76%) in 2009 (Figure 2-14). In contrast, the difference between workers with and without children in the rest of the United States (78% vs. 79%) is not statistically significant. For family and personal business stops, workers without children were more likely to make such stops than those with children in both NYS (16% vs. 8%) and the rest of the United States (14% vs. 7%), and these differences are statistically significant. Social and recreational stops were infrequent

across all groups and years, with no statistically significant differences observed between those with and without children in either region.

Short stops before work were more common among workers with children in both 2009 and 2017, likely reflecting the need to drop off children at school. In 2009, the share was significantly higher among those with children in the rest of the United States (11%) compared to those without children (3%). In 2017, this difference persisted in both regions. In NYS, 12% of workers with children made short stops which statistically significant higher than 3% of those without children. Similarly, in the rest of the United States, the share was 15% for those with children and 3% for those without, also statistically significant.

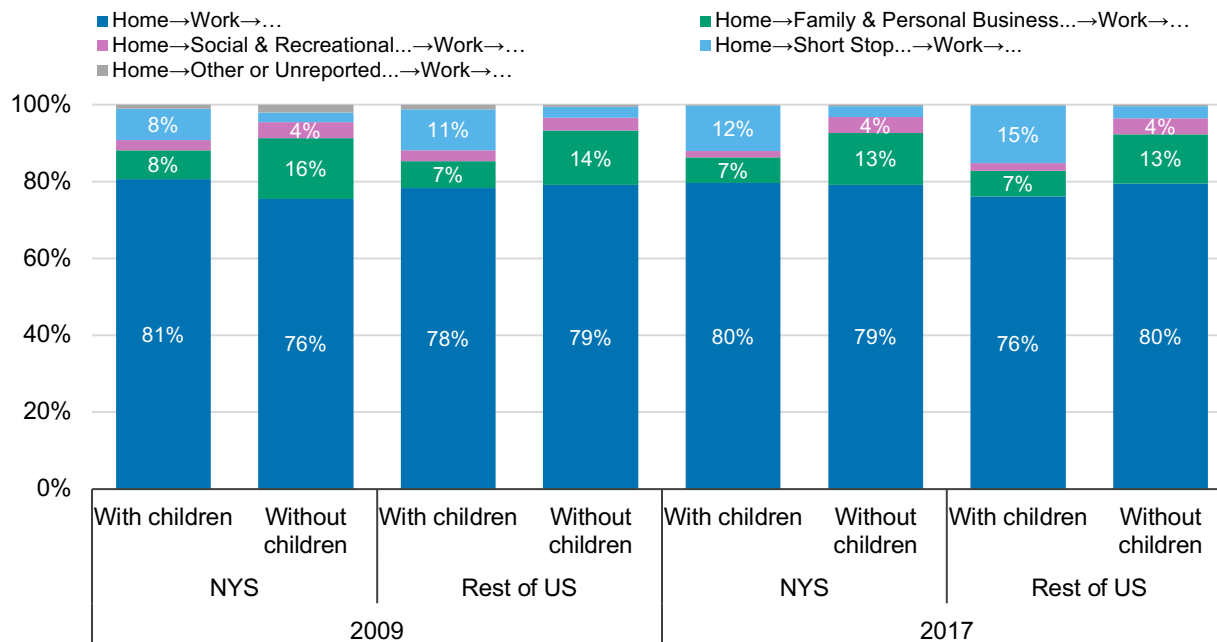


Figure 2-14. Pre-Work Activity Patterns by Lifecycle in NYS and Rest of the United States

2.4.2 Post-Work Trip Chaining

In both NYS and the rest of the United States, individuals with children were more likely to return directly home after work compared to those without children (Figure 2-15). In 2009, 78% of NYS respondents with children went straight home, statistically significantly higher than 64% of those without children. The pattern held in 2017 as well, though with a slight decline; 73% of NYS respondents with children went home directly, compared to only 62% of those without children. A similar but slightly more stable pattern can be observed in the rest of the United States, where about 75% of those with children returned directly home in both years, while the share among those without children was 66% in 2009 and 65% in 2017.

Personal and family-related errands were the most common type of post-work stops among those without children. In NYS, 28% of individuals without children made such stops in both 2009 and 2017, statistically significantly higher than just 14% and 16% among those with children. This indicates that individuals without children may have more flexibility to engage in additional activities after work. Recreational stops were also slightly more common among those without children. In 2017, 8% of childless individuals in NYS made social or recreational stops after work, compared with 6% among those with children. The rest of the United States shows slightly lower rates overall, with smaller differences between the two lifecycle groups.

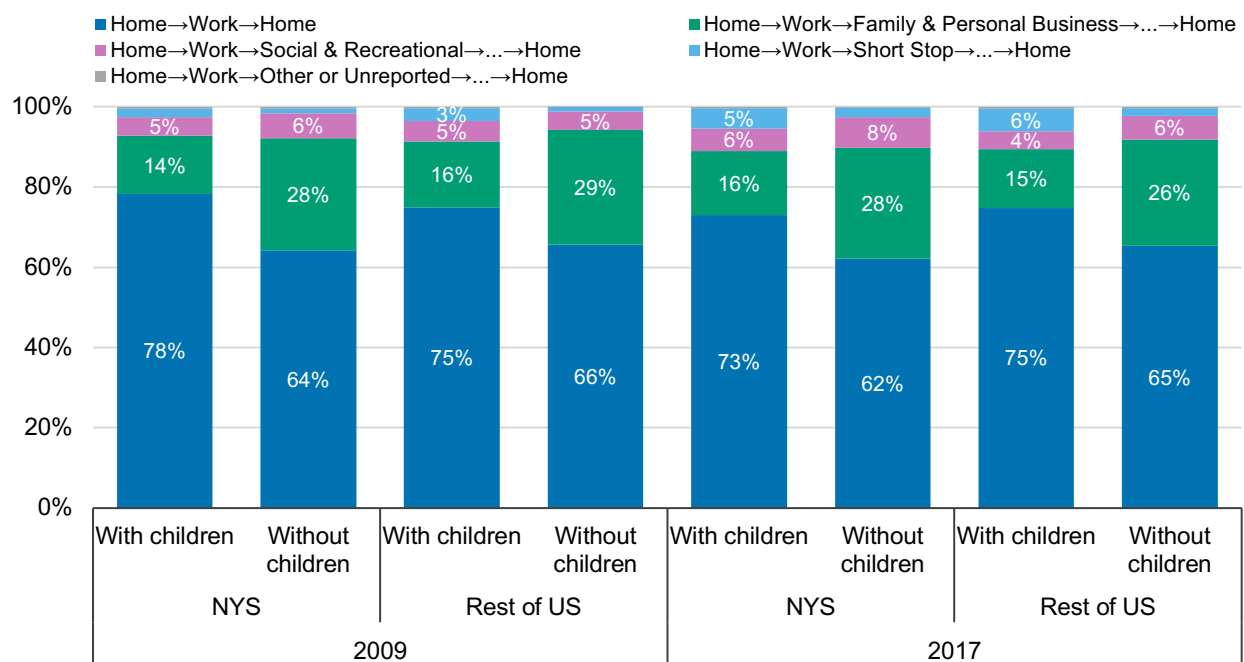


Figure 2-15. Post-Work Activity Patterns by Lifecycle in NYS and Rest of the United States: Home→Work Trip Chains

2.4.3 Non-Work Days Trip Chaining

NYS residents without children were statistically significantly more likely to begin their day with family and personal business activities (69%) than those with children (55%) in 2009 (Figure 2-16). A similar statistically significant gap is observed in the rest of the United States, where 68% of childless individuals started with family or personal business trips compared to 50% of those with children.

Social and recreational trips showed an opposite pattern. In 2009, individuals with children in NYS were statistically significantly more likely to begin the day with these activities (32%) compared to those without children (25%). The rest of the United States followed the same pattern (32% with children vs. 24% without), and the difference is statistically significant. Short stops were more common among individuals with children in both regions. In NYS, 12% of those with children made short stops, compared to 5% without children, a statistically significant difference. The rest of the United States also showed a significant difference (16% with children vs. 6% without). In 2017, similar trends persist.

2.4.4 Transportation Mode Transition

In 2009, 12% of those with children and 15% without switched to public transport from POV, and both groups had similar shares for nonmotorized and other modes (Figure 2-17). In the rest of the United States, however, individuals without children were statistically significantly more likely to switch from POV to nonmotorized modes (88%) compared with those with children (77%). By 2017, in NYS, transitions from POV to public transport rose significantly for those with children (from 12% in 2009 to 29% in 2017) while the increase for those without children (from 15% to 14%) was not statistically significant. Meanwhile, nonmotorized mode use dropped statistically significantly among those with children in NYS (from 76% to 68%) and remained relatively stable for those without children.

The share of other or unreported transitions increased among NYS individuals with children (13% to 21%), a statistically significant rise, while it remained low for those without children (9%). In the rest of

the United States, this share also rose among parents (19% to 28%) but stayed relatively low for those without children (7% to 11%), with the difference between the two groups in 2017 statistically significant.

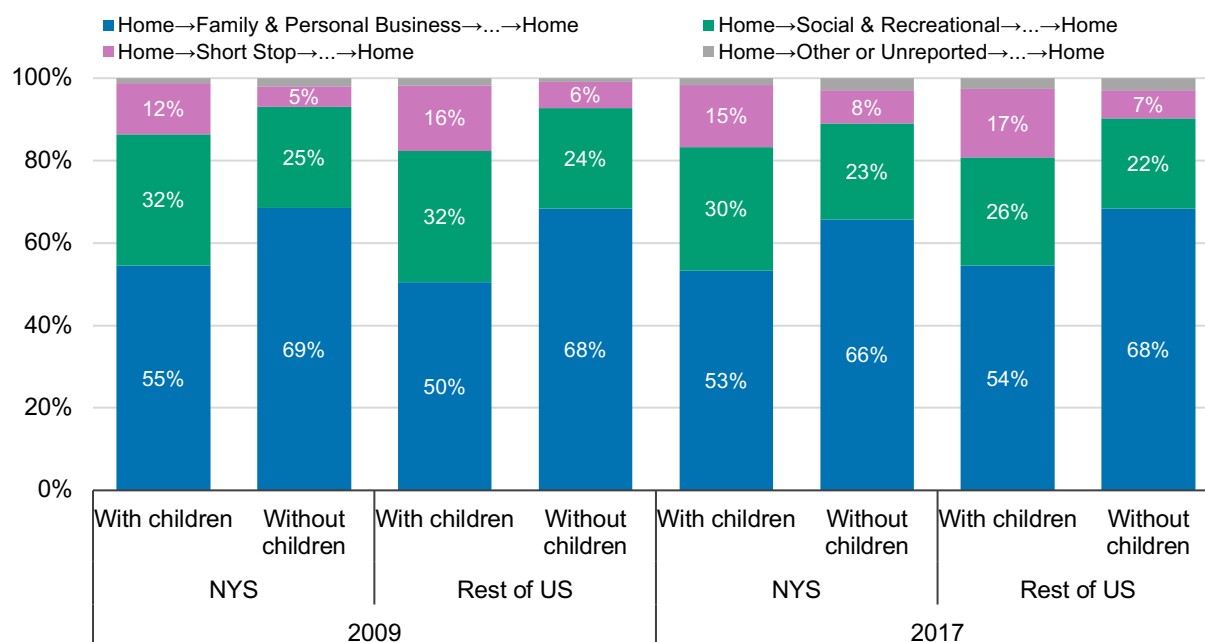


Figure 2-16. Activity Patterns on Non-Work Days by Lifecycle in NYS and Rest of the United States

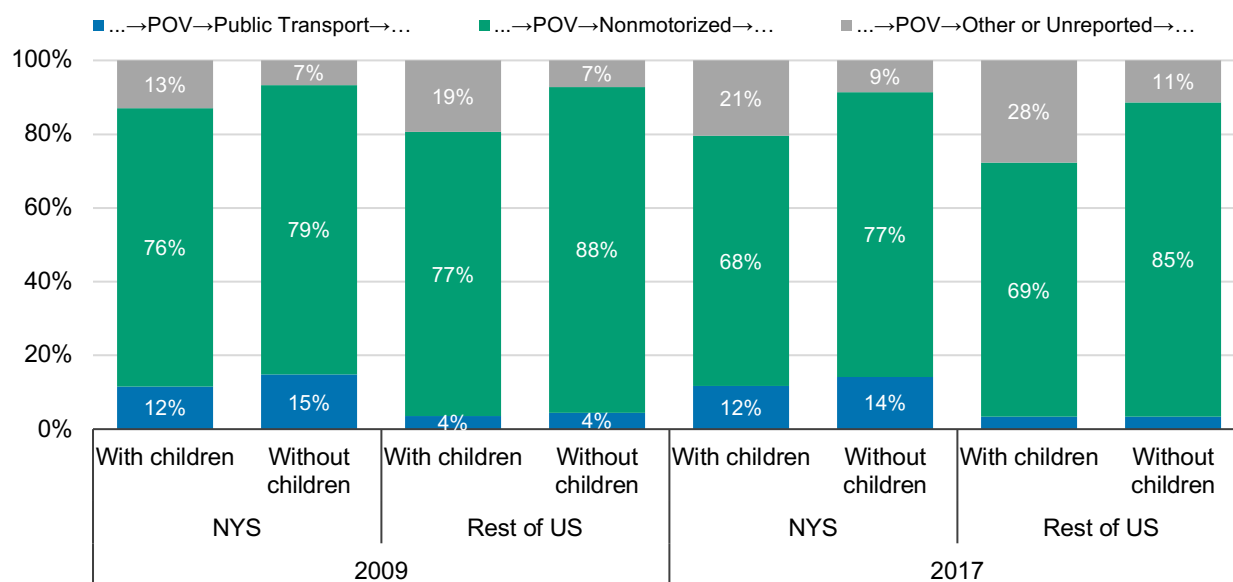


Figure 2-17. Transportation Mode Transition Following a POV Trip by Lifecycle in NYS and Rest of the United States

2.5 TRIP CHAINING BY EDUCATION LEVEL

2.5.1 Pre-Work Trip Chaining

In 2017, 84% of workers in NYS with a high school diploma or less went to work directly from home, compared to 76% of those with a college degree or higher (Figure 2-18). This difference is statistically significant. The same pattern holds in the rest of the United States, with 82% of non-college-educated workers versus 75% of college-educated workers going directly to work in 2017, also statistically significant.

Family or personal business activities were slightly more common among college-educated individuals than those with non-college-educated individuals. The differences are not statistically significant in any region or year. Short stops were also slightly more common among college-educated workers. In NYS in 2017, 9% of college-educated individuals made short stops before work compared with 5% among those with lower education. This difference is statistically significant. Social and recreational stops were rare across all groups.

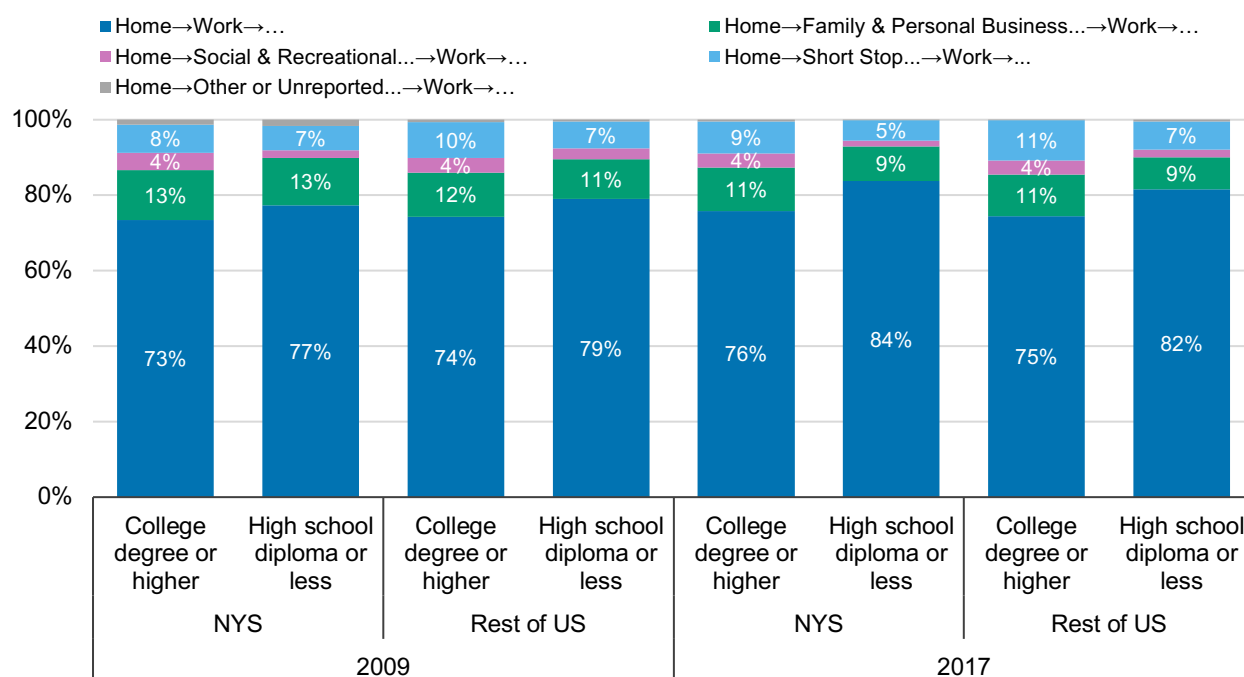


Figure 2-18. Pre-Work Activity Patterns by Education Level in NYS and Rest of the United States

2.5.2 Post-Work Trip Chaining

In 2017, 75% of workers in NYS with a high school diploma or less went home directly after work, compared with 64% of those with a college degree or higher (Figure 2-19). A similar difference appears in the rest of the United States (77% vs. 66%). College-educated workers were more likely to stop for family or personal business. In NYS, this share was 25% in both 2009 and 2017, compared with 20% and 17% for non-college-educated workers. The 2009 difference is statistically significant; the 2017 one is not. In the rest of the United States, the same pattern appears with 28% of college-educated workers making such stops in 2009 and 24% in 2017, both significantly higher than among those with lower education.

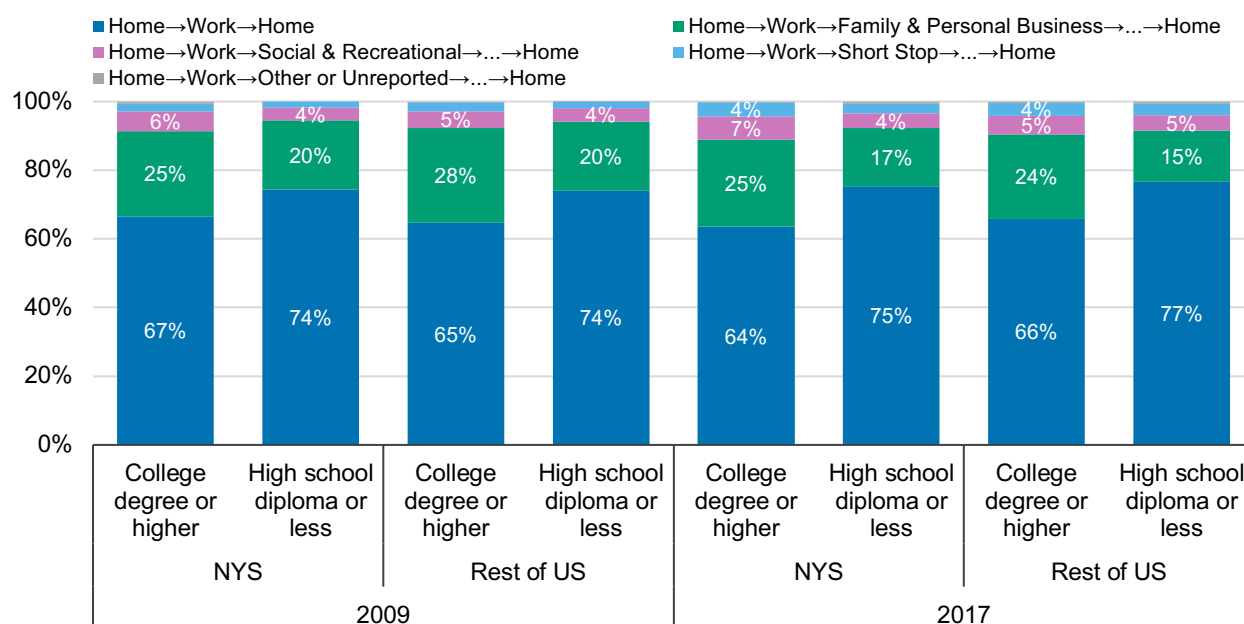


Figure 2-19. Post-Work Activity Patterns by Education Level in NYS and Rest of the United States: Home→Work Trip Chains

In 2009, individuals with a college degree or higher were statistically significantly more likely to make three or more trips after work than those with a high school diploma or less (Table 2-8). In NYS, 33% of college-educated individuals made 3+ trips, compared with 26% among non-college-educated people. The same pattern holds in the rest of the United States (37% vs. 26%), with both differences statistically significant. Conversely, non-college-educated individuals were significantly more likely to report no trips after work. In NYS, 42% of individuals with a high school diploma or less made no trips, compared to 33% of those with a college degree or higher. A similar statistically significant gap exists in the rest of the United States (41% vs. 30%).

Table 2-8. Distribution of Number of Trips After Work by Education Level in NYS and Rest of the United States

NHTS year	Region	Education level	No trip	One trip	Two trips	3+ trips
2009	NYS	College degree or higher	33%	10%	24%	33%
		High school diploma or less	42%	9%	22%	26%
	Rest of US	College degree or higher	30%	10%	23%	37%
		High school diploma or less	41%	11%	21%	27%
2017	NYS	College degree or higher	41%	16%	19%	24%
		High school diploma or less	53%	15%	16%	16%
	Rest of US	College degree or higher	39%	15%	21%	26%
		High school diploma or less	53%	13%	18%	16%

By 2017, the differences in NYS became more pronounced. Over half (53%) of individuals with a high school diploma or less made no trips after work, statistically significantly higher than the 41% among college-educated individuals. Meanwhile, only 16% of the non-college-educated group made 3+ trips, compared to 24% among the higher-education group, a statistically significant difference. The rest of the United States in 2017 showed similar trends. Those with less education were more likely to make no trips

(53% vs. 39%) and less likely to make 3+ trips (16% vs. 26%), with both differences statistically significant.

2.5.3 Non-Work Days Trip Chaining

In both 2009 and 2017, individuals with a high school diploma or less were statistically significantly more likely to begin their day with a family or personal business activity compared with those with a college degree or higher in both NYS and the rest of the United States (Figure 2-20). For instance, in 2017 in NYS, 65% of non-college-educated individuals followed this pattern, compared to 59% of those with higher education. In the rest of the States, the gap was similar (66% vs. 62%), and the difference is statistically significant. In NYS in 2017, individuals with a college degree or higher were statistically significantly more likely to start their day with a short stop compared to non-college-educated people (13% vs. 9%).

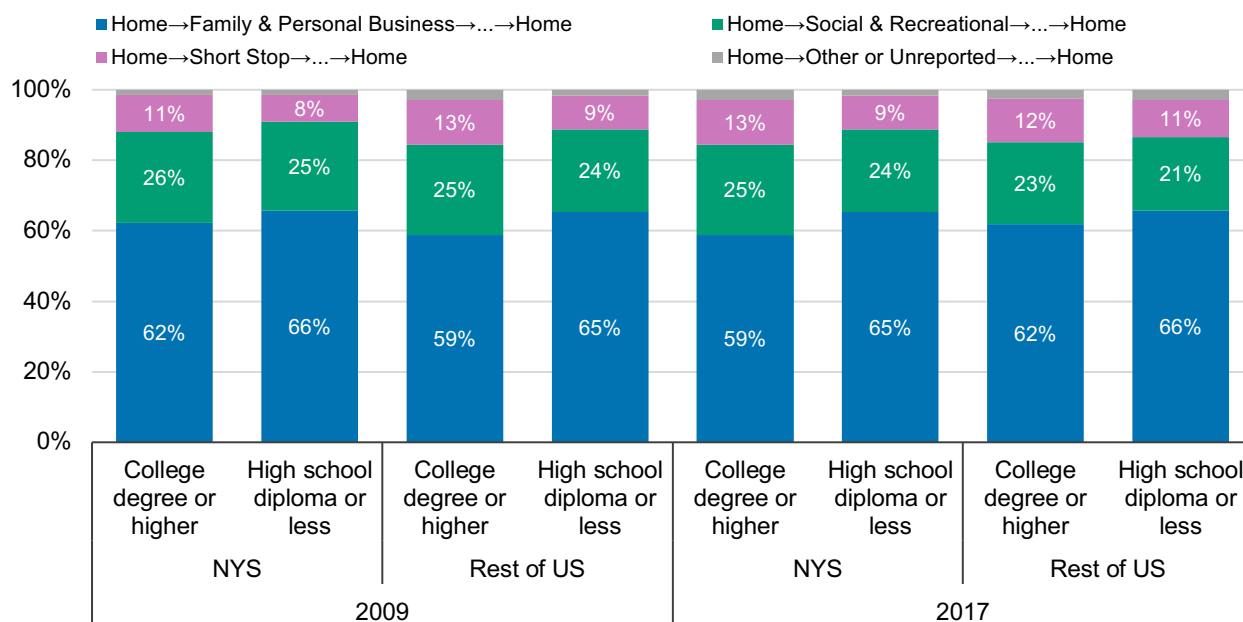


Figure 2-20. Activity Patterns on Non-Work Days by Education Level in NYS and Rest of the United States

2.5.4 Transportation Mode Transition

College-educated individuals in NYS were statistically significantly more likely to switch from POV to public transport than those in the rest of the United States (14% vs. 4%) in 2009 (Figure 2-21). Within NYS, the difference between education groups (14% for college-educated vs. 11% for those with a high school diploma or less) was not statistically significant. For the POV to nonmotorized modes transfer, rates were high across all groups, with no statistically significant differences by education level or region. In 2017, the overall patterns remained consistent, but differences became more pronounced in NYS. College-educated individuals in NYS continued to be statistically significantly more likely to transfer from POV to public transport than those in the rest of the United States (15% vs. 4%). Individuals with a high school diploma or less were statistically significantly more likely to transfer to these less common or unreported modes (21%) than college-educated individuals (9%).

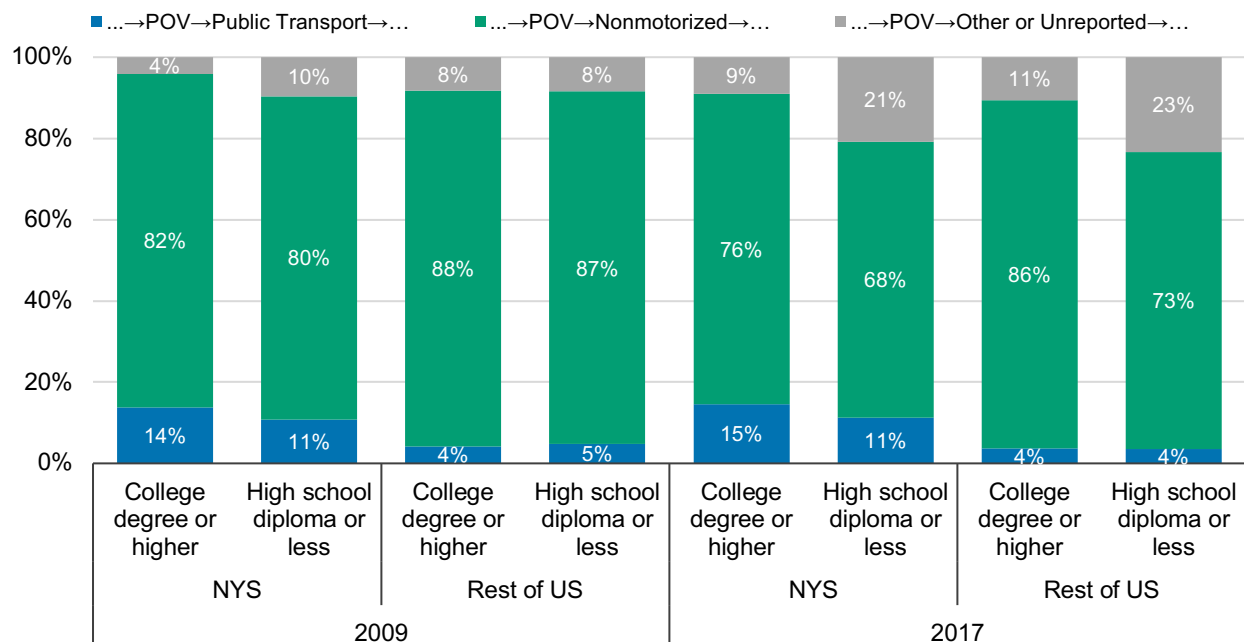


Figure 2-21. Transportation Mode Transition Following a POV Trip by Education Level in NYS and Rest of the United States

2.6 TRIP CHAINING BY HOME OWNERSHIP

In 2009, renters were more likely than owners to have no trips recorded in both NYS and the rest of the United States (Table 2-9). In NYS, renters (16%) had a statistically significantly higher share of no-trip days than owners (13%). In the United States, the same pattern held (16% renters vs. 12% owners), also statistically significant. When looking at higher activity levels (3+ types), in NYS, homeowners had a higher share (14%) compared to renters (9%), a statistically significant difference. In the United States, the difference was also present (15% for owners vs. 11% for renters), though this is not statistically significant. These findings suggest that owners were generally more active than renters in 2009, with fewer zero-activity days and more multi-activity days. By 2017, the pattern of no recorded trips rose across all groups. Both owners and renters in NYS saw a statistically significant increase to 19% on no-trip days, up from 13% and 16%, respectively, in 2009.

Table 2-9. Daily Number of Activity Types by Home Ownership in NYS and Rest of the United States

NHTS year	Region	Home ownership	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	Own	13%	39%	35%	14%
		Rent	16%	42%	34%	9%
	Rest of US	Own	12%	37%	36%	15%
		Rent	16%	41%	32%	11%
2017	NYS	Own	19%	40%	30%	11%
		Rent	19%	38%	32%	10%
	Rest of US	Own	19%	39%	31%	11%
		Rent	19%	40%	30%	11%

2.6.1 Pre-Work Trip Chaining

In 2009, the majority of workers across all groups traveled directly to work without making any prior stops (Figure 2-22). In NYS, renters (80%) were slightly more likely than owners (78%) to travel directly to work. However, this difference was not statistically significant. Similarly, in the rest of the United States, 80% of renters and 78% of owners went directly to work; again, with no statistically significant difference. In 2017, the pre-work activity patterns remained largely consistent with 2009, with most individuals still traveling directly to work. Slight increases in short stops were noted between 2009 and 2017. Among owners in NYS, the share increased from 7% to 8%. In the rest of the United States, owners and renters both saw an increase to 10%. However, none of these changes are statistically significant.

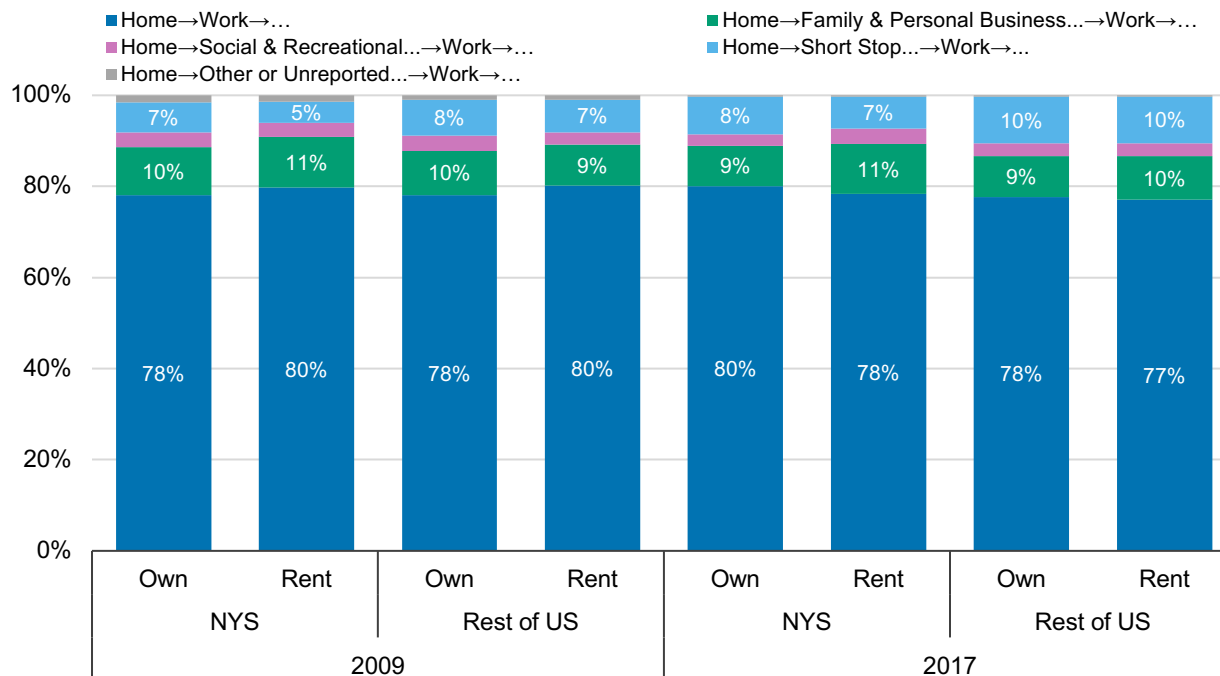


Figure 2-22. Pre-Work Activity Patterns by Home Ownership in NYS and Rest of the United States

2.6.2 Post-Work Trip Chaining

In 2009, most individuals returned home directly after work regardless of ownership status (Figure 2-23). In NYS, renters (75%) were slightly more likely than owners (72%) to return directly home, though the difference was not statistically significant. In the United States overall, a similar pattern was observed: 76% of renters vs. 70% of owners returned home directly. This was statistically significant. Family & personal business stops were more common among owners than renters in both regions, though differences were modest and not statistically significant. By 2017, clearer differences emerged, particularly in NYS. Renters in NYS were statistically significantly less likely to return directly home (63%) compared with owners (72%). In the rest of the United States, both owners and renters exhibited similar patterns in 2017.

In 2017, renters in NYS whose first trip after work was for family or personal business had a median departure time from work of 2:37 p.m. (Figure 2-24). Their final trip of the day, however, typically began around 6:26 p.m., suggesting they often engaged in multiple activities in between, possibly returning to work after a short family or personal errand before finally heading home.

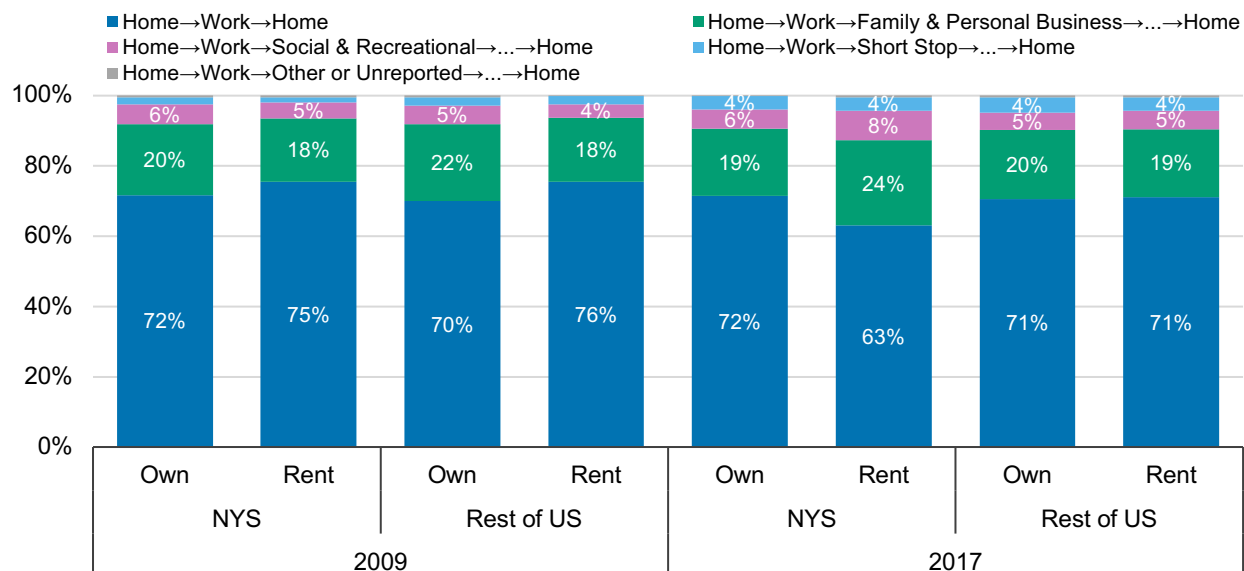


Figure 2-23. Post-Work Activity Patterns by Home Ownership in NYS and Rest of the United States: Home→Work Trip Chains

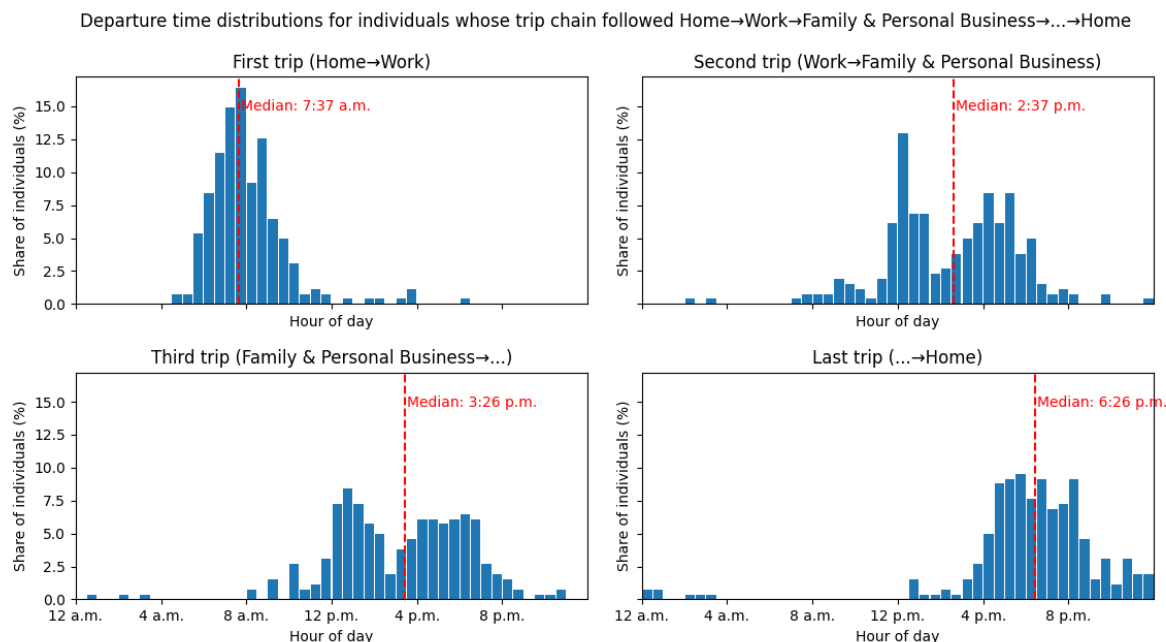


Figure 2-24. Departure Time Distribution of Individuals Who Rented and the First Trip After Work is Family & Personal Business Activity

2.6.3 Non-Work Days Trip Chaining

Renters were statistically significantly more likely to start their day with a family and personal business trip chain (67%) than homeowners (58%) in NYS (Figure 2-25). Renters in NYS were also less likely to make short stop trips (6%) than owners (10%), with a statistically significant gap. In the rest of the United States, homeowners and renters had similar activity patterns overall. However, renters were slightly more

likely to begin the day with family and personal business trips (62%) compared to owners (58%), though this is not statistically significant. By 2017, differences between renters and owners in NYS diminished, though renters showed slight increases in short stop and social and recreational trips.

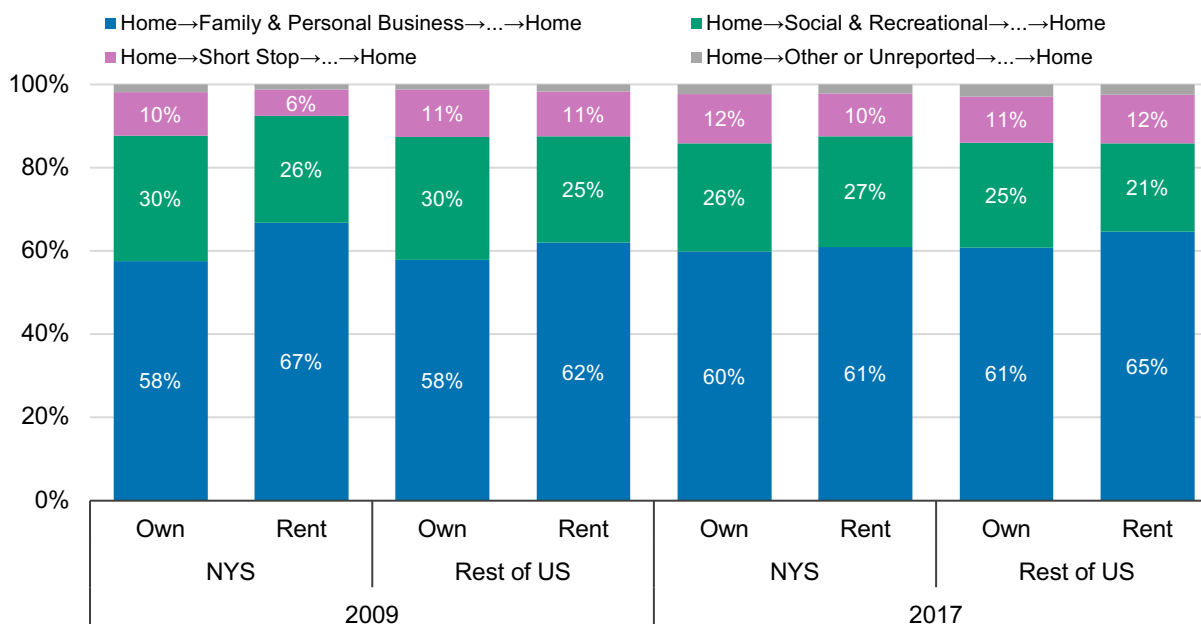


Figure 2-25. Activity Patterns on Non-Work Days by Home Ownership in NYS and Rest of the United States

2.6.4 Transportation Mode Transition

Figure 2-26 shows that in 2009, renters in NYS were statistically significantly less likely than homeowners to switch from public transport to a POV (13% vs. 29%) and statistically significantly more likely to switch to nonmotorized modes (79% vs. 66%). These differences are statistically significant. A similar pattern appears in the rest of the United States, where renters were less likely to use a POV (30%) than homeowners (45%) and more likely to use nonmotorized modes (65% vs. 48%), both differences statistically significant. By 2017, the gap widened in NYS. Homeowners' use of POV after public transport dropped from 29% to 19%, while renters' use dropped further from 13% to just 7%, a statistically significant difference. Meanwhile, renters in NYS were significantly more likely than homeowners to continue by nonmotorized modes (82% vs. 76%). The "other or unreported" category rose slightly for renters (11%), though the difference was not statistically significant.

In the rest of the United States in 2017, similar patterns held: renters were less likely than owners to use POVs after transit (17% vs. 36%) and more likely to walk or bike (69% vs. 54%). These differences remained statistically significant. In both years and regions, renters consistently relied more on nonmotorized modes and less on cars after taking public transportation compared to homeowners. The contrast was particularly strong in NYS in 2017, where POV use among renters was exceptionally low.

2.7 TRIP CHAINING BY VEHICLE OWNERSHIP

Table 2-10 shows that, in NYS, individuals in zero vehicle households were statistically significantly more likely to record no activities (20%) compared to those with vehicles (13%). This may suggest that lack of vehicle access limits participation in out-of-home activities. In the rest of the United States, the contrast was even more pronounced: 28% of zero-vehicle individuals reported no activities compared to just 12% of those with vehicles, a statistically significant difference. In 2017, in NYS, the share of people

reporting no activities increased for both groups, but the gap narrowed: 21% for zero-vehicle households and 19% for those with vehicles. However, this difference is not statistically significant in 2017.

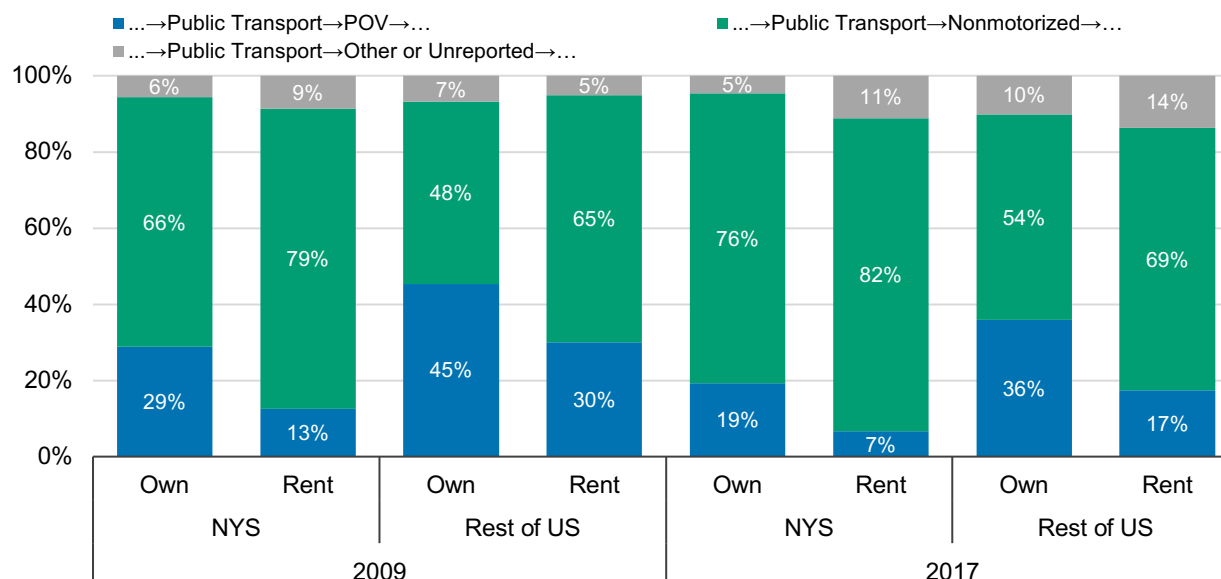


Figure 2-26. Transportation Mode Transition Following a Public Transport Trip by Home Ownership in NYS and Rest of the United States

Table 2-10. Share of Daily Number of Activity Types by Vehicle Ownership in NYS and Rest of the United States

NHTS year	Region	Vehicle ownership	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYS	Not zero vehicle	13%	39%	35%	13%
		Zero vehicle	20%	42%	31%	8%
	Rest of US	Not zero vehicle	12%	38%	36%	14%
		Zero vehicle	28%	44%	23%	5%
2017	NYS	Not zero vehicle	19%	39%	31%	12%
		Zero vehicle	21%	41%	30%	8%
	Rest of US	Not zero vehicle	18%	39%	31%	11%
		Zero vehicle	29%	45%	22%	4%

2.7.1 Pre-Work Trip Chaining

A large majority of workers in both groups went directly to work without any other stops (78% for vehicle owners vs. 80% for non-owners) in NYS (Figure 2-27). The differences across categories were generally small, and none were statistically significant. However, in 2009, zero vehicle households were slightly less likely to make short stops before work (3% vs. 7%) and slightly more likely to make "other or unreported" stops (4% vs. 1%), though again, not significantly different. In the rest of the United States, similar patterns were observed. In NYS in 2017, both groups showed similar shares of direct home-to-work commutes (79% vs. 80%). However, vehicle owners were statistically significantly more likely to make short stops before work (9% vs. 5%). This suggests that owning a vehicle may offer more flexibility to incorporate quick errands or tasks into the morning commute. In the rest of the United

States, most workers still went directly to work regardless of vehicle ownership (77% for owners vs. 79% for non-owners).

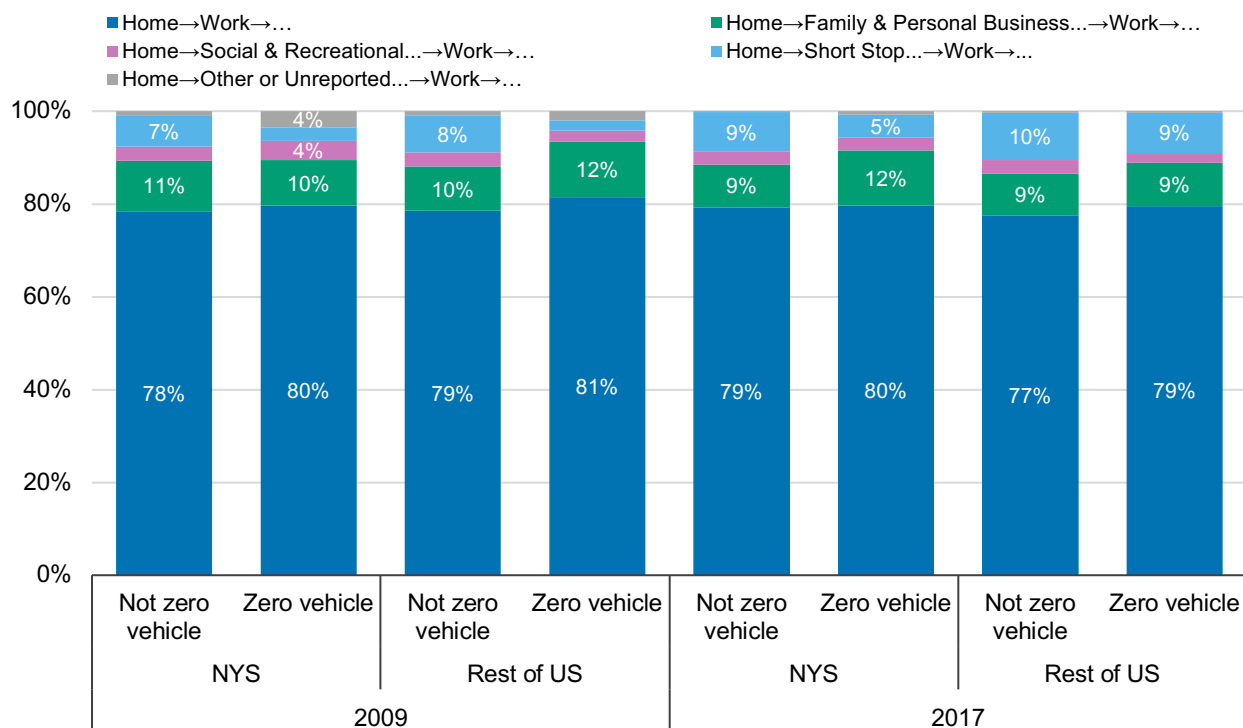


Figure 2-27. Pre-Work Activity Patterns by Vehicle Ownership in NYS and Rest of the United States

2.7.2 Post-Work Trip Chaining

In NYS, individuals with access to a vehicle were statistically significantly more likely to return directly home after work (74%) compared to those without a vehicle (66%) in 2009 (Figure 2-28). Conversely, zero vehicle individuals were more likely to engage in family & personal business (25% vs. 18%) and social and recreational activities (7% vs. 5%), although the differences were not statistically significant. In the rest of the United States, the differences were smaller. Both groups regardless of vehicle access were most likely to return home after work, with those without a vehicle actually showing slightly higher rates of direct return (77% vs. 71%). By 2017, disparities in NYS became more pronounced. Individuals without a vehicle were significantly less likely to return home directly after work (56%) compared to those with a vehicle (71%). They were also statistically significantly more likely to engage in family & personal business activities after work (29% vs. 19%) and social & recreational activities (11% vs. 5%). These differences reflect changes in travel behavior or constraints experienced by zero vehicle workers, such as combining errands into commute trips or relying on shared/public modes.

2.7.3 Non-Work Days Trip Chaining

On non-work days in NYS, Individuals in zero vehicle households were statistically significantly more likely to begin their day with family & personal business trips (66%) compared to those with a vehicle (60%) in 2009 (Figure 2-29). Conversely, vehicle owners were more likely to engage in short stop trips (11% vs. 2%), a statistically significant difference. In the rest of the United States, zero vehicle households were significantly more likely to start the day with family & personal business activities (70%) compared with vehicle owners (58%). In 2017, in NYS, patterns were more similar between groups. Those in zero vehicle households were still more likely to begin the day with family & personal

business (66%) than vehicle owners (58%), though the difference was not statistically significant. In the rest of the United States, individuals in zero vehicle households continued to be statistically significantly more likely to begin their day with family & personal business (73% vs. 61% for vehicle owners).

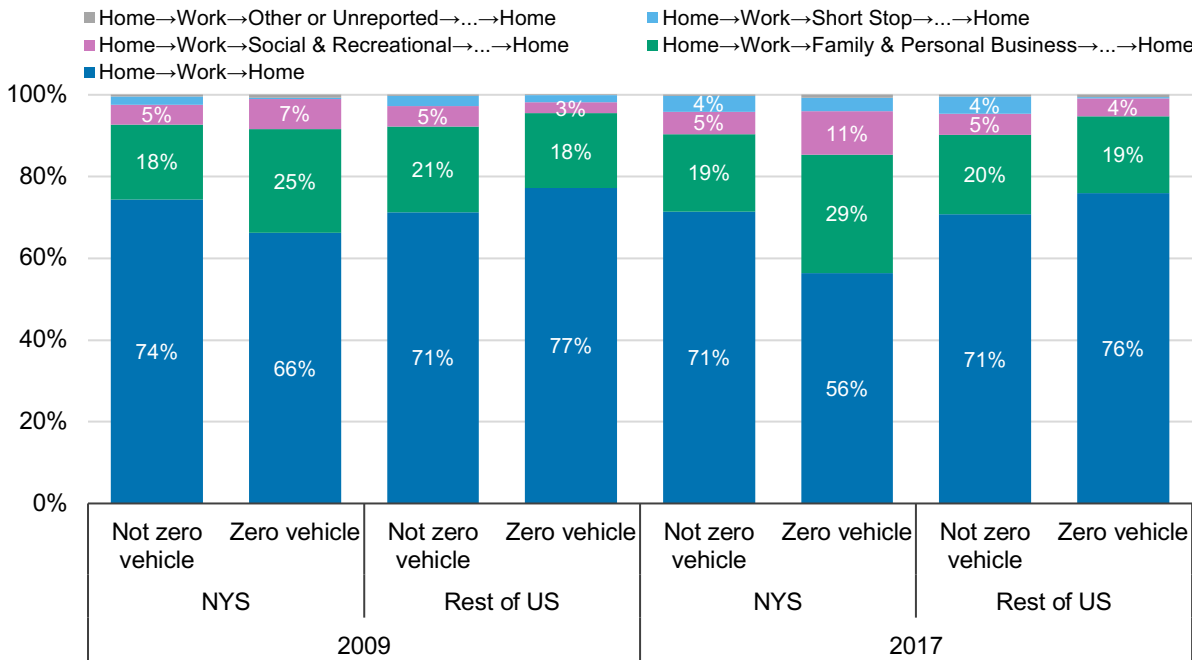


Figure 2-28. Post-Work Activity Patterns by Vehicle Ownership in NYS and Rest of the United States: Home→Work Trip Chains

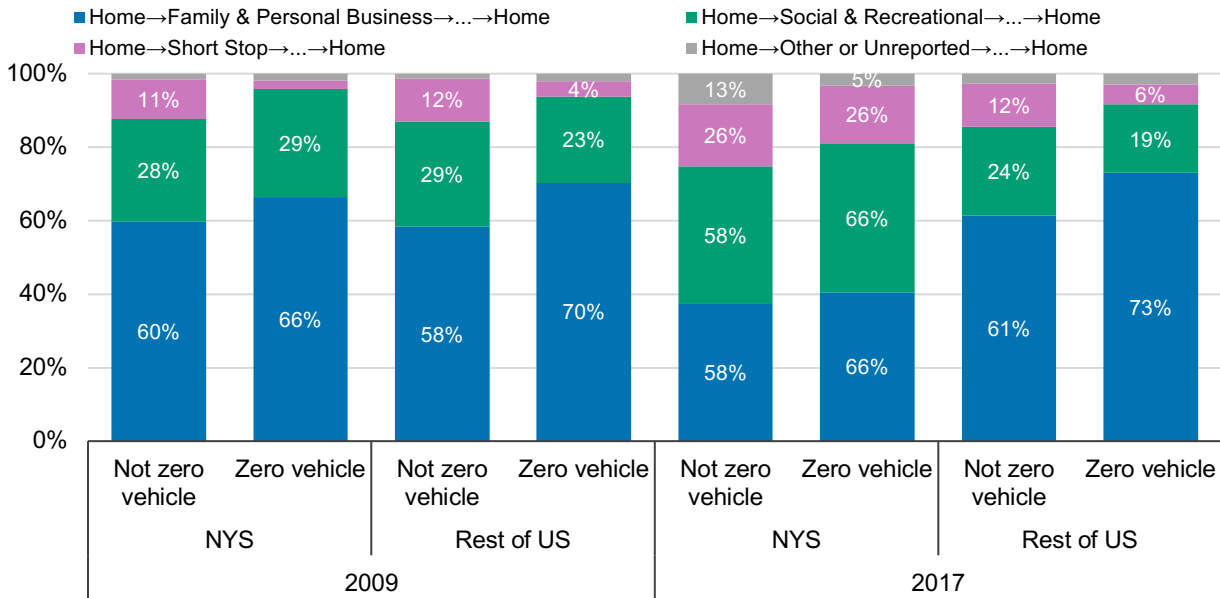


Figure 2-29. Activity Patterns on Non-Work Days by Home Ownership in NYS and Rest of the United States

2.7.4 Transportation Mode Transition

In NYS, individuals in zero vehicle households were significantly more likely to shift from public transport to nonmotorized modes (85%) compared to those in households with vehicles (62%) in 2009 (Figure 2-30). Conversely, those with vehicles were significantly more likely to transfer to POV (31% vs. 7%), indicating vehicle access enabled multimodal trips using both transit and private vehicles. A similar pattern was observed in the rest of the United States. In 2017, the patterns were similar in both regions.

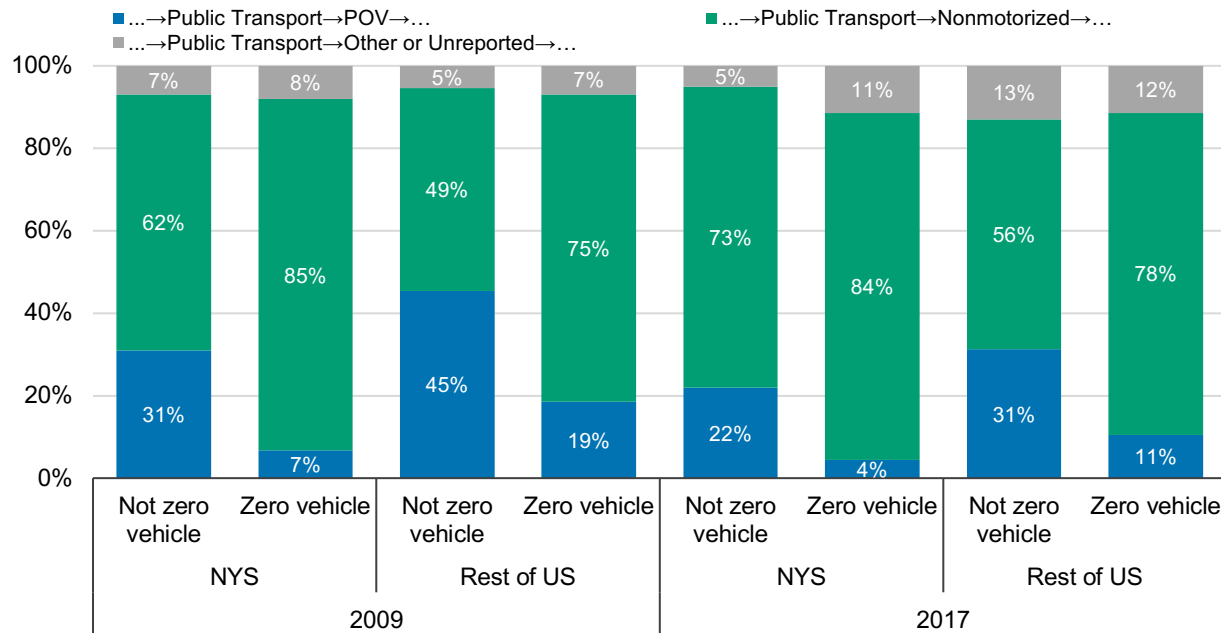


Figure 2-30. Transportation Mode Transition Following a Public Transport Trip by Vehicle Ownership in NYS and Rest of the United States

3. TRIP CHAINING BEHAVIOR BETWEEN NYC AND REST OF NYS

3.1 TRIP CHAINING BY OCCUPATION

In both years, across regions and occupations, the most common categories were engaging in either one or two types of activities per day (Table 3-1). In 2009, clerical or professional workers in NYC had 44% reporting two types of activities, statistically significantly higher than the share reporting one activity (31%) or three or more (14%). This pattern persisted in 2017. Manufacturing workers in NYC showed a distinct pattern compared to other groups. In 2009, 48% reported one type of activity, statistically significantly higher than those reporting two (33%) or three or more (9%). This pattern remained in 2017, with 38% reporting one type and 35% reporting two types of activities, with no statistically significant difference between the two shares in 2017. Notably, the share of manufacturing workers in NYC reporting no trips increased from 10% in 2009 to 17% in 2017, a statistically significant rise. Sales or service workers in NYC tended to have higher shares of two or more activity types compared to manufacturing workers.

In the rest of NYS, patterns were generally similar to that of NYC regardless of groups. For example, clerical or professional workers in the rest of NYS in both years had statistically significantly higher shares of two activity types (41% in 2009, 37% in 2017) compared to one activity type or three or more. No significant differences were observed between NYC and the rest of NYS.

Table 3-1. Share of Daily Number of Activity Types by Occupation in NYS and Rest of the United States

NHTS year	Region	Occupation	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	Clerical or professional	11%	31%	44%	14%
		Manufacturing, construction, maintenance, or farming	10%	48%	33%	9%
		Sales or service	7%	40%	38%	15%
	Rest of NYS	Clerical or professional	7%	33%	41%	19%
		Manufacturing, construction, maintenance, or farming	10%	39%	39%	12%
		Sales or service	7%	38%	37%	17%
2017	NYC	Clerical or professional	9%	35%	41%	15%
		Manufacturing, construction, maintenance, or farming	17%	38%	35%	9%
		Sales or service	11%	41%	37%	10%
	Rest of NYS	Clerical or professional	11%	35%	37%	17%
		Manufacturing, construction, maintenance, or farming	17%	40%	32%	12%
		Sales or service	16%	40%	31%	13%

3.1.1 Pre-Work Trip Chaining

Across both years, manufacturing, construction, maintenance, and farming workers in NYC were statistically significantly more likely to go directly from home to work without any intermediate stops compared to clerical or professional and sales or service workers (Figure 3-1). In 2009, 85% of NYC manufacturing workers went straight to work, statistically significantly higher than both clerical or professional workers (74%) and sales or service workers (74%). However, in 2017, this share for manufacturing workers in NYC dropped to 73%, no longer statistically significantly higher than clerical

or professional workers (77%) or sales or service workers (76%), suggesting a shift in pre-work behavior among this group.

Sales or service workers in the rest of NYS consistently had a statistically significantly higher share of direct home-to-work trips compared to their NYC counterparts in 2017 (80% in the rest of NYS versus 76% in NYC), though this difference was not statistically significant in 2009 when both stood at 74%.

Family and personal business stops before work were statistically significantly more common among manufacturing workers in NYC in 2017 compared to 2009. The share increased from 3% in 2009 to 22% in 2017, a statistically significant rise. This made manufacturing workers in NYC statistically significantly more likely to make family and personal business stops before work than their counterparts in the rest of NYS in 2017 (22% versus 14%).

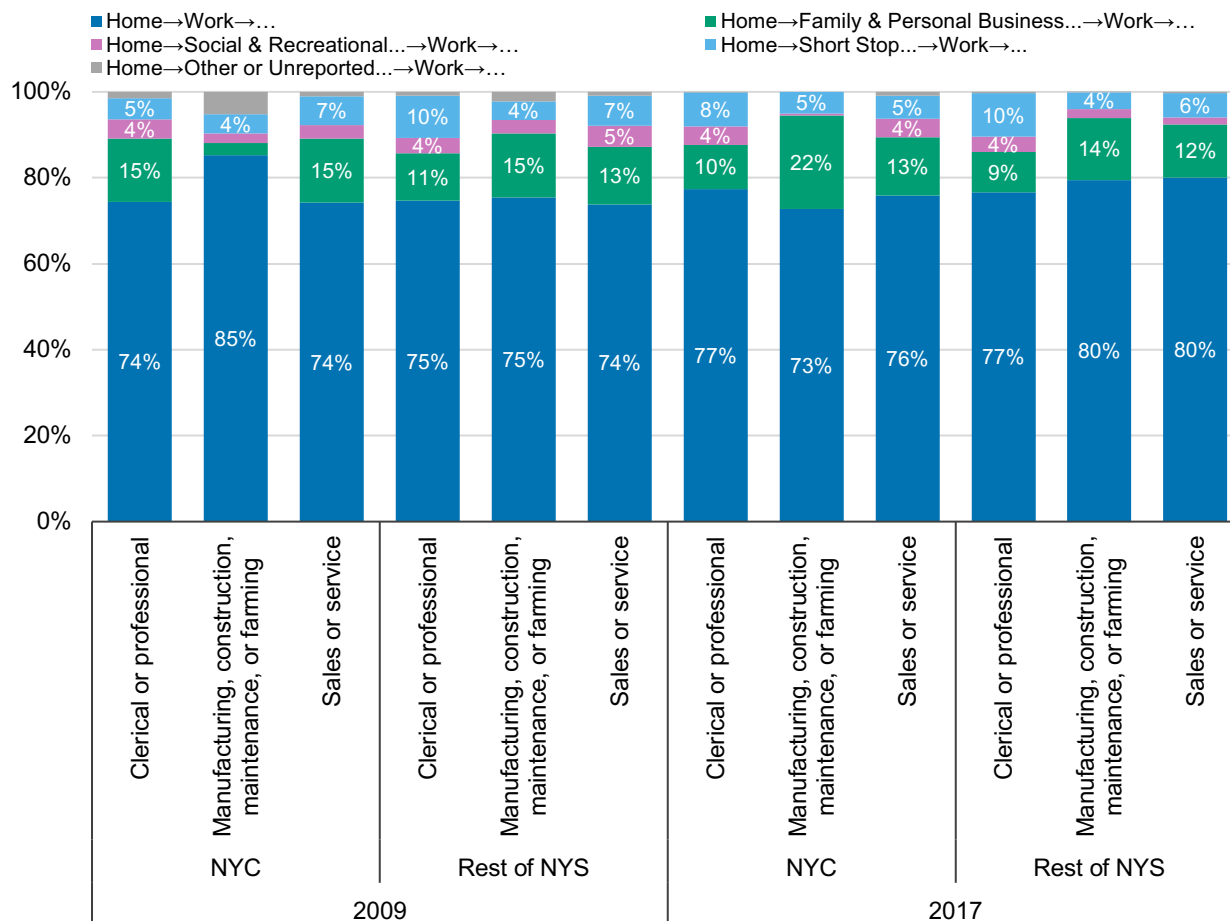


Figure 3-1. Pre-Work Activity Patterns by Occupation in NYC and Rest of NYS

3.1.2 Post-Work Trip Chaining

In both years, manufacturing, construction, maintenance, and farming workers in NYC were statistically significantly more likely to go directly home after work compared to clerical or professional and sales or service workers (Figure 3-2). For example, in 2017, 80% of NYC manufacturing workers went directly home, statistically significantly higher than both clerical or professional workers (57%) and sales or service workers (66%). Clerical or professional workers in NYC were statistically significantly more likely to make family and personal business stops after work compared to manufacturing workers in both

years. In 2009, 34% of clerical or professional workers made such stops, statistically significantly higher than 15% of manufacturing workers. The same pattern held in 2017, with 32% of clerical or professional workers versus 17% of manufacturing workers. Comparing NYC and the rest of NYS, clerical or professional workers in the rest of NYS were statistically significantly more likely to go directly home after work than their NYC counterparts in both years. In 2017, 66% of clerical or professional workers in the rest of NYS went directly home, statistically significantly higher than the 57% in NYC. In contrast, manufacturing workers showed no statistically significant differences between NYC and the rest of NYS, with both groups maintaining high shares of direct-home trips (76-80%) in both years.

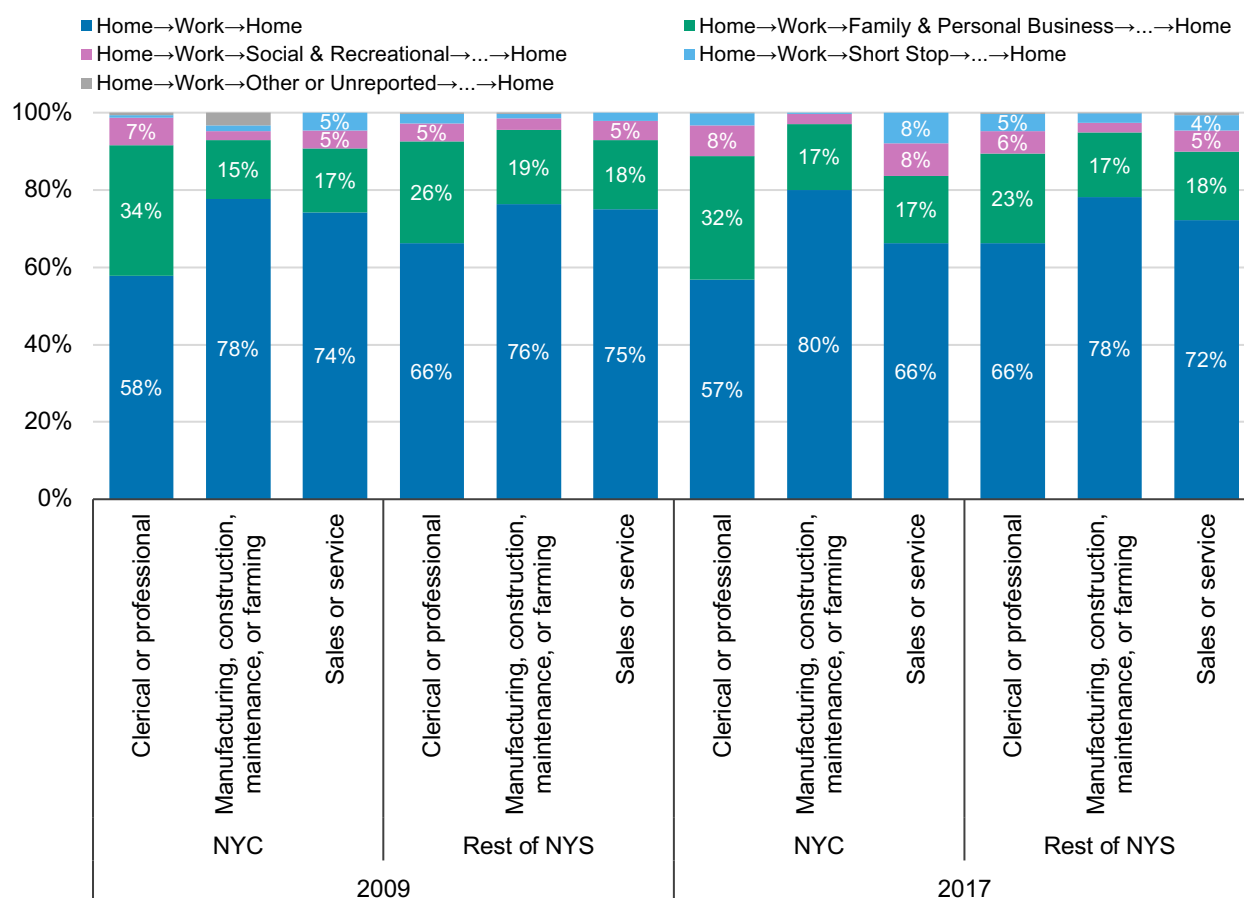


Figure 3-2. Post-Work Activity Patterns by Occupation in NYC and Rest of NYS: Home→Work Trip Chains

3.1.3 Non-Work Days Trip Chaining

In both years, family and personal business trips were the most common activity type on non-work days across all occupations and regions (Figure 3-3). In 2009, manufacturing workers in NYC were statistically significantly more likely to make family and personal business trips (74%) compared to clerical or professional workers (67%) and sales/service workers (75%) in NYC. However, the difference between manufacturing and sales/service workers was not statistically significant in NYC in 2009. In 2017, manufacturing workers in NYC again showed a statistically significantly higher share making family and personal business trips (69%) compared to sales or service workers (49%). Additionally, NYC manufacturing workers were statistically significantly more likely to make such trips compared to manufacturing workers in the rest of NYS (60%).

For sales or service workers in NYC, there was a statistically significant decline in family and personal business trips from 75% in 2009 to 49% in 2017, suggesting a shift in behavior on non-work days for this group. No similar statistically significant declines were observed for sales or service workers in the rest of NYS. NYC sales or service workers had a statistically significantly higher share of social and recreational trips in 2017 (32%) compared to 2009 (18%).

Short stops remained a minor share of non-work day activities across all groups. However, in 2017, NYC sales/service workers had a notably higher percentage (13%) compared to other groups, and this share represented a statistically significant increase from 4% in 2009.

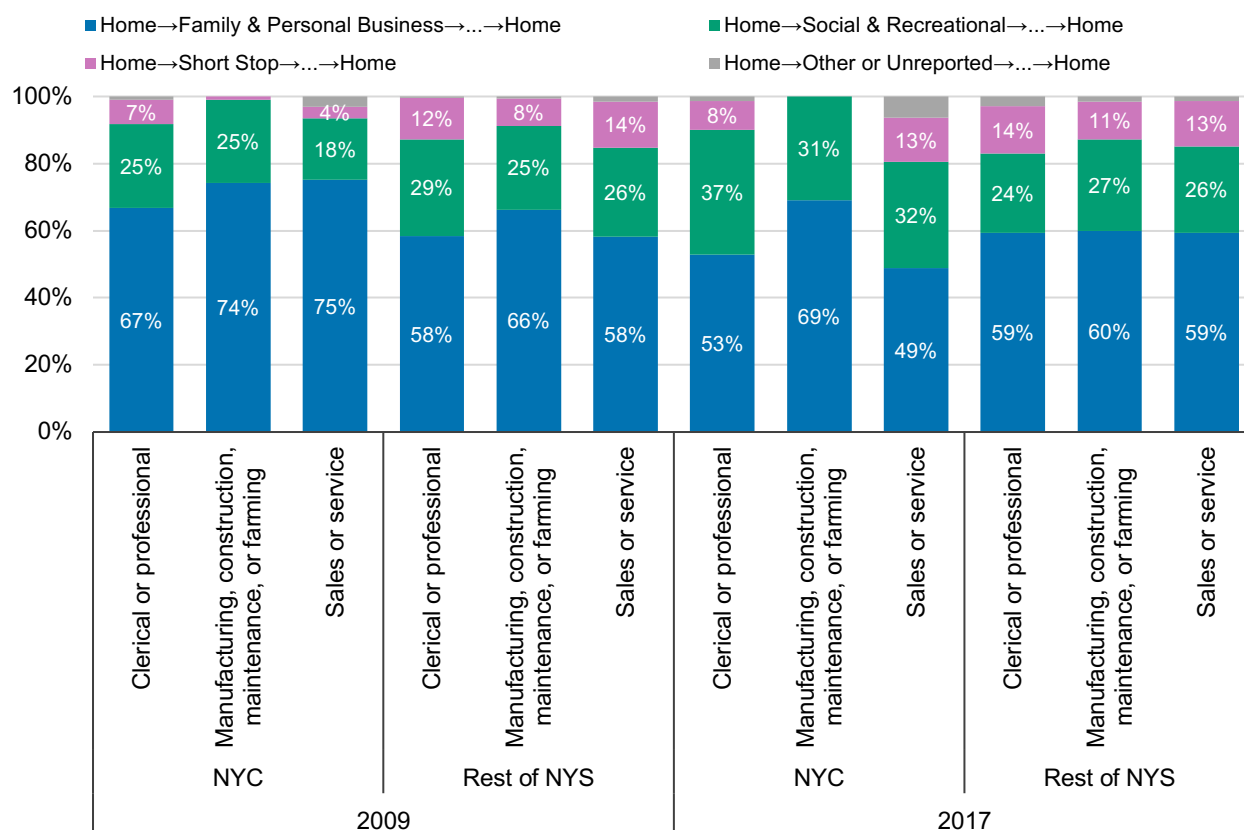


Figure 3-3. Activity Patterns on Non-Work Days by Occupation in NYC and Rest of NYS

3.2 TRIP CHAINING BY OPTION OF WFH

In both years, NYC workers who could not WFH were statistically significantly more likely to report only one activity type per day compared to WFH-capable workers (Table 3-2). In 2009, 38% of NYC non-WFH workers engaged in one activity type, statistically significantly higher than the 26% of WFH-capable workers. This difference persisted in 2017, with 40% of non-WFH workers versus 30% of WFH-capable workers reporting one activity type.

Conversely, NYC WFH-capable workers were statistically significantly more likely to report two or more types of activities per day. In 2009, 27% of WFH-capable workers reported three or more activity types, statistically significantly higher than the 11% of non-WFH workers. Although the share of WFH-capable workers reporting three or more activities decreased slightly to 17% in 2017, it remained statistically significantly higher than the 13% of non-WFH workers.

In the rest of NYS, the differences between WFH-capable and non-WFH workers were generally smaller. For example, in 2017, 39% of non-WFH workers in the rest of NYS reported one activity type, statistically significantly higher than 34% of WFH-capable workers. Meanwhile, 18% of WFH-capable workers reported three or more activity types, statistically significantly higher than 14% of non-WFH workers.

Table 3-2. Share of Daily Number of Activity Types by Option of WFH in NYC and Rest of NYS

NHTS year	Region	Option of WFH	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	Can WFH	7%	26%	40%	27%
		Cannot WFH	10%	38%	41%	11%
	Rest of NYS	Can WFH	7%	33%	39%	21%
		Cannot WFH	7%	36%	40%	17%
2017	NYC	Can WFH	8%	30%	45%	17%
		Cannot WFH	9%	40%	38%	13%
	Rest of NYS	Can WFH	9%	34%	39%	18%
		Cannot WFH	13%	39%	34%	14%

3.2.1 Pre-Work Trip Chaining

In both years, workers who could not WFH were statistically significantly more likely to go directly from home to work compared to those who could WFH, especially in NYC (Figure 3-4). In 2009, 80% of NYC non-WFH workers went directly to work, statistically significantly higher than the 60% of WFH-capable workers. This difference narrowed in 2017 but remained statistically significant, with 78% of non-WFH workers and 76% of WFH-capable workers going directly to work. NYC workers with WFH options were statistically significantly more likely to make stops for social and recreational or family and personal business activities before work. For example, in 2017, the share of WFH-capable NYC workers making family and personal business stops was 11%, statistically significantly higher than non-WFH workers.

In the rest of NYS, differences between WFH-capable and non-WFH workers were smaller and generally not statistically significant. For example, in 2017, 73% of WFH-capable workers and 80% of non-WFH workers went directly to work, a difference not statistically significant. Pre-work activity shares for family and personal business, social and recreational, and short stops remained relatively similar between the two groups, with few statistically significant differences across both years.

3.2.2 Post-Work Trip Chaining

In both years, workers in NYC who could not WFH were statistically significantly more likely to go directly home after work compared to those who could WFH (Figure 3-5). In 2009, 69% of NYC workers without WFH options went directly home, statistically significantly higher than the 43% of those with WFH options. This pattern remained in 2017, with 62% of non-WFH workers going directly home compared to 51% of WFH-capable workers. NYC workers who could WFH were statistically significantly more likely to make family and personal business stops after work compared to those who could not. In 2009, 47% of NYC WFH-capable workers made such stops, statistically significantly higher than 23% of non-WFH workers. This difference remained statistically significant in 2017, though the share dropped to 34% for WFH workers and slightly increased to 26% for non-WFH workers.

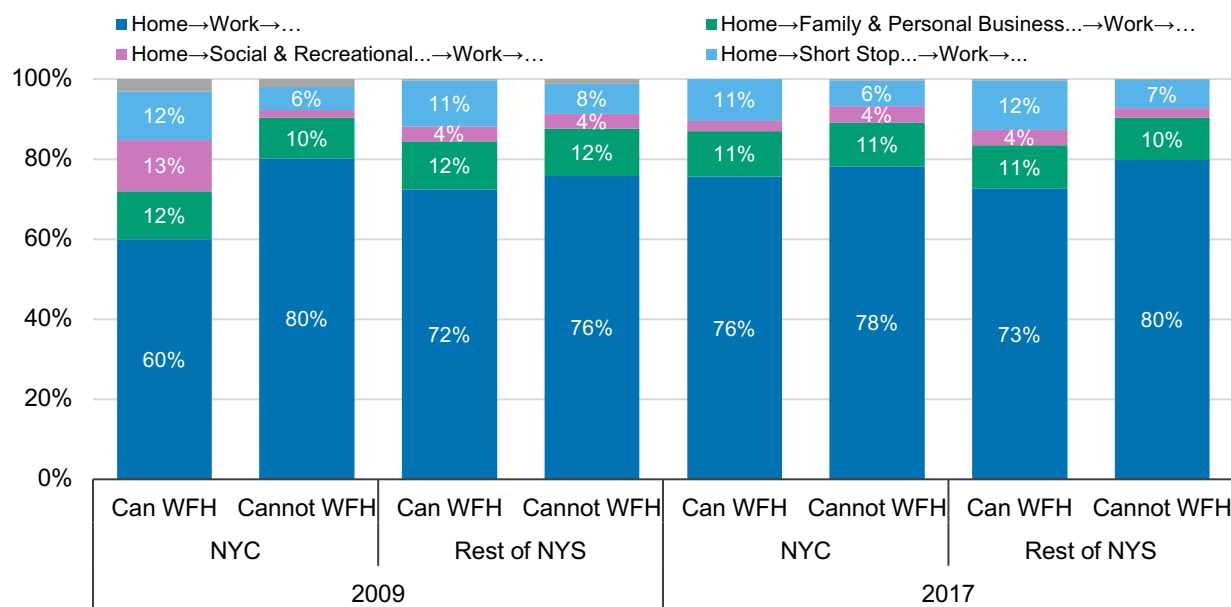


Figure 3-4. Pre-Work Activity Patterns by Option of WFH in NYC and Rest of NYS

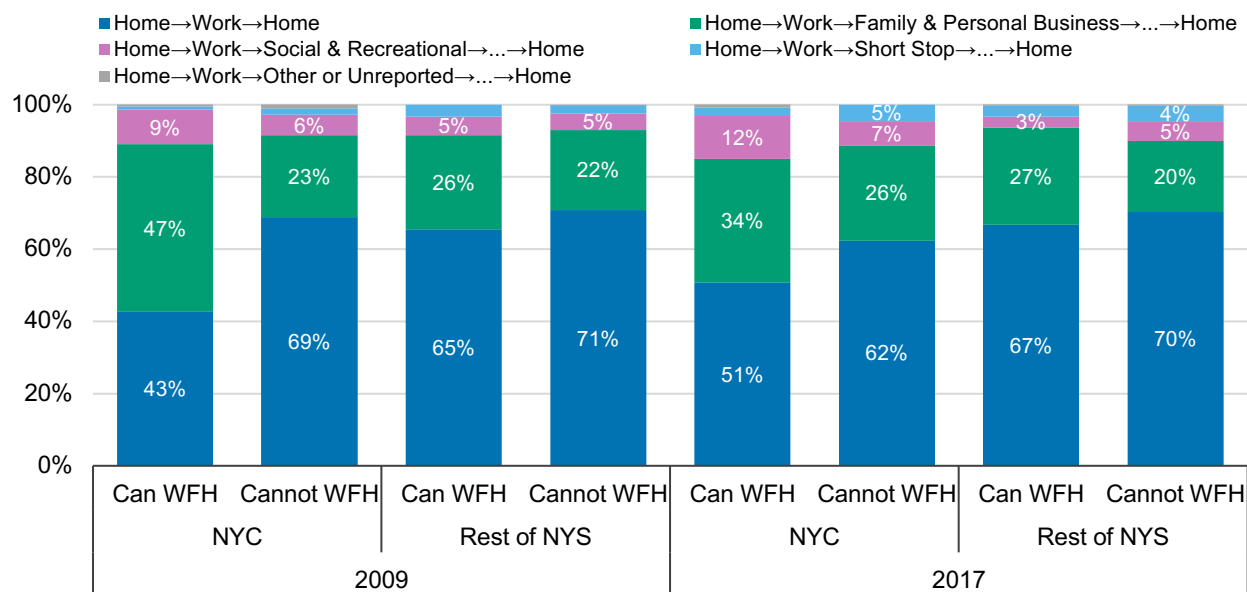


Figure 3-5. Post-Work Activity Patterns by Option of WFH in NYC and Rest of NYS: Home→Work Trip Chains

In the rest of NYS, differences between WFH and non-WFH workers were smaller. In both years, workers in the rest of NYS who could not WFH were slightly more likely to go directly home after work than those who could, but the differences were generally not statistically significant. For example, in 2017, 70% of non-WFH workers went directly home compared to 67% of WFH workers.

Social and recreational stops after work were more common among NYC WFH-capable workers compared to non-WFH workers in both years. In 2017, 12% of WFH workers made such stops in NYC, statistically significantly higher than the 7% of non-WFH workers. In the rest of NYS, the differences in

social and recreational stops between WFH and non-WFH workers were small and generally not statistically significant.

Table 3-3 further breaks down the data and shows the number of trips made after work before returning home. In general, WFH-capable workers were more likely to make three or more post-work trips than non-WFH workers in both years. In 2009, 39% of NYC WFH-capable workers made three or more post-work trips, statistically significantly higher than the 27% of non-WFH workers. Although this share slightly decreased to 31% in 2017, it remained statistically significantly higher than the 19% of non-WFH workers making three or more trips. In the rest of NYS, WFH-capable workers were more likely to make three or more post-work trips than non-WFH workers in both years. In 2009, 42% of WFH-capable workers made three or more trips, statistically significantly higher than the 30% of non-WFH workers. In 2017, this gap persisted, with 29% of WFH-capable workers versus 23% of non-WFH workers making three or more trips.

Table 3-3. Distribution of Number of Trips After Work by Option of WFH in NYC and Rest of NYS

NHTS year	Region	Option of WFH	No trip	One trip	Two trips	3+ trips
2009	NYC	Can WFH	25%	13%	22%	39%
		Cannot WFH	41%	8%	24%	27%
	Rest of NYS	Can WFH	26%	8%	25%	42%
		Cannot WFH	36%	11%	23%	30%
2017	NYC	Can WFH	31%	22%	16%	31%
		Cannot WFH	45%	18%	18%	19%
	Rest of NYS	Can WFH	35%	16%	21%	29%
		Cannot WFH	46%	14%	17%	23%

3.2.3 Non-Work Days Trip Chaining

In both years, family and personal business trips were the most common activity type on non-work days across all groups (Figure 3-6). In 2009, there were no statistically significant differences between NYC workers with and without WFH options for family and personal business trips (69% vs. 70%). Similarly, in the rest of NYS, both groups reported nearly identical shares (61%). However, by 2017, differences emerged in NYC. NYC workers without WFH options were statistically significantly more likely to make family and personal business trips (55%) compared to those with WFH options (49%). In the rest of NYS, no statistically significant difference was observed in 2017, though WFH-capable workers showed a slight decrease in this activity type compared to 2009.

Short stops showed some differences by region and year. In NYC, WFH-capable workers were more likely to make short stops than non-WFH workers in both years, though the differences were not consistently statistically significant. In the rest of NYS, WFH-capable workers were statistically significantly more likely to make short stops in 2017 (19%) compared to non-WFH workers (12%).

3.2.4 Transportation Mode Transition

Workers who could not WFH were statistically significantly more likely to transfer from public transport to a personal vehicle compared to WFH-capable workers in 2009 (17% vs. 8%) in NYC (Figure 3-7). This difference narrowed by 2017, with both groups showing similarly low shares (10% for non-WFH and 8% for WFH-capable workers), with no statistically significant difference in that year. The dominant transfer mode in NYC for both groups was nonmotorized, accounting for over 77% of transfers in both years. No statistically significant differences were observed between WFH-capable and non-WFH

workers in either year for nonmotorized transfers. However, NYC WFH-capable workers consistently had slightly higher shares of transferring to nonmotorized modes, though not statistically significant.

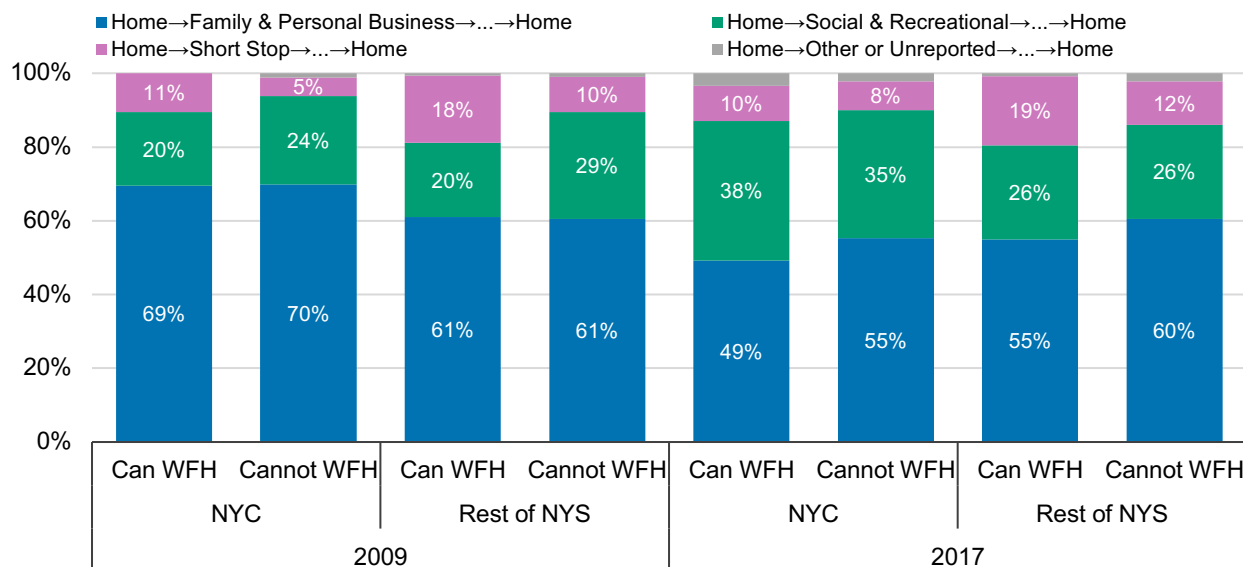


Figure 3-6. Activity Patterns on Non-Work Days by Option of WFH in NYC and Rest of NYS

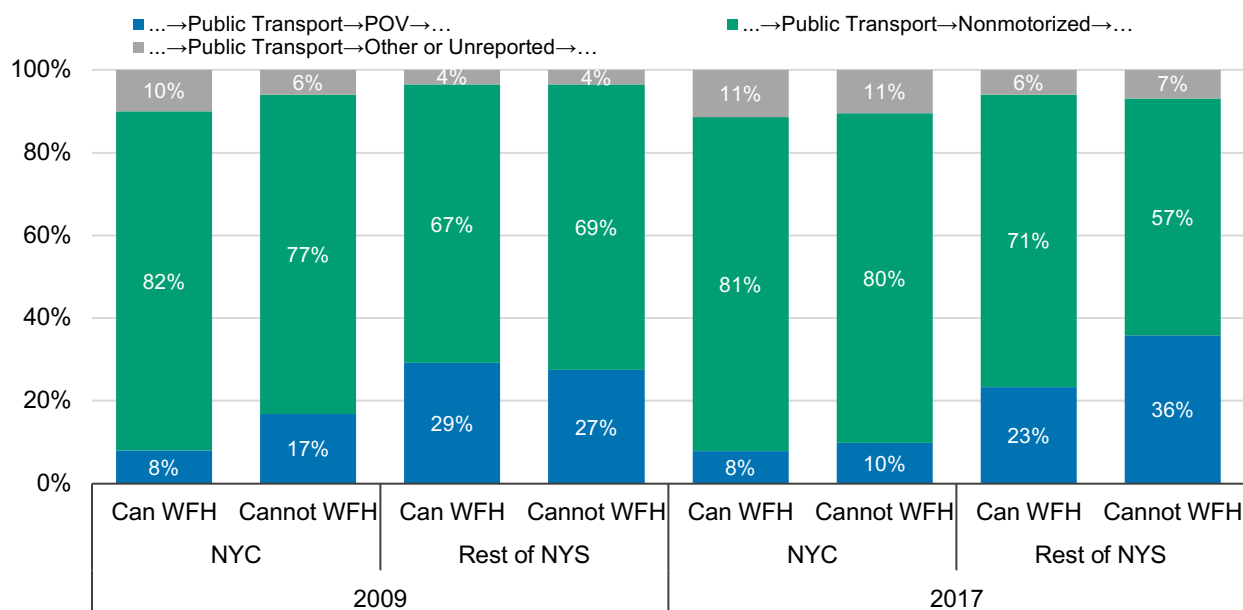


Figure 3-7. Transportation Mode Transition Following a Public Transport Trip by Option of WFH in NYC and Rest of NYS

In the rest of NYS, the share of individuals transferring from public transport to a personal vehicle was generally higher than in NYC, particularly among WFH-capable workers. In 2009, 29% of WFH-capable workers in the rest of NYS transferred to POV, statistically significantly higher than the 27% of non-WFH workers. In 2017, this gap widened with WFH-capable workers at 23% and non-WFH workers at 36%, though the change over time among WFH-capable workers (from 29% to 23%) was not statistically

significant. Nonmotorized transfers in the rest of NYS were consistently more common among WFH-capable workers compared to non-WFH workers in both years. In 2017, WFH-capable workers in the rest of NYS had a statistically significantly higher share transferring to nonmotorized modes (71%) compared to non-WFH workers (57%).

3.3 TRIP CHAINING BY SEX

In 2009, there were no statistically significant differences between females and males in NYC across all activity levels (Table 3-4). Both sexes were most likely to report either one or two trips per day, with about 34% engaging in two trips and approximately 40–43% in one trip. However, the rest of NYS showed more variation by sex. Females were statistically significantly more likely than males to make no trips (15% vs. 11%) and to engage in three or more activities (15% vs. 12%). Males, by contrast, were statistically significantly more likely to report two trips (36% vs. 33%). Comparing NYC and the rest of NYS in 2009 reveals clear regional differences. NYC residents of both sexes were statistically significantly less likely to report three or more trips compared to their counterparts in the rest of NYS. For instance, only 8–10% of NYC residents made three or more trips, compared to 12–15% in the rest of NYS.

By 2017, NYC females had a statistically significantly higher share of no-trip days (20%) than in 2009 (16%). For NYC males, the share of three or more trips increased significantly from 8% to 11%. In the rest of NYS in 2017, both males and females experienced statistically significant increases in no-trip days compared to 2009. Among females, the no-trip rate increased significantly from 15% to 21%, and for males, from 11% to 19%. At the same time, the share of individuals undertaking two trips declined statistically significantly for both sexes (from 33% to 29% for females and from 36% to 30% for males).

Table 3-4. Share of Daily Number of Activity Types by Sex in NYC and Rest of NYS

NHTS year	Region	Sex	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	Female	16%	40%	34%	10%
		Male	15%	43%	34%	8%
	Rest of NYS	Female	15%	37%	33%	15%
		Male	11%	40%	36%	12%
2017	NYC	Female	20%	38%	34%	9%
		Male	17%	41%	31%	11%
	Rest of NYS	Female	21%	38%	29%	13%
		Male	19%	41%	30%	11%

3.3.1 Pre-Work Trip Chaining

In 2009, the majority of both males and females in NYC and the rest of NYS commuted directly from home to work, with 77–81% following this pattern (Figure 3-8). There were no statistically significant differences between sexes or regions in this regard. Secondary stop types, such as family & personal business, social and recreational, and short stops, were relatively infrequent across all groups, generally under 10%. However, NYC females were statistically significantly more likely than NYC males to make a short stop before work (6% vs. 3%). Similarly, females in the rest of NYS were more likely than males to do so (8% vs. 5%), though not at a statistically significant level. By 2017, direct commutes remained dominant, especially among males (81% in both NYC and the rest of NYS), with no statistically significant change from 2009. NYC females showed a slight decline in direct commuting from 78% to 77%, though this change was not statistically significant.

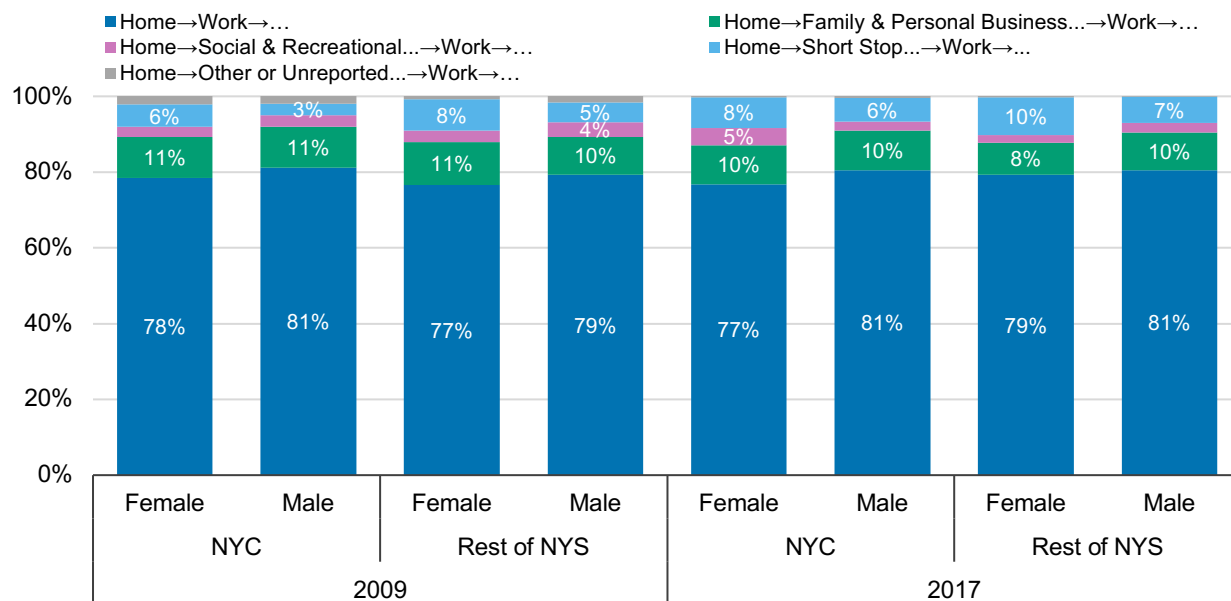


Figure 3-8. Pre-Work Activity Patterns by Sex in NYC and Rest of NYS

Across all groups and both years, the dominant pattern was making no trips before work, especially among males (Table 3-5). In 2009, the majority of workers in both NYC and the rest of NYS made no trips before work, with rates ranging from 77% to 81%. By 2017, patterns began to diverge more notably, particularly among NYC females. The share of NYC females making one trip rose sharply from 9% to 16%, a statistically significant increase. NYC females also showed a statistically significant decrease in the share making three or more trips, dropping from 6% to just 2%. NYC males, by contrast, maintained a largely stable pattern. The share making no trips remained at 81%, and changes in the distribution of trips were small and not statistically significant. Males in the rest of NYS also showed little change. Females in the rest of NYS experienced a small but statistically significant increase in one-trip behavior from 11% to 14%, and a decrease in two-trip days from 6% to 4%. However, their overall profile remained relatively stable, with nearly 80% still making no trips before work.

Table 3-5. Distribution of Number of Trips Before Work by Sex in NYC and Rest of NYS

NHTS year	Region	Sex	No trip	One trip	Two trips	3+ trips
2009	NYC	Female	78%	9%	6%	6%
		Male	81%	6%	6%	6%
	Rest of NYS	Female	77%	11%	6%	6%
		Male	79%	11%	7%	3%
2017	NYC	Female	77%	16%	5%	2%
		Male	81%	9%	7%	3%
	Rest of NYS	Female	79%	14%	4%	3%
		Male	81%	12%	4%	3%

3.3.2 Post-Work Trip Chaining

Figure 3-9 shows post-work activity patterns among workers who began their day at home and ended at home. In 2009, both males and females in the rest of NYS were more likely to follow a direct return-home pattern after work compared to their NYC counterparts. For example, 76% of males and 73% of

females in the rest of NYS returned straight home, compared to 74% of NYC males and just 67% of NYC females. The difference between NYC and rest of NYS females was statistically significant, indicating that NYC females were more likely to stop for other purposes after work. NYC females also had the highest share engaging in family and personal business stops (24%), significantly higher than NYC males (18%) and their female counterparts in the rest of NYS (19%).

By 2017, the share of NYC females returning directly home after work dropped from 67% to 56%, a statistically significant decrease. This shift was accompanied by a statistically significant increase in post-work family and personal business stops (rising from 24% to 32%). NYC males also showed a decrease in direct returns home (from 74% to 66%) but the decline was not statistically significant. However, their share making social & recreational stops increased from 6% to 9%, a statistically significant rise, suggesting growing engagement in leisure after work.

Comparing across regions in 2017, NYC females were statistically significantly less likely to go directly home after work (56%) than any other group. They were also significantly more likely to stop for family and personal business (32%), highlighting a distinct post-work activity pattern not seen among males or among females in less urbanized areas.

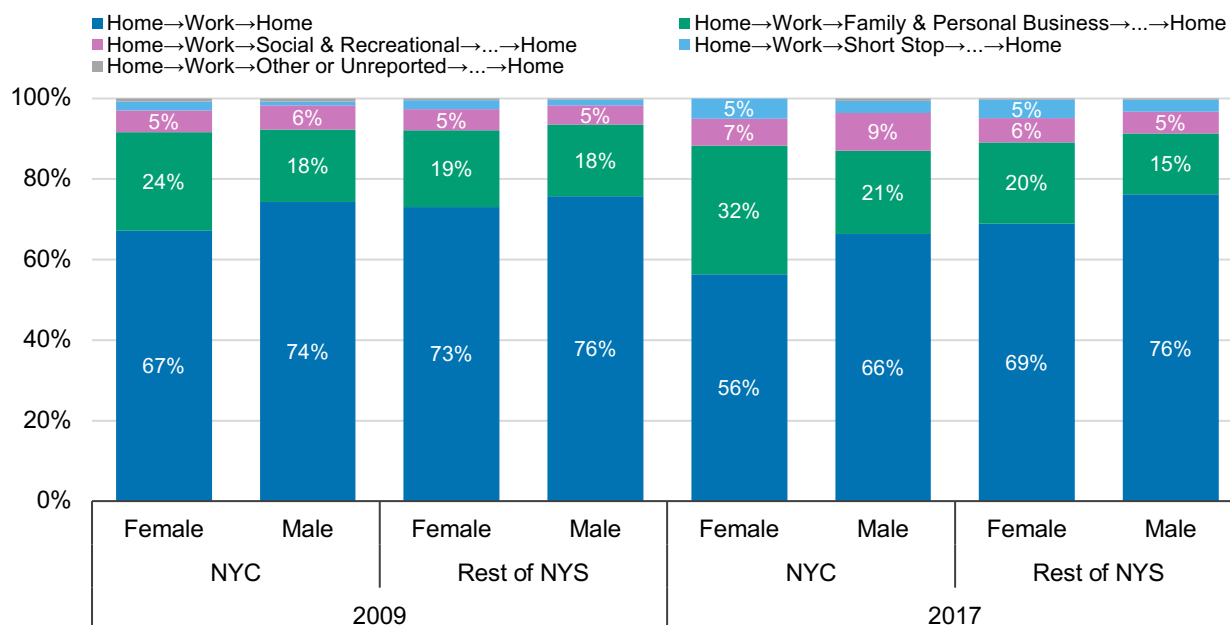


Figure 3-9. Post-Work Activity Patterns by Sex in NYC and Rest of NYS: Home→Work Trip Chains

Figure 3-10 shows the departure time distributions for women in NYC in 2017 whose first trip after work is for family and personal business. The four panels correspond to successive trips in the chain: the morning commute to work, the post-work trip to family or personal business, the continuation from there, and finally the return home. The third trip (after the family & personal business activity) has a median of 3:45 p.m., just half an hour after the second trip. This indicates that most family or personal business stops might be short in duration, often involving quick errands. While there is a small secondary spread into the evening, the activity is strongly centered around mid-to-late afternoon.

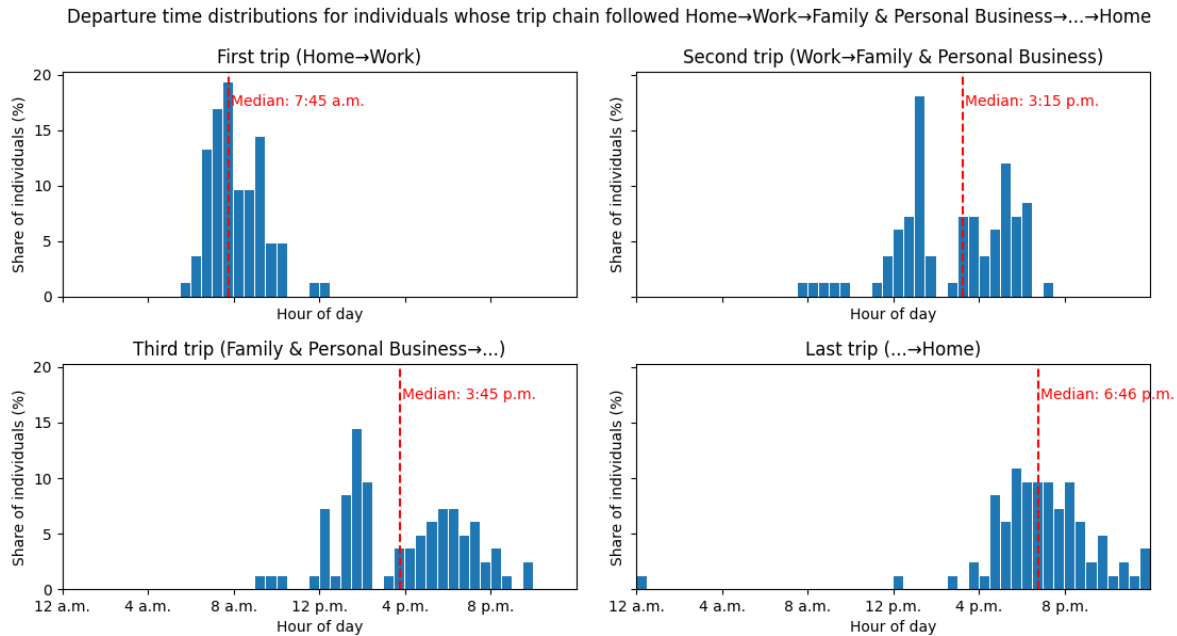


Figure 3-10. Departure Time Distributions for Females in NYC in 2017 Whose First Trip After Work is for Family and Personal Business

3.3.3 Non-Work Days Trip Chaining

In 2009, females in NYC were statistically significantly more likely than males to undertake trips for family and personal business (65% vs. 64%) and less likely to engage in social and recreational activities (25% vs. 32%), although the difference in recreation is not statistically significant (Figure 3-11). NYC males had the highest share of social and recreational trips, statistically significantly higher than females in both NYC and the rest of NYS. Meanwhile, rest of NYS residents, both male and female, had lower rates of family and personal business trips and higher shares of short stops than their NYC counterparts, with females more likely than males to make short stops (12% vs. 10%), though not significantly so.

By 2017, NYC females experienced a statistically significant decline in family and personal business activity (from 65% to 61%) and a corresponding, though not significant, rise in social and recreational trips (28% vs. 25%). In the rest of NYS, the most notable change was a statistically significant increase in family and personal business trips (from 58% to 62%), reversing the prior gap with NYC males.

3.3.4 Transportation Mode Transition

Figure 3-12 shows how individuals transferred from a privately owned vehicle (POV) trip to other modes: public transport, nonmotorized modes (such as walking or biking), or other/unreported types. In 2009, NYC females were significantly less likely than NYC males to switch from a POV to public transport (17% vs. 26%), and significantly more likely to transition to nonmotorized modes (79% vs. 69%). The opposite pattern was seen for NYC males, who had the highest share shifting to public transport. In contrast, women and men in the rest of NYS were much less likely to switch to public transport—only 6% and 8%, respectively—and significantly more likely to rely on nonmotorized modes, with 81% of women and 77% of men doing so. These differences highlight the more multimodal environment of NYC compared to the rest of the state.

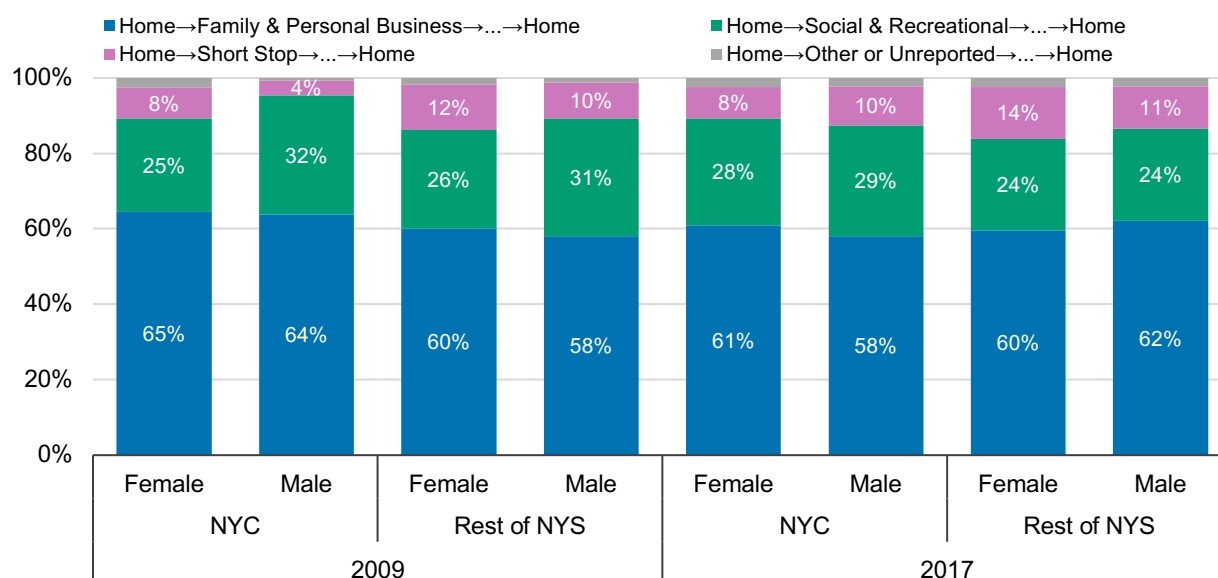


Figure 3-11. Activity Patterns on Non-Work Days by Sex in NYC and Rest of NYS

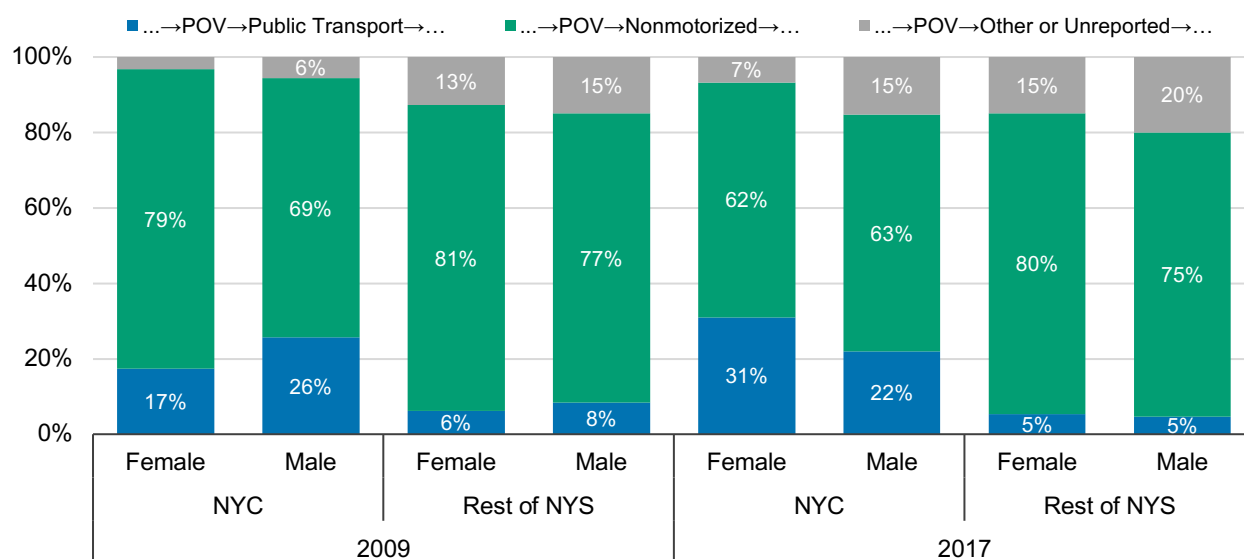


Figure 3-12. Transportation Mode Transition Following a POV Trip by Sex in NYC and Rest of NYS

By 2017, the most dramatic shift occurred among NYC females, whose transfers from POV to public transport jumped from 17% to 31%—a statistically significant increase—while their use of nonmotorized modes declined to 62%. This suggests greater integration of transit into travel chains involving POV, possibly due to changing accessibility or service improvements. NYC males, however, saw a statistically significant decline in public transport transfers (from 26% to 22%) and a statistically significant increase in the share ending in other or unreported modes (from 6% to 15%). In the rest of NYS, mode transition patterns remained largely unchanged between 2009 and 2017.

Figure 3-13 presents how individuals transferred from a public transport trip to other modes. In 2009, NYC residents of both sexes were highly likely to continue their travel by nonmotorized modes following

public transport: 76% for both males and females. However, NYC males were statistically significantly more likely than females to transfer to a POV (20% vs. 14%). In contrast, residents in the rest of NYS were far more reliant on POV following transit. A statistically significantly higher share of females in the rest of NYS transferred to POV (34%) compared to NYC females (14%). The same pattern held for males. By 2017, NYC females became even more likely to continue with nonmotorized modes, rising from 76% to 85%, a statistically significant increase. Their reliance on POV dropped to 8%, now significantly lower than all other groups.

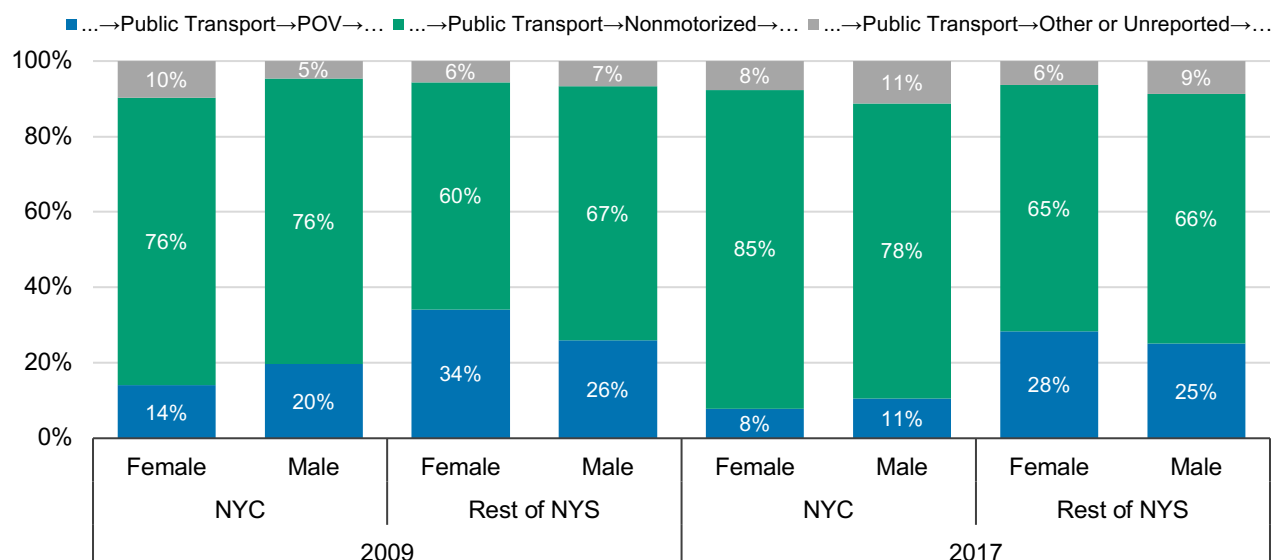


Figure 3-13. Transportation Mode Transition Following a Public Transport Trip by Sex in NYC and Rest of NYS

3.4 TRIP CHAINING BY HOUSEHOLD SIZE

In NYC, individuals from three-or-more-person households were statistically significantly more likely to report just one activity type (45%) compared with those from single-person (38%) or two-person households (35%) in 2009 (Table 3-6). This pattern held in 2017, with 40% of three-or-more-person households in NYC reporting one activity type, though the differences with smaller households were narrower and not consistently statistically significant.

The share of individuals reporting no trips varied by household size. In both NYC and the rest of NYS, in 2009, individuals from larger households (three or more people) were statistically significantly less likely to report no trips compared to those from single-person households. For example, in NYC in 2009, 13% of three-or-more-person households reported no trips, statistically significantly lower than the 17% of single-person households. By 2017, however, this difference narrowed or reversed in NYC, with 21% of individuals from three-or-more-person households reporting no trips; higher than in 2009 and comparable to other household sizes. In the rest of NYS, patterns were similar. In both years, single-person households consistently showed the highest share of no-trip days, although the differences were small and often not statistically significant.

Table 3-6. Share of Daily Number of Activity Types by Household Size in NYC and Rest of NYS

NHTS year	Region	Household size	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	One person	17%	38%	34%	11%
		Two people	20%	35%	34%	11%
		3+ people	13%	45%	34%	8%
	Rest of NYS	One person	20%	36%	34%	10%
		Two people	15%	39%	37%	9%
		3+ people	12%	39%	34%	16%
2017	NYC	One person	16%	36%	37%	11%
		Two people	16%	40%	34%	10%
		3+ people	21%	40%	30%	10%
	Rest of NYS	One person	18%	37%	32%	13%
		Two people	19%	39%	31%	11%
		3+ people	20%	40%	29%	12%

3.4.1 Pre-Work Trip Chaining

Across both years, workers from larger households (three or more people) were statistically significantly more likely to go directly from home to work without any stops compared to those from smaller households (Figure 3-14). In 2009, 82% of NYC workers from households with three or more people went directly to work, statistically significantly higher than 75% of two-person households and 77% of single-person households. This pattern persisted in 2017, with 78% of workers from three-or-more-person households in NYC going directly to work, statistically significantly higher than the 70% among single-person households. In the rest of NYS, the pattern was similar.

Family and personal business stops before work were more common among single-person and two-person households, particularly in NYC. In 2009, 16% of NYC single-person households made such stops, statistically significantly higher than the 9% of three-or-more-person households. In 2017, this pattern remained, with 22% of single-person households making family/personal business stops compared to 8% of three-or-more-person households, a statistically significant difference. In the rest of NYS, the differences between household sizes for family/personal business stops were smaller but followed a similar trend, with single-person households generally showing higher shares than larger households.

Short stops before work showed some variation, particularly in 2017. In NYC, workers from households with three or more people had a higher share of short stops (11%) compared to those from smaller households. In the rest of NYS, differences in short stops were minor and not statistically significant.

3.4.2 Post-Work Trip Chaining

Across both years, in both regions, workers from larger households (three or more people) were statistically significantly more likely to go directly home after work compared to those from smaller households (Figure 3-15). In 2009, in NYC, 75% of workers from households with three or more people went directly home, statistically significantly higher than 67% of those from two-person households and 53% of those from single-person households. This pattern persisted in 2017, though the overall shares declined slightly, with 67% for three or more people households still statistically significantly higher than 57% for two-person and 50% for single-person households.

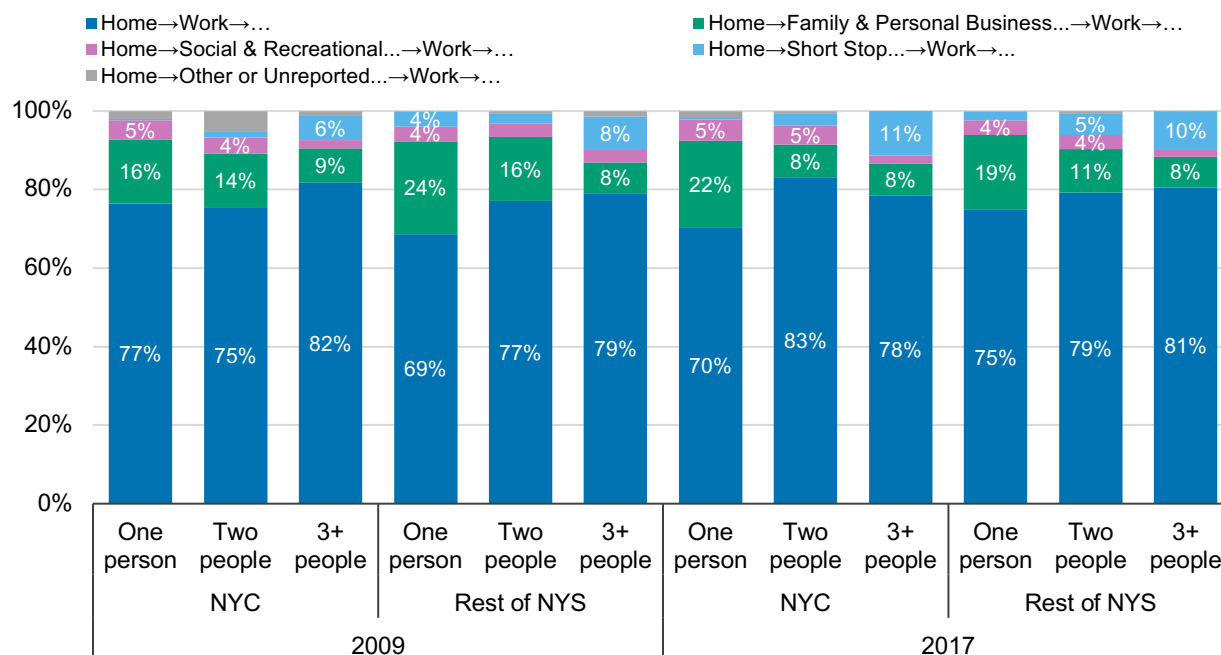


Figure 3-14. Pre-Work Activity Patterns by Household Size in NYC and Rest of NYS

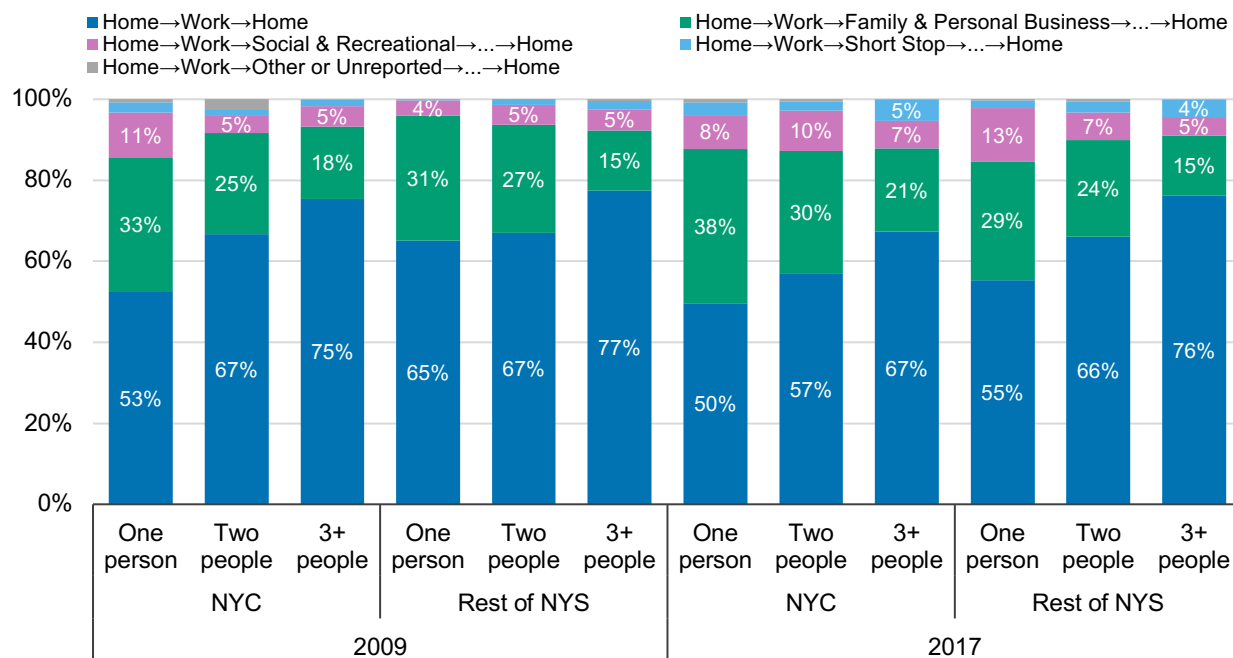


Figure 3-15. Post-Work Activity Patterns by Household Size in NYC and Rest of NYS: Home→Work Trip Chains

Conversely, single-person households in both regions were statistically significantly more likely to engage in family and personal business activities after work. In NYC, 33% of single-person households in 2009 made such stops, statistically significantly higher than 25% of two-person and 18% of three-or-more-person households. This difference remained statistically significant in 2017, despite a slight

decline. In the rest of NYS, a similar pattern was observed, though the differences were generally smaller and sometimes not statistically significant.

Social and recreational stops were also more common among single-person households, particularly in NYC. In 2009, 11% of NYC single-person households made such stops, statistically significantly higher than the 5% or lower observed among larger households. This trend remained in 2017, though with some narrowing of the gap.

3.4.3 Non-Work Days Trip Chaining

In both years, family and personal business trips were the most common activity on non-work days across all household sizes and regions (Figure 3-16). In NYC, single-person households were statistically significantly more likely to make family/personal business trips compared to larger households in 2009. For example, 70% of single-person households engaged in such trips, statistically significantly higher than the 63% of two-person households and the 64% of three-or-more-person households. In 2017, this difference narrowed, with single-person and two-person households in NYC reporting similar shares (63% and 64%, respectively), both statistically significantly higher than three-or-more-person households (55%). In the rest of NYS, similar trends were observed. In 2017, the highest share was among single-person households (72%), statistically significantly higher than three-or-more-person households (57%).

Social and recreational activities were the second most common non-work day activity in both regions. In NYC, the share making such trips was relatively similar across household sizes, with no consistently statistically significant differences in 2009. However, in 2017, three-or-more-person households in NYC showed a slightly higher share of social/recreational trips (31%) compared to other groups, though differences were not consistently statistically significant. In the rest of NYS, three-or-more-person households showed a statistically significantly higher share of social/recreational trips in both years compared to single-person and two-person households.

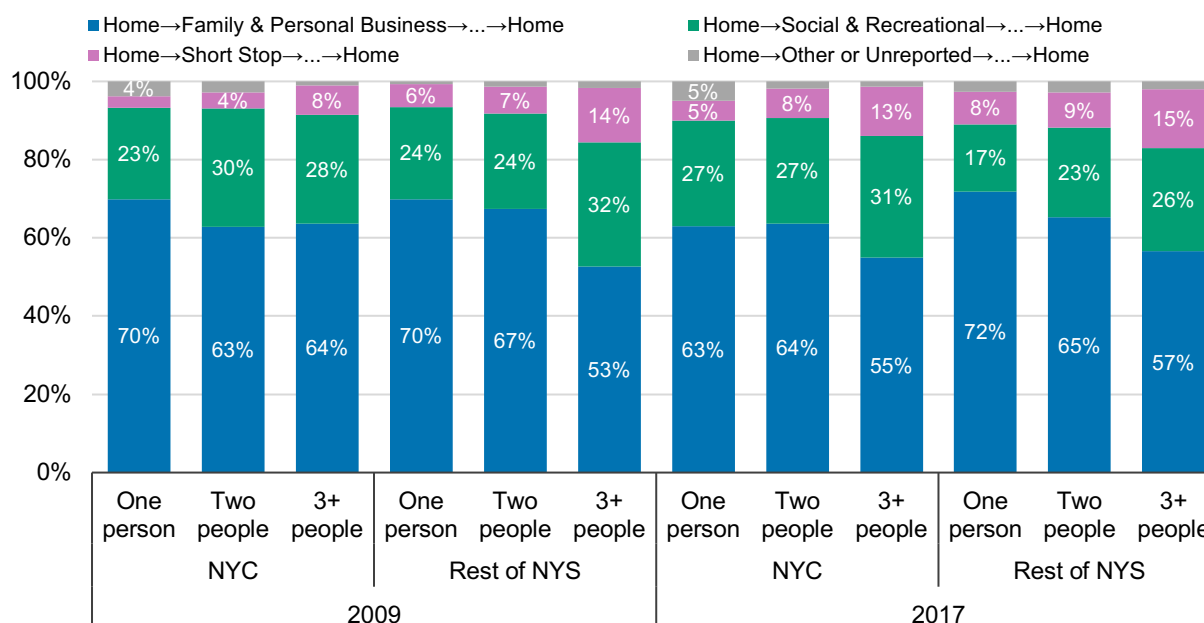


Figure 3-16. Activity Patterns on Non-Work Days by Household Size in NYC and Rest of NYS

3.5 TRIP CHAINING BY HOME OWNERSHIP

One or two activity types per day were the most common patterns for both homeowners and renters (Table 3-7). In NYC in 2009, the shares of individuals reporting one or two activity types were nearly identical between homeowners and renters, with no statistically significant differences: 41% of homeowners and 42% of renters engaged in one activity type, while 33% of homeowners and 34% of renters reported two activity types. In the rest of NYS in 2009, a similar pattern held.

By 2017, some changes emerged, particularly in NYC. Homeowners in NYC had a higher share of individuals reporting one activity type (43%) compared to renters (36%), a difference that was statistically significant. Conversely, renters in NYC were statistically significantly more likely to engage in two activity types (36%) than homeowners (28%) in 2017. In the rest of NYS in 2017, differences between homeowners and renters remained small, with no consistently statistically significant differences across categories. Both groups most commonly engaged in one activity type, followed by two activity types, similar to the patterns observed in 2009.

Table 3-7. Share of Daily Number of Activity Types by Home Ownership in NYC and Rest of NYS

NHTS year	Region	Home ownership	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	Own	15%	41%	33%	11%
		Rent	16%	42%	34%	9%
	Rest of NYS	Own	13%	38%	35%	15%
		Rent	17%	41%	32%	10%
2017	NYC	Own	19%	43%	28%	10%
		Rent	18%	36%	36%	10%
	Rest of NYS	Own	19%	39%	30%	12%
		Rent	22%	39%	28%	11%

3.5.1 Pre-Work Trip Chaining

In NYC, in 2009, 81% of homeowners went directly to work compared to 79% of renters, a difference not statistically significant (Figure 3-17). In 2017, the shares remained close, with 79% for homeowners and 78% for renters. In the rest of NYS, direct home-to-work trips were consistently common for both groups, with little change over time. For example, in 2017, 80% of homeowners and 78% of renters went directly to work, with no statistically significant difference observed.

Family and personal business stops before work were slightly more common among renters than homeowners in both NYC and the rest of NYS across both years, but these differences were generally small and often not statistically significant. For example, in NYC in 2017, 11% of renters made family/personal business stops compared to 9% of homeowners.

Short stops before work were made by a small share of workers across all groups. In 2017, homeowners in NYC showed a slightly higher share of short stops (8%) compared to renters (6%), but this difference was not statistically significant. In the rest of NYS, both groups reported similar levels of short stops in both years.

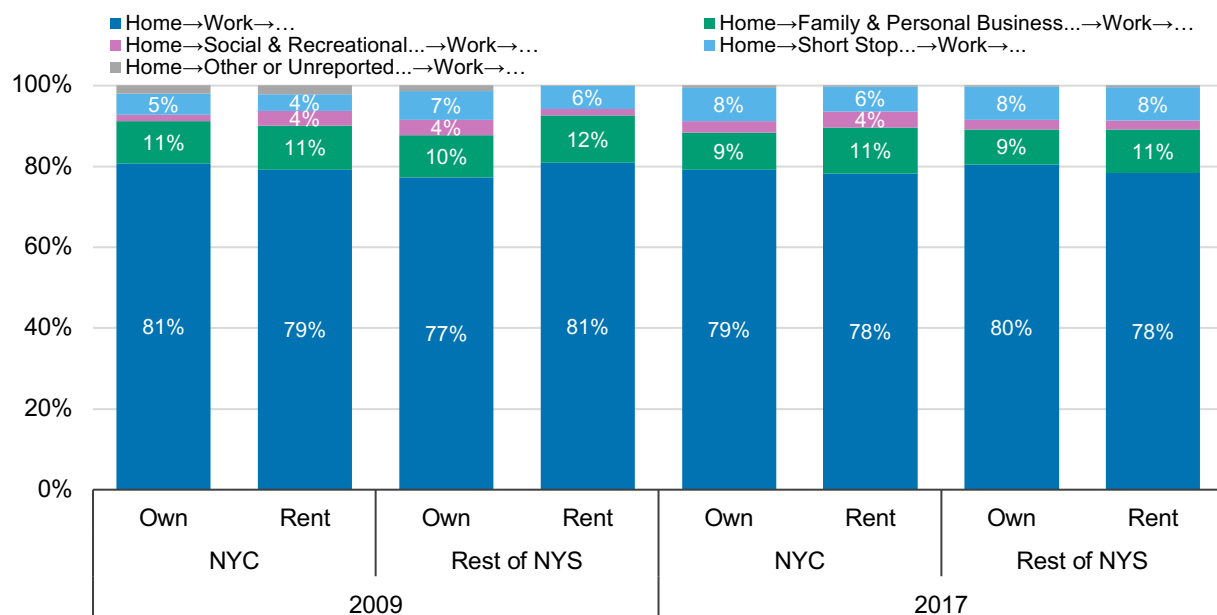


Figure 3-17. Pre-Work Activity Patterns by Home Ownership in NYC and Rest of NYS

3.5.2 Post-Work Trip Chaining

Homeowners in both NYC and the rest of NYS were statistically significantly more likely to go directly home after work compared to renters (Figure 3-18). In 2009, 69% of NYC homeowners went directly home, statistically significantly higher than 72% of NYC renters. However, in this case, the difference was small and not statistically significant. In the rest of NYS, the difference was more pronounced: 72% of homeowners versus 83% of renters in 2009, a statistically significant difference favoring renters for direct-home trips. By 2017, the gap widened in NYC. The share of NYC homeowners going directly home after work remained relatively high at 67%, while it dropped to 57% for renters. This difference was statistically significant, indicating that NYC renters in 2017 were more likely to make post-work stops compared to homeowners. In the rest of NYS, homeownership continued to be associated with a higher likelihood of going directly home after work in both years, with minimal change over time. In 2017, 73% of homeowners and 72% of renters in the rest of NYS went directly home, with no statistically significant difference observed that year.

Family and personal business stops were slightly more common among renters than homeowners in both regions and years, though differences were not always statistically significant. In 2017, NYC renters had a statistically significantly higher share of family and personal business stops (29%) compared to NYC homeowners (22%). This suggests renters in NYC may have more diverse post-work activity patterns.

Social and recreational stops after work were also more frequent among NYC renters, especially in 2017. In that year, 9% of NYC renters made social and recreational stops compared to 7% of homeowners. The difference was statistically significant.

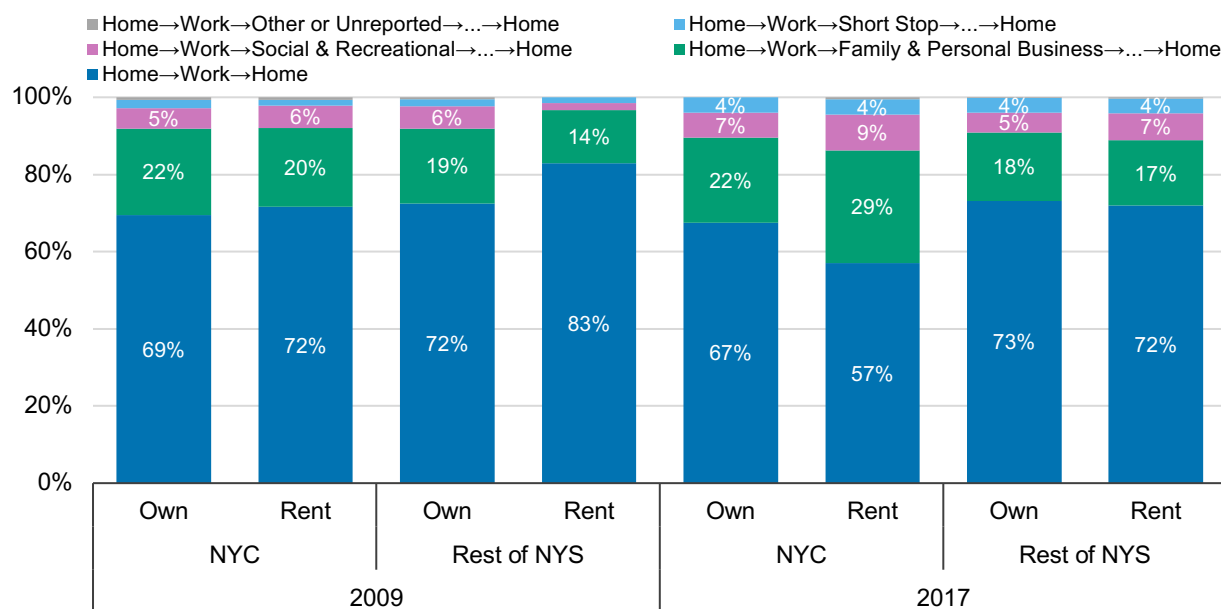


Figure 3-18. Post-Work Activity Patterns by Home Ownership in NYC and Rest of NYS: Home→Work Trip Chains

3.5.3 Non-Work Days Trip Chaining

In NYC, renters were statistically significantly more likely to make family/personal business trips (68%) compared to homeowners (57%) in 2009 (Figure 3-19). However, by 2017, this difference narrowed, with 61% of homeowners and 58% of renters making such trips, a difference that was not statistically significant. In the rest of NYS, renters also showed a statistically significantly higher share of family/personal business trips than homeowners in 2009 (64% vs. 58%). This difference persisted in 2017, though the gap was not statistically significant (64% for renters vs. 59% for homeowners).

Social and recreational activities showed mixed patterns. In NYC in 2009, homeowners were statistically significantly more likely to make social/recreational trips (33%) than renters (25%). However, by 2017, renters slightly surpassed homeowners in NYC (30% vs. 28%), though the difference was not statistically significant. In the rest of NYS, homeowners consistently showed similar or slightly higher shares of social/recreational trips compared to renters.

3.5.4 Transportation Mode Transition

In NYC, homeowners were statistically significantly more likely to transfer from public transport to a personal vehicle compared to renters in both years (Table 3-8). In 2009, 25% of NYC homeowners switched to POV after transit, statistically significantly higher than the 12% of renters. Although the share decreased in 2017 for both groups, the difference remained statistically significant: 16% for homeowners versus 5% for renters.

Conversely, NYC renters were statistically significantly more likely to switch to nonmotorized modes after public transport. In 2009, 79% of NYC renters transferred to nonmotorized modes compared to 69% of homeowners. This difference persisted in 2017, with 84% of renters versus 79% of homeowners making this transition, again statistically significant. In the rest of NYS, the pattern was similar but with higher overall shares of transfers to POV among homeowners. In 2009, 43% of homeowners transferred from public transport to POV, statistically significantly higher than the 14% of renters. This difference

remained in 2017, with 32% of homeowners versus 18% of renters switching to POV, both statistically significant differences.

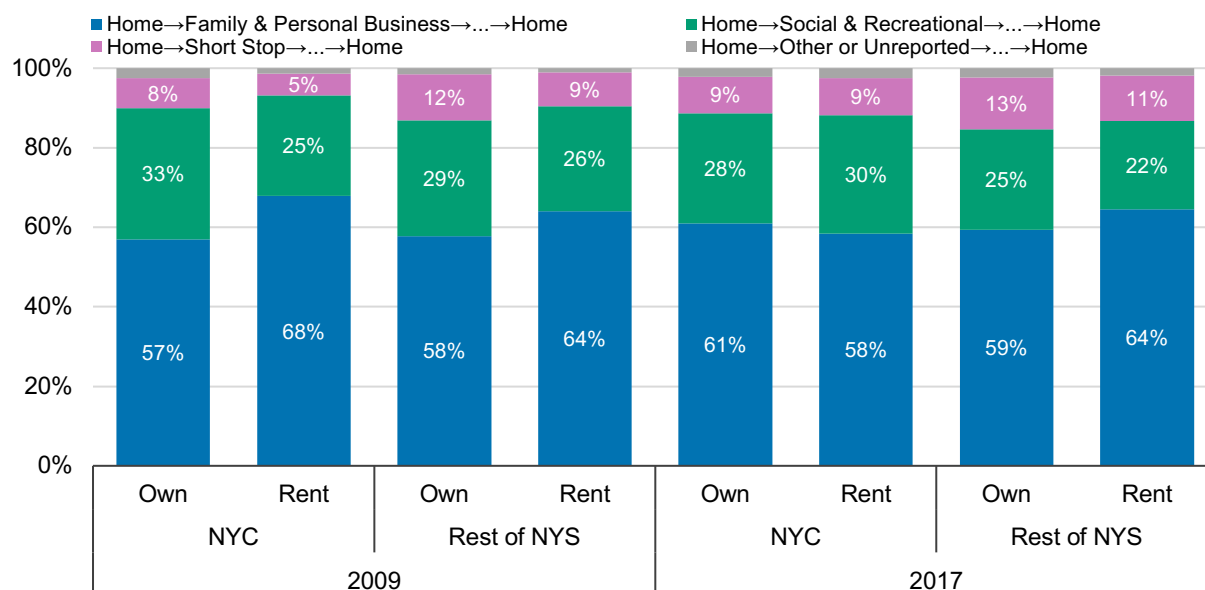


Figure 3-19. Activity Patterns on Non-Work Days by Home Ownership in NYC and Rest of NYS

Table 3-8. Transportation Mode Transition Following a Public Transport Trip by Home Ownership in NYC and Rest of NYS

NHTS year	Region	Home ownership	...→Public Transport→POV→...	...→Public Transport→Nonmotorized→...	...→Public Transport→ Other or Unreported→...
2009	NYC	Own	25%	69%	6%
		Rent	12%	79%	9%
	Rest of NYS	Own	43%	53%	5%
		Rent	14%	78%	8%
2017	NYC	Own	16%	79%	5%
		Rent	5%	84%	11%
	Rest of NYS	Own	32%	66%	2%
		Rent	18%	68%	13%

Nonmotorized transfers in the rest of NYS were more common among renters in both years. In 2009, 78% of renters transferred to nonmotorized modes compared to 53% of homeowners, a statistically significant gap. This pattern persisted in 2017 with 68% of renters compared to 66% of homeowners, though the difference narrowed slightly and was less consistently statistically significant.

3.6 TRIP CHAINING BY VEHICLE OWNERSHIP

In NYC, in both years, individuals from households with at least one vehicle (non-zero vehicle) were statistically significantly less likely to have no trips compared to those from zero-vehicle households (Table 3-9). In 2009, 13% of vehicle owners reported no trips versus 18% of zero-vehicle individuals, a statistically significant difference. This pattern remained in 2017, with 18% for vehicle owners and 20%

for zero-vehicle households, though the difference narrowed and may not be statistically significant. In the rest of NYS, differences by vehicle ownership were more pronounced, particularly in the share of individuals reporting no trips. In both years, zero-vehicle households were statistically significantly more likely to report no trips compared to vehicle owners. In 2009, 30% of zero-vehicle individuals reported no trips versus 12% of vehicle owners. In 2017, this gap persisted, with 26% of zero-vehicle individuals and 19% of vehicle owners reporting no trips, a statistically significant difference in both years.

For one activity type per day, zero-vehicle households in the rest of NYS consistently had a higher share than vehicle owners. In 2017, 50% of zero-vehicle individuals reported one activity type, statistically significantly higher than the 38% of vehicle owners. Conversely, vehicle owners in the rest of NYS were statistically significantly more likely to engage in two or more activity types per day compared to zero-vehicle individuals. In 2017, 30% of vehicle owners reported two activity types versus only 20% of zero-vehicle individuals. The difference in engaging in three or more activity types was also statistically significant, 12% of vehicle owners versus 5% of zero-vehicle individuals.

Table 3-9. Share of Daily Number of Activity Types by Vehicle Ownership in NYC and Rest of NYS

NHTS year	Region	Vehicle ownership	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	Not zero vehicle	13%	42%	35%	10%
		Zero vehicle	18%	41%	32%	8%
	Rest of NYS	Not zero vehicle	12%	38%	35%	14%
		Zero vehicle	30%	45%	21%	4%
2017	NYC	Not zero vehicle	18%	40%	32%	11%
		Zero vehicle	20%	39%	33%	9%
	Rest of NYS	Not zero vehicle	19%	38%	30%	12%
		Zero vehicle	26%	50%	20%	5%

3.6.1 Pre-Work Trip Chaining

In both years, the majority of workers regardless of vehicle ownership went directly from home to work, though patterns varied between NYC and the rest of NYS (Figure 3-20). In NYC, there was no statistically significant difference between vehicle-owning and zero-vehicle households in the likelihood of going directly to work without stops in either year. In 2009, 81% of vehicle owners and 78% of zero-vehicle workers went directly to work. This pattern held in 2017, with both groups at 79%. Family and personal business stops before work were similar between vehicle owners and non-owners in NYC, with no consistently statistically significant differences. For example, in 2017, 9% of vehicle owners and 12% of zero-vehicle workers made family and personal business stops, a difference that was not statistically significant. Short stops before work in NYC showed some variation but no consistent statistically significant patterns. In 2017, 9% of vehicle owners made short stops compared to 4% of zero-vehicle workers, though this difference may not be statistically significant.

In the rest of NYS, workers without a vehicle were statistically significantly more likely to go directly to work in 2009. In that year, 92% of zero-vehicle workers in the rest of NYS reported direct home-to-work trips, statistically significantly higher than the 77% of vehicle owners. This difference narrowed by 2017, with 85% of zero-vehicle workers and 80% of vehicle owners reporting direct trips, though the gap remained statistically significant. Pre-work stops for other purposes, including family and personal business and short stops, were consistently less common among zero-vehicle workers in the rest of NYS compared to vehicle owners, with several statistically significant differences observed in both years. For

example, in 2017, vehicle owners in the rest of NYS were statistically significantly more likely to make short stops (8%) compared to zero-vehicle workers (7%), though the difference was small.

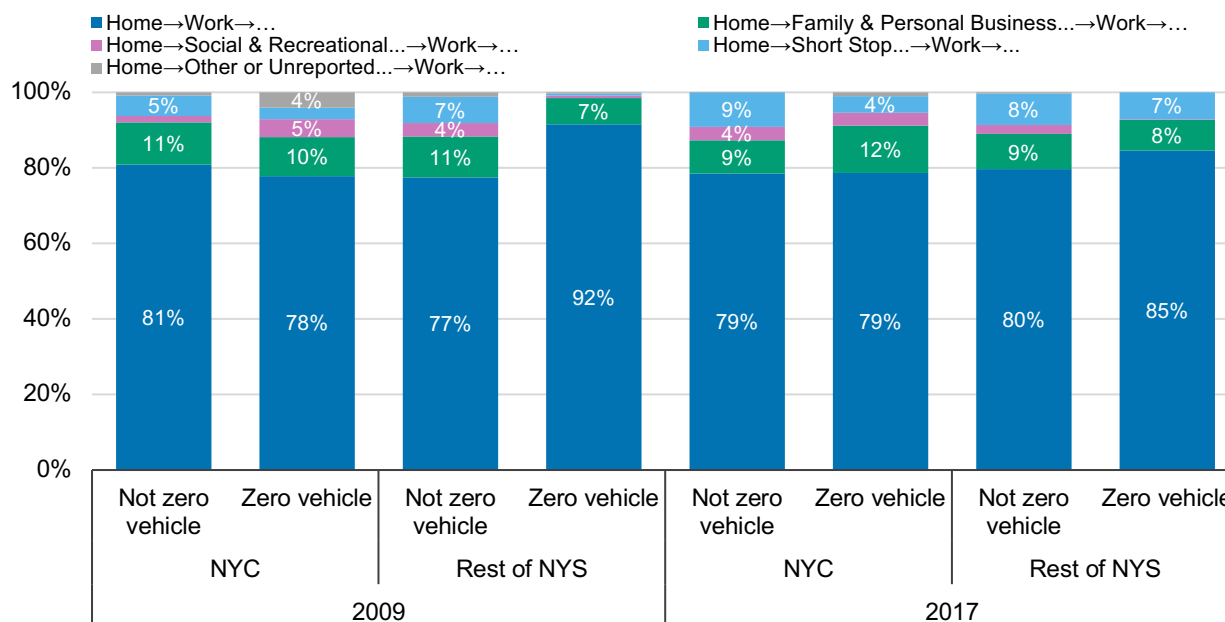


Figure 3-20. Pre-Work Activity Patterns by Vehicle Ownership in NYC and Rest of NYS

3.6.2 Post-Work Trip Chaining

Workers with at least one vehicle (non-zero vehicle households) were statistically significantly more likely to go directly home after work compared to zero vehicle households in NYC (Figure 3-21). In NYC in 2009, 75% of workers with a vehicle went straight home, statistically significantly higher than the 63% of zero vehicle workers. This gap widened in 2017, with 69% of workers with a vehicle versus 52% of those without a vehicle making direct-home trips, a statistically significant difference. In the rest of NYS, the opposite patterns were observed.

Family and personal business stops after work were consistently more common among zero vehicle households in NYC. In 2009, 28% of zero vehicle households made such stops, statistically significantly higher than the 17% of non-zero vehicle households. This difference persisted in 2017, with 33% for zero vehicle households versus 21% for those with a vehicle, again statistically significant. In the rest of NYS, differences in family and personal business stops by vehicle ownership were smaller. In both years, the shares of workers making these stops were generally similar between vehicle-owning and zero vehicle households, with no consistently statistically significant differences.

Social and recreational stops showed a similar pattern, particularly in NYC. Zero-vehicle households in NYC were statistically significantly more likely to make social and recreational stops after work in both years. In 2017, 11% of NYC zero vehicle workers made social and recreational stops, statistically significantly higher than the 6% of workers with a vehicle.

Figure 3-22 shows the sequence of departure times for people in NYC who live in zero-vehicle households and whose first trip after work is for family and personal business. Each panel tracks one stage of their daily travel, with distributions and median departure times marked. It in general shows how zero-vehicle households in NYC structure their days: a highly regular morning work trip, followed by mid-afternoon family or personal business stops, and a more flexible homebound trip in the evening.

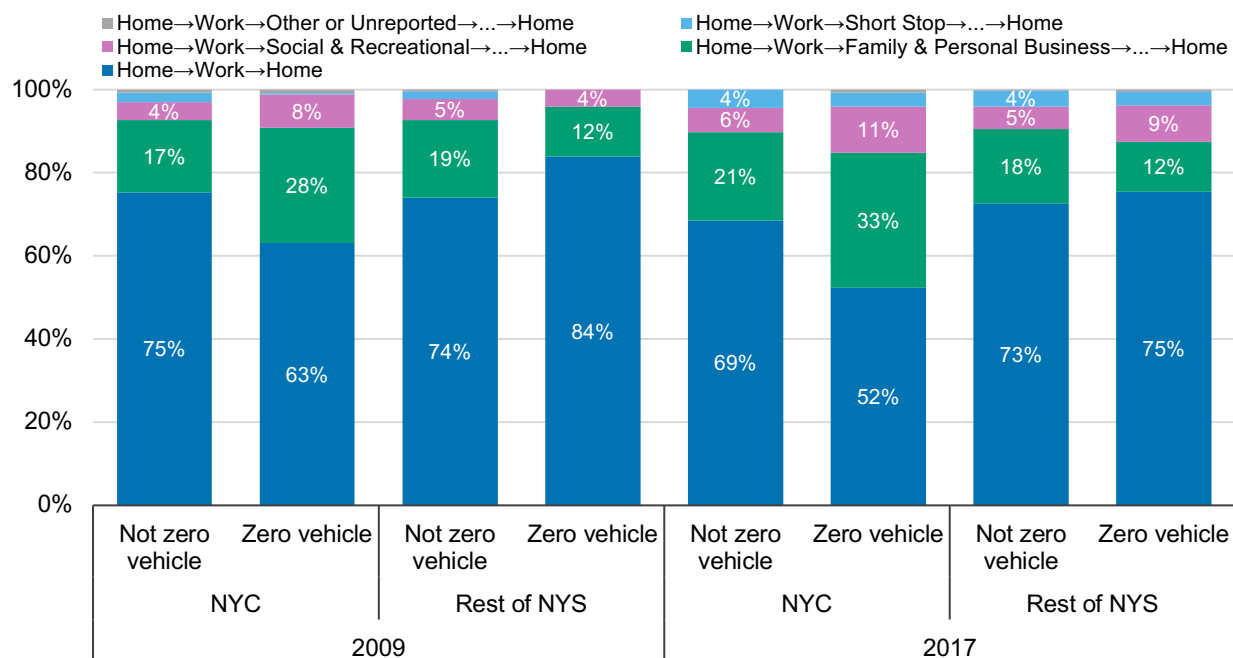


Figure 3-21. Post-Work Activity Patterns by Vehicle Ownership in NYC and Rest of NYS: Home→Work Trip Chains

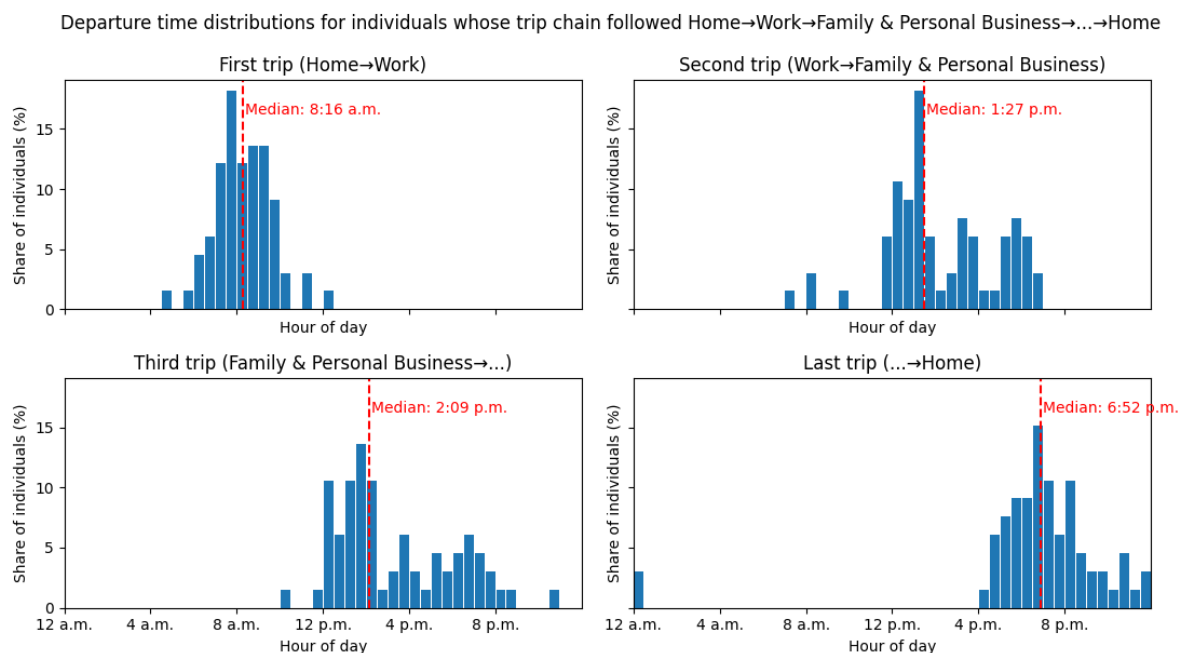


Figure 3-22. Departure Time Distributions for Zero Vehicle Individuals in NYC in 2017 Whose First Trip After Work is for Family and Personal Business

3.6.3 Non-Work Days Trip Chaining

In NYC, the differences between those with and without vehicles were generally small (Figure 3-23). Short stops were more common among vehicle owners in NYC, particularly in 2017, when 12% of vehicle owners reported short stops compared to 6% of zero-vehicle individuals. This difference is statistically significant, suggesting that vehicle ownership could be associated with slightly more short-stop activity in NYC.

In the rest of NYS, differences by vehicle ownership were more pronounced, especially for family and personal business trips. In both years, zero-vehicle individuals in the rest of NYS were statistically significantly more likely to make family/personal business trips than those with vehicles. In 2009, 71% of zero-vehicle individuals made such trips compared to 58% of vehicle owners. In 2017, this difference grew: 80% of zero-vehicle individuals compared to 59% of vehicle owners, a statistically significant gap. In 2017, vehicle owners reported 25% social and recreational trips, while zero-vehicle individuals reported 15%, a statistically significant gap.

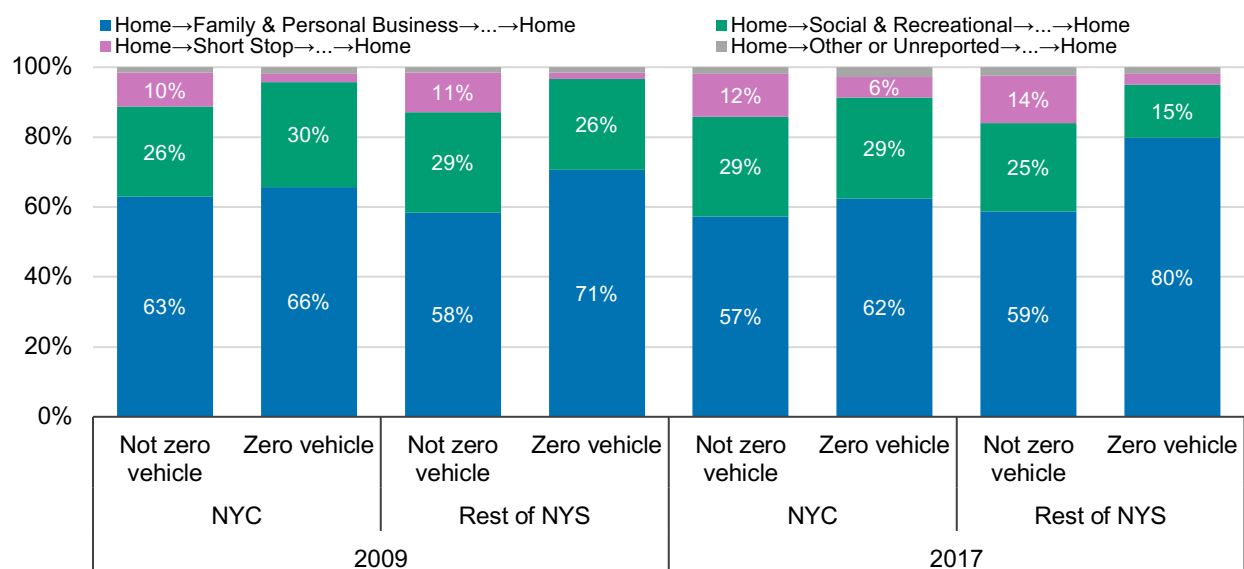


Figure 3-23. Activity Patterns on Non-Work Days by Vehicle Ownership in NYC and Rest of NYS

3.7 TRIP CHAINING BY ALICE

This section analyzed the trip chaining patterns by ALICE. ALICE is an acronym for Asset Limited, Income Constrained, Employed, and represents the growing number of families who are unable to afford the basics. According to NHTS data, in both NYC and the rest of NYS, ALICE individuals generally exhibited fewer daily activity types than non-ALICE individuals (Table 3-10). In 2009 NYC, ALICE individuals were statistically significantly more likely to make only one type of activity (48% vs. 34%) and less likely to report no trip (9% vs. 19%) or 3+ types of activities (8% vs. 12%) than non-ALICE. By 2017 NYC, the share of ALICE individuals with no trips rose to 22%, becoming statistically significantly higher than the 14% among non-ALICE. Meanwhile, ALICE individuals were less likely to make two types of activities (28% vs. 37%), indicating reduced daily travel engagement compared to non-ALICE. In the rest of NYS, ALICE individuals in the rest of NYS had a higher share with no trips (23% vs. 18%) and lower shares with two or more types of activities than non-ALICE.

Table 3-10. Share of Daily Number of Activity Types by ALICE in NYS and rest of the United States

NHTS year	Region	ALICE category	No trip	One type of activity	Two types of activities	3+ types of activities
2009	NYC	ALICE	9%	48%	35%	8%
		Non-ALICE	19%	34%	35%	12%
	Rest of NYS	ALICE	10%	46%	33%	11%
		Non-ALICE	14%	35%	36%	15%
2017	NYC	ALICE	22%	42%	28%	8%
		Non-ALICE	14%	38%	37%	12%
	Rest of NYS	ALICE	23%	42%	26%	9%
		Non-ALICE	18%	38%	31%	13%

3.7.1 Pre-Work Trip Chaining

ALICE and Non-ALICE workers in NYC had similar rates of going directly from home to work (79–80%) in 2009 (Figure 3-24). ALICE workers were statistically significantly less likely to stop for social or recreational purposes than non-ALICE workers (2% vs. 4%). By 2017, the share of workers going straight to work remained similar between ALICE and Non-ALICE groups (around 78–80%), with no statistically significant differences across stop types. In the rest of NYS, ALICE and non-ALICE workers also had similar pre-work patterns across both years. One difference in 2009 was that non-ALICE workers were statistically significantly more likely to make short stops before work (8% vs. 5%). However, this difference diminished by 2017.

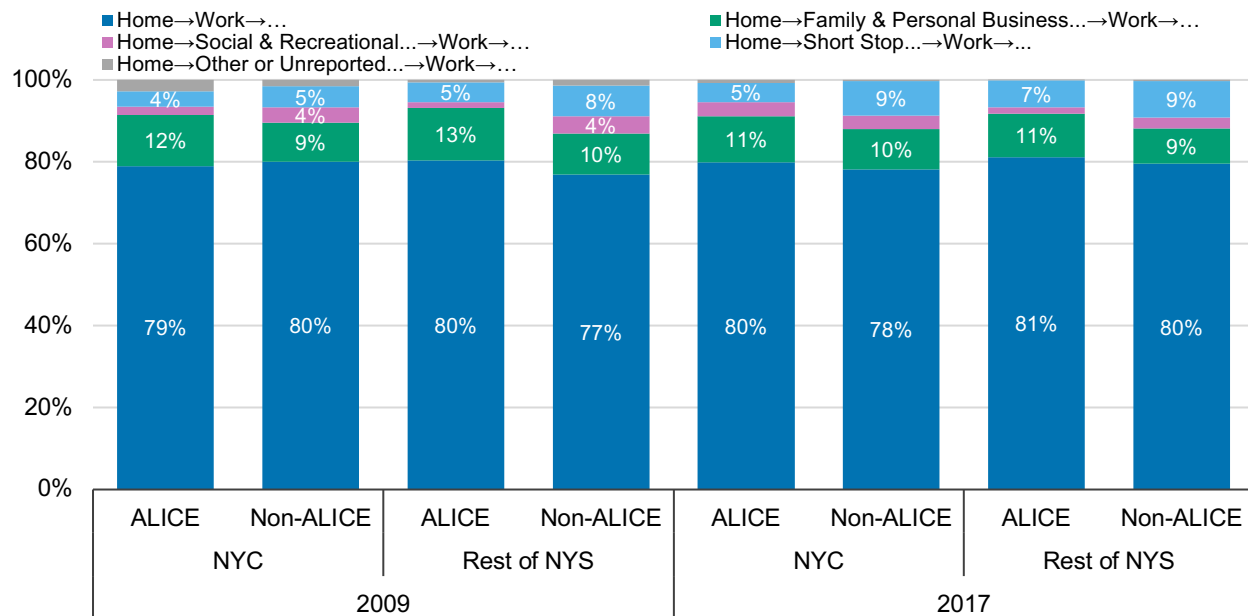


Figure 3-24. Pre-Work Activity Patterns by ALICE in NYC and Rest of NYS

3.7.2 Post-Work Trip Chaining

ALICE workers in NYC were statistically significantly more likely than Non-ALICE workers to return directly home after work (76% vs. 65%) and statistically significantly less likely to stop for family or personal business (17% vs. 26%) in 2009 (Figure 3-25). By 2017, the share of ALICE workers going

directly home declined to 64%, and the difference with non-ALICE workers (60%) was not statistically significant. ALICE workers in NYC also showed a statistically significant increase in stopping for short errands after work—from 2% in 2009 to 7% in 2017—a change not observed among non-ALICE workers. In the rest of NYS, both ALICE and non-ALICE workers maintained similar post-work activity patterns in both years. The share going directly home remained between 73–77%, and differences between groups were not statistically significant. These findings suggest post-work activity complexity increased among ALICE workers in NYC between 2009 and 2017.

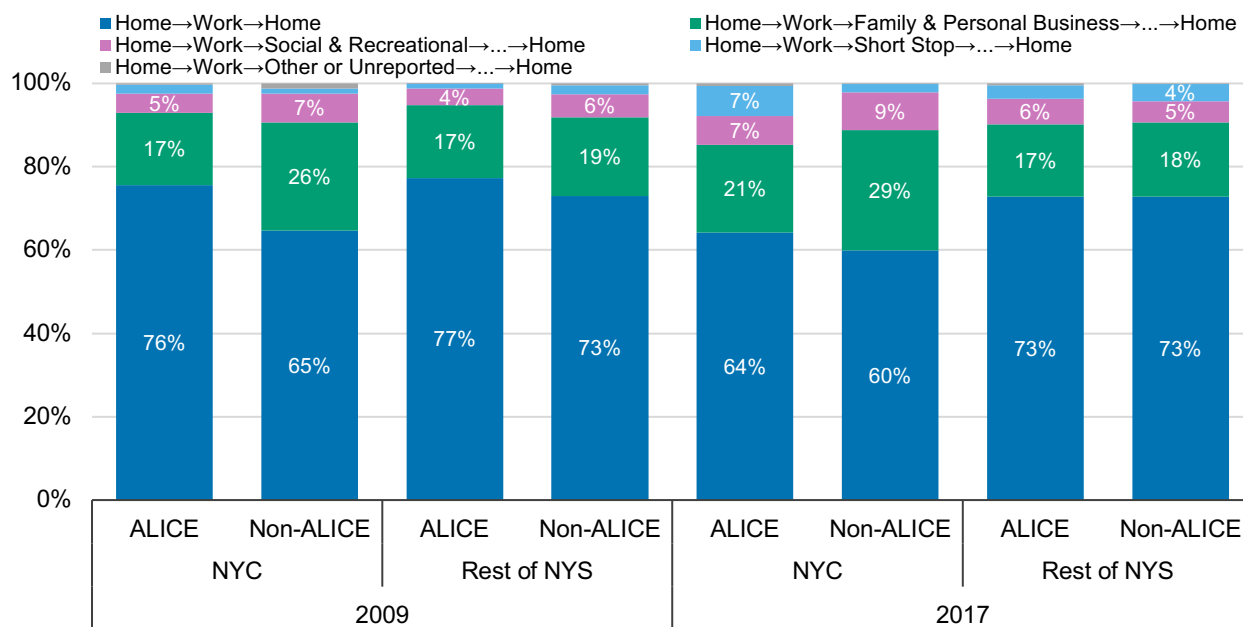


Figure 3-25. Post-Work Activity Patterns by ALICE in NYC and Rest of NYS: Home→Work Trip Chains

3.7.3 Non-Work Days Trip Chaining

On non-work days, ALICE individuals in both NYC and the rest of NYS consistently showed a higher share of family and personal business activity chains than non-ALICE individuals across 2009 and 2017 (Figure 3-26). In 2017 NYC, ALICE individuals were statistically significantly more likely than non-ALICE to engage in family and personal business trips (66% vs. 54%) and less likely to engage in social and recreational activities (25% vs. 33%). In the rest of NYS, similar differences were observed: in both years, ALICE individuals were statistically significantly more likely than non-ALICE to report family and personal business trips (e.g., 67% vs. 57% in 2017), and less likely to report social and recreational trips (20% vs. 27% in 2017). Over time, family and personal business trips remained stable for ALICE groups in both regions, while non-ALICE groups in NYC experienced a statistically significant decrease in such trips (from 59% to 54%) and an increase in social and recreational activities (from 30% to 33%).

3.7.4 Transportation Mode Transition

In 2009, both ALICE and non-ALICE individuals had similar shares transferring from public transport to nonmotorized modes (75%) in NYC (Figure 3-27). Their transitions to POV or other modes were also statistically similar. In 2009 rest of NYS, non-ALICE individuals were statistically significantly more likely to transfer from public transport to POV (35% vs. 24%) and less likely to continue with nonmotorized modes (58% vs. 69%) than ALICE individuals. By 2017 NYC, ALICE individuals were statistically significantly more likely to transfer to nonmotorized modes (85% vs. 80%) and less likely to use POV (9% vs. 8%, though not statistically significant). Non-ALICE individuals had a statistically

significantly higher share transferring to other or unreported modes (12% vs. 6%). In 2017 rest of NYS, non-ALICE individuals remained more likely than ALICE individuals to transfer to POV (33% vs. 23%), and less likely to continue with nonmotorized modes (63% vs. 65%), though the gaps narrowed slightly from 2009. The share using other or unreported modes was statistically significantly lower for non-ALICE (4%) compared to ALICE (12%).

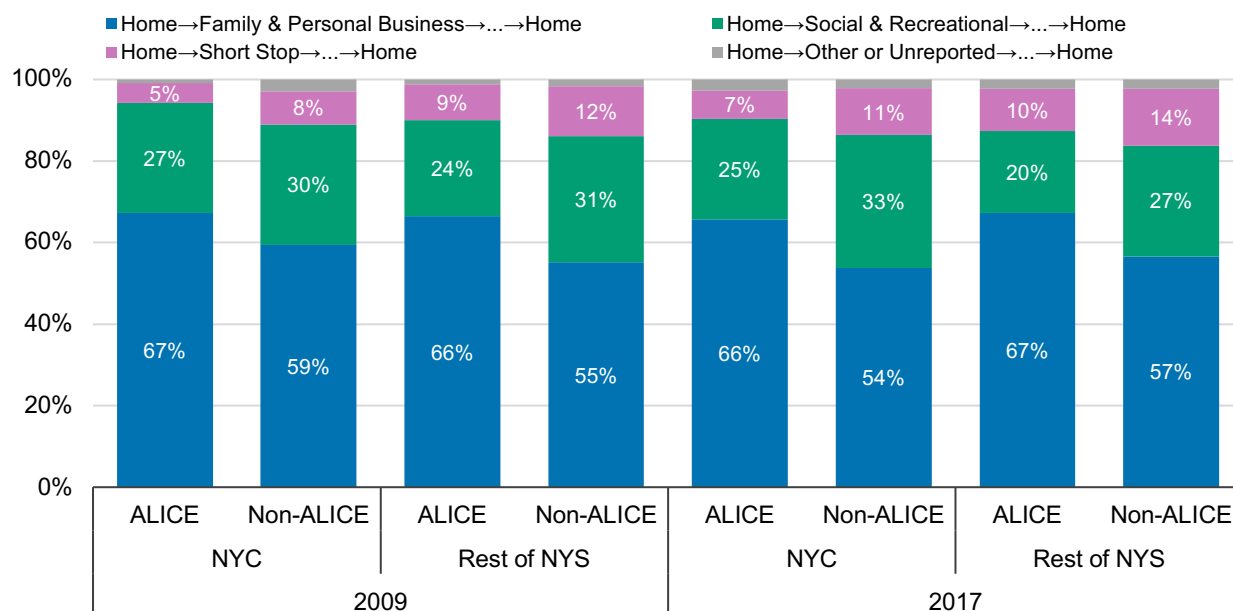


Figure 3-26. Activity Patterns on Non-Work Days by ALICE in NYC and Rest of NYS

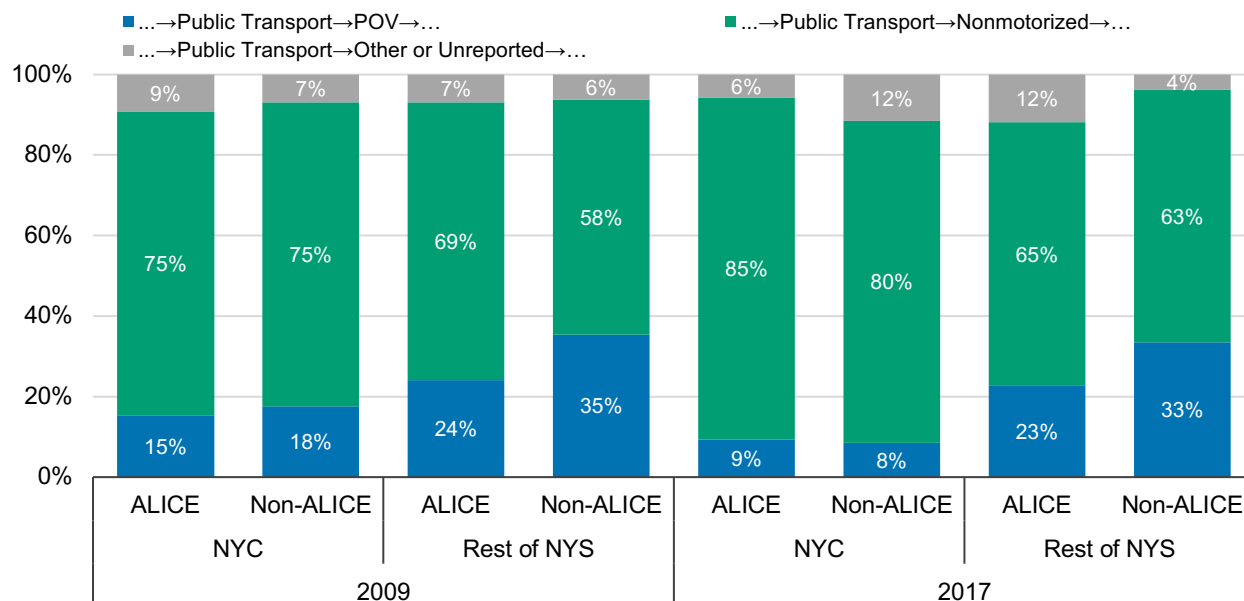


Figure 3-27. Transportation Mode Transition Following a Public Transport Trip by ALICE in NYC and Rest of NYS

4. TRIP CHAINING BEHAVIOR BY POPULATION DENSITY IN NYS

Table 4-1 provides population and population density statistics for NYS and NYC in 2010 and 2020. These figures highlight the stark difference in urban form between NYC and the rest of the state, providing important context for interpreting travel behavior patterns by population density in NYS.

Table 4-1. Population Density in NYS and NYC (US Census Bureau 2021)

Region	Year	Resident population	Population density (people per square mile)
NYS	2010	19,378,102	411.2
	2020	20,201,249	428.7
NYC	2010	8,175,133	27,012.4
	2020	8,804,190	29,303.2

We categorized population density of the entire NYS into four groups—urban [0, 2000) people/sq mile, urban [2000, 4000) people/sq mile, urban [4000, ∞) people/sq mile, and Rural—using two NHTS variables: *URBRUR* (urban/rural classification of the household) and *HBPPOPDN* (population density category of the household's census block group). This categorization balances the need for detailed urban differentiation with practical relevance for analyzing trip patterns across diverse geographic contexts.

- **[0, 2000) people/sq mile.** Represents low-density areas but still considered as urban where private vehicle usage is typically dominant, and public transit access is limited or requires longer first/last mile connections.
- **[2000, 4000) people/sq mile.** Captures medium-density urban areas that often have moderate public transit availability and a mix of travel modes, including walking, biking, and personal vehicles.
- **[4000, ∞) people/sq mile.** Reflects high-density urban areas where public transportation, walking, and micromobility are more viable due to proximity to services and denser transit networks.
- **Rural.** Encompasses areas outside designated urbanized zones, where travel patterns are distinct, typically characterized by car dependency, longer trip distances, and limited multimodal options.

4.1 TRIP CHAINING BY OPTION OF WFH

4.1.1 Pre-Work Trip Chaining

Workers who could WFH were generally less likely to go directly to work than those who could not, especially in high-density areas (4,000+ people/sq mi), where the gap was statistically significant (63% vs. 78%) in 2009 (Figure 4-1). WFH-capable workers in these dense areas were also statistically more likely to make short stops before work (14% vs. 7%) and to engage in social and recreational activities (9% vs. 3%). In mid-density areas (2,000–4,000), WFH workers had a higher short stop rate (14% vs. 5%) and a slightly lower direct-to-work rate, both statistically significant. In rural and low-density areas, the differences were smaller and generally not statistically significant.

By 2017, some of these patterns persisted, but the gaps narrowed in rural and mid-density areas. In low-density urban areas (<2,000), WFH workers were statistically less likely to go directly to work (67% vs. 79%) and more likely to make short stops (16% vs. 6%) or engage in social and recreational activities (7% vs. 3%). In mid-density areas, WFH workers again had a higher short stop share (13% vs. 7%) and a slightly lower direct-to-work share (75% vs. 79%), with the short stop difference being statistically significant. In high-density areas, WFH workers were still somewhat more likely to make short stops (10% vs. 7%), though this was not statistically significant in 2017.

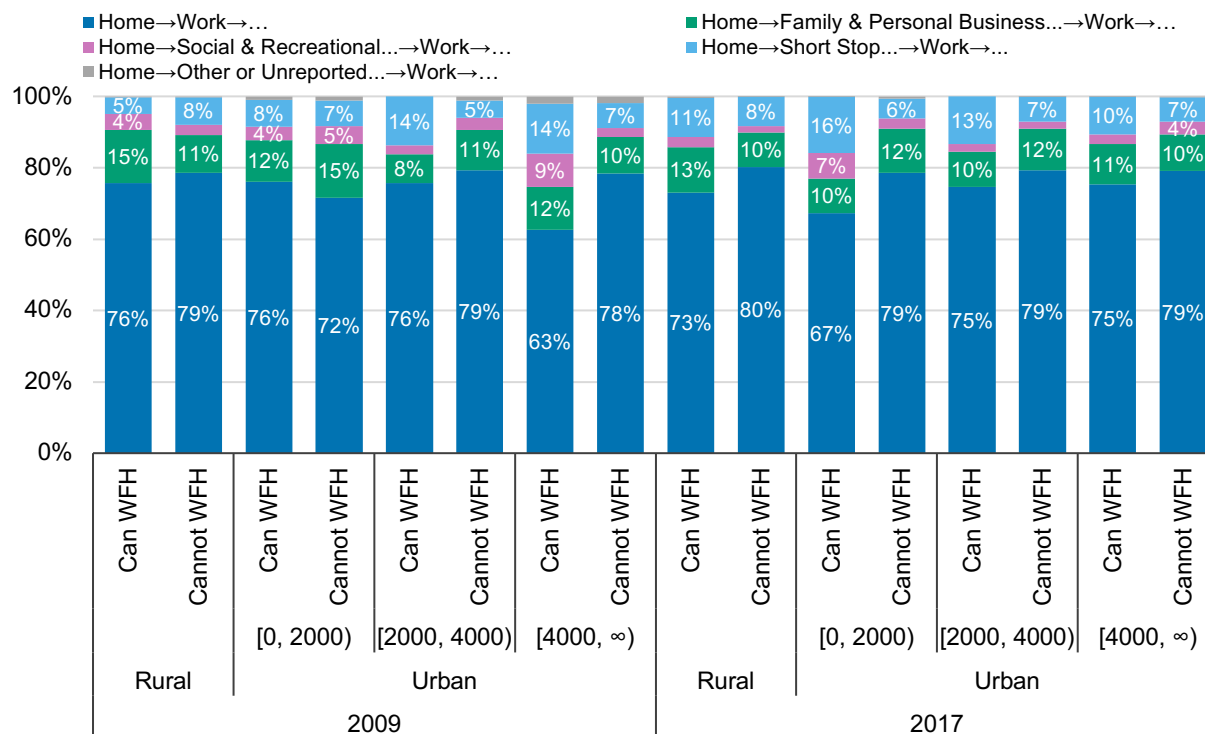


Figure 4-1. Pre-Work Activity Patterns by Option of WFH and Population Density

4.1.2 Post-Work Trip Chaining

Workers who could WFH were consistently less likely to go directly home after work than those who could not in 2009 (Figure 4-2). This difference was statistically significant in rural areas (51% vs. 69%), low-density urban areas (<2,000 people/sq mi; 64% vs. 73%), and high-density areas (4,000+; 57% vs. 70%). Those with WFH flexibility were also statistically more likely to stop for family and personal business in all density categories, with the largest gap in rural areas (45% vs. 23%). Social and recreational stops were also higher among WFH-capable workers in some areas, such as low-density urban areas (13% vs. 6%), a statistically significant difference.

By 2017, the differences narrowed but still appeared in some contexts. In rural areas, direct-to-home rates between WFH and non-WFH workers (65% vs. 68%) were not statistically significant, and the gap in family/personal business stops also diminished (20% vs. 22%). However, in mid-density urban areas (2,000–4,000 people/sq mi), WFH workers remained statistically less likely to go directly home (64% vs. 71%) and statistically more likely to make family/personal business stops (30% vs. 20%). High-density areas still showed lower direct-to-home shares for WFH workers (56% vs. 65%), but this difference was not statistically significant in 2017.

In both 2009 and 2017, high-density urban areas had the lowest direct-to-home shares and relatively higher social/recreational participation for both WFH and non-WFH workers. Conversely, rural areas had the highest direct-to-home shares for non-WFH workers, but in 2009, WFH workers in rural settings had the most pronounced diversion toward errands.

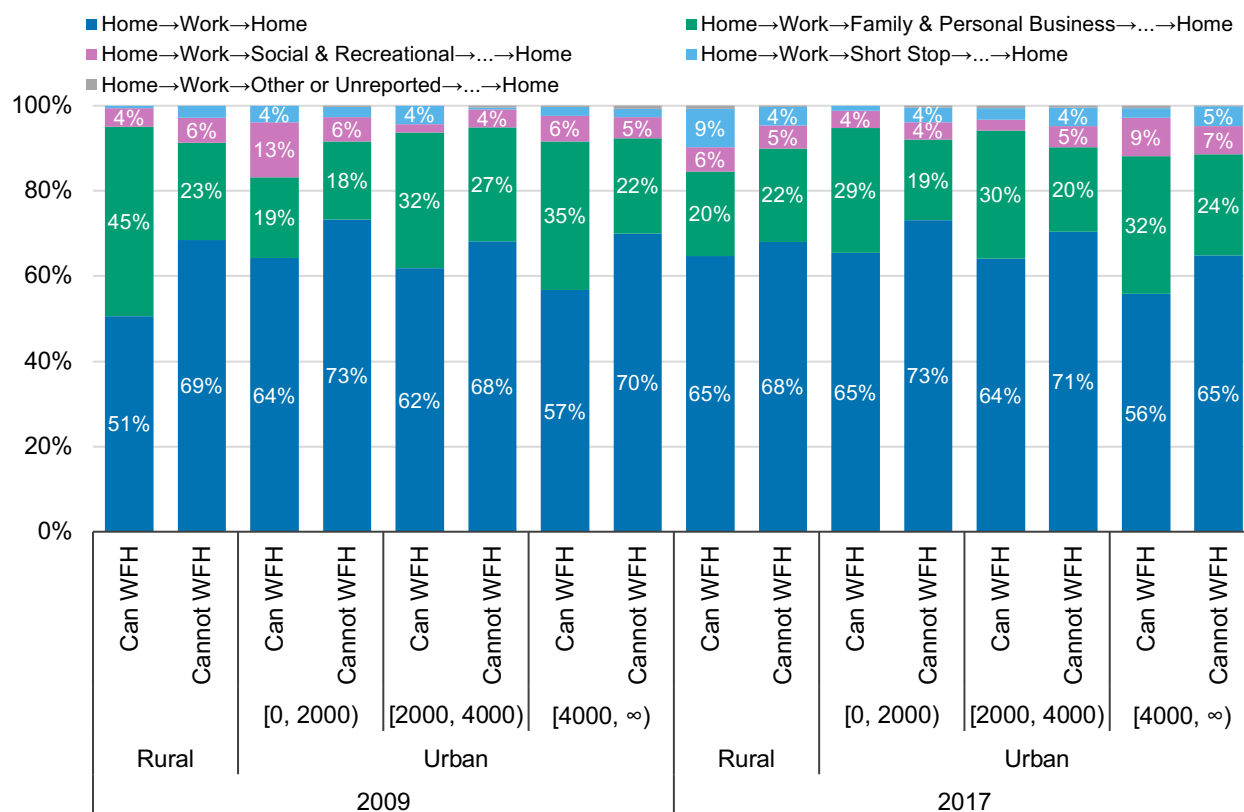


Figure 4-2. Post-Work Activity Patterns by Option of WFH and Population Density: Home→Work Trip Chains

4.1.3 Transportation Mode Transition

In 2009, rural WFH workers were statistically less likely than non-WFH workers to transition to non-motorized modes (60% vs. 43%) and more likely to continue by POV (40% vs. 47%), though the POV difference was not statistically significant (Figure 4-3). In low-density urban areas (<2,000 people/sq mi), the patterns diverged sharply: WFH workers were statistically more likely to switch to POV (68% vs. 33%) and statistically less likely to walk/bike (30% vs. 65%). This suggests that in less dense settings, WFH-capable workers using transit were more dependent on cars for onward travel.

In mid-density areas (2,000–4,000 people/sq mi) in 2009, the reverse occurred; WFH workers had a much lower POV share (9% vs. 29%) and statistically significantly higher nonmotorized share (78% vs. 67%). In high-density areas (4,000+ people/sq mi), nonmotorized travel dominated for both groups (82% WFH vs. 77% non-WFH), with no statistically significant difference.

By 2017, many of these differences shifted. In rural areas, WFH workers were now statistically more likely to use POV after transit (39% vs. 21%) and less likely to walk/bike (58% vs. 60%), reversing the 2009 pattern. In low-density urban areas, differences in POV and nonmotorized shares narrowed and were not statistically significant.

However, in mid-density areas, a dramatic change occurred. In 2009, WFH workers overwhelmingly walked or biked after transit, but in 2017, non-WFH workers had a much higher POV share (60% vs. 2%), a statistically significant shift. In high-density areas, nonmotorized travel remained dominant for both groups, but WFH workers had a slightly higher share (80% vs. 77%) with no significant gap.

In summary, low-density urban areas consistently saw higher POV use, especially among WFH workers in 2009. Mid-density areas had volatile differences between years, flipping the WFH/non-WFH relationship. High-density areas maintained high nonmotorized continuation rates for all workers, regardless of WFH status.

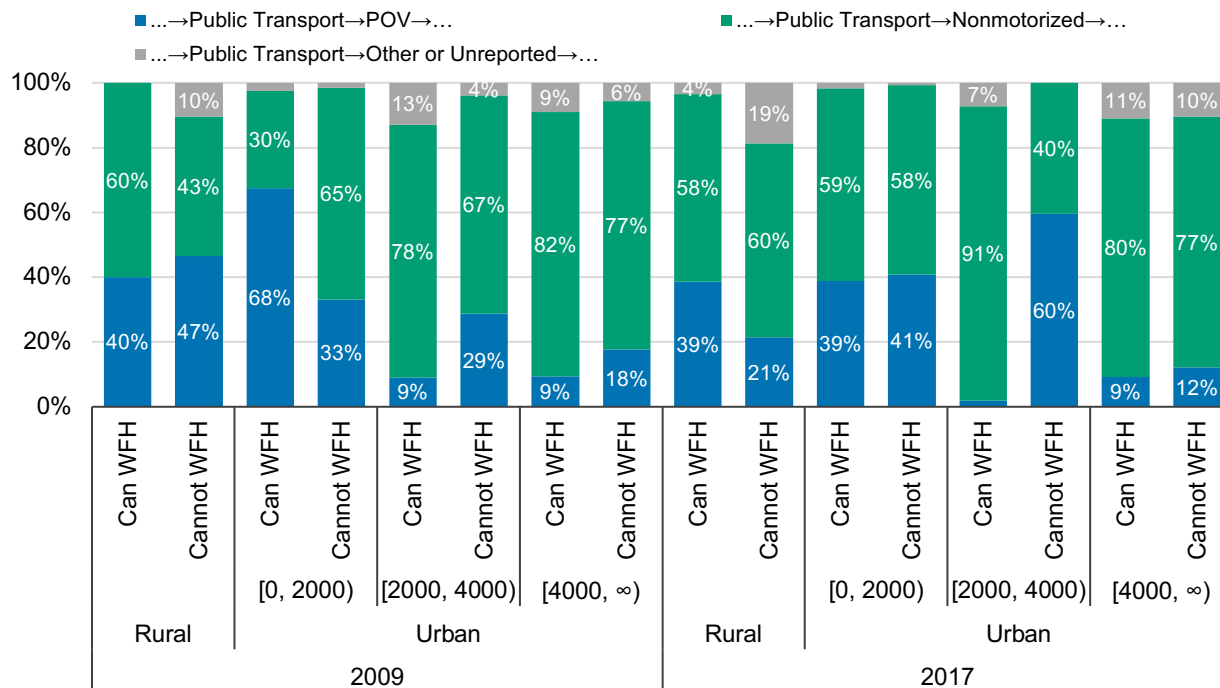


Figure 4-3. Transportation Mode Transition Following a Public Transport Trip by Option of WFH and Population Density

4.2 TRIP CHAINING BY SEX

4.2.1 Pre-Work Trip Chaining

In both 2009 and 2017, the share of workers going directly from home to work was high across all groups, generally between 75% and 83% (Figure 4-4). Differences between men and women within the same density category were mostly not statistically significant. However, in 2017 mid-density urban areas (2,000–4,000 people per square mile), men were statistically more likely to go directly to work than women (83% vs. 78%). In other density categories, the male/female gap was smaller and not statistically significant.

Family and personal business activities before work were consistently uncommon, but in several cases, women had slightly higher rates than men, particularly in lower-density areas. For example, in 2009 low-density urban areas (<2,000 people/sq mi), women reported such stops at 14% compared to 12% for men, but the difference was not statistically significant. By 2017, these shares had converged even further, with no statistically significant gaps across any density group. Short stops (e.g., quick errands) showed more variation by density than by sex. In 2017 low- and mid-density urban areas, women had higher rates than men (10% vs. 6% in low-density and 10% vs. 6% in mid-density) but only the low-density difference was statistically significant. In rural areas, women also reported more short stops than men in both years, but these differences were not statistically significant.

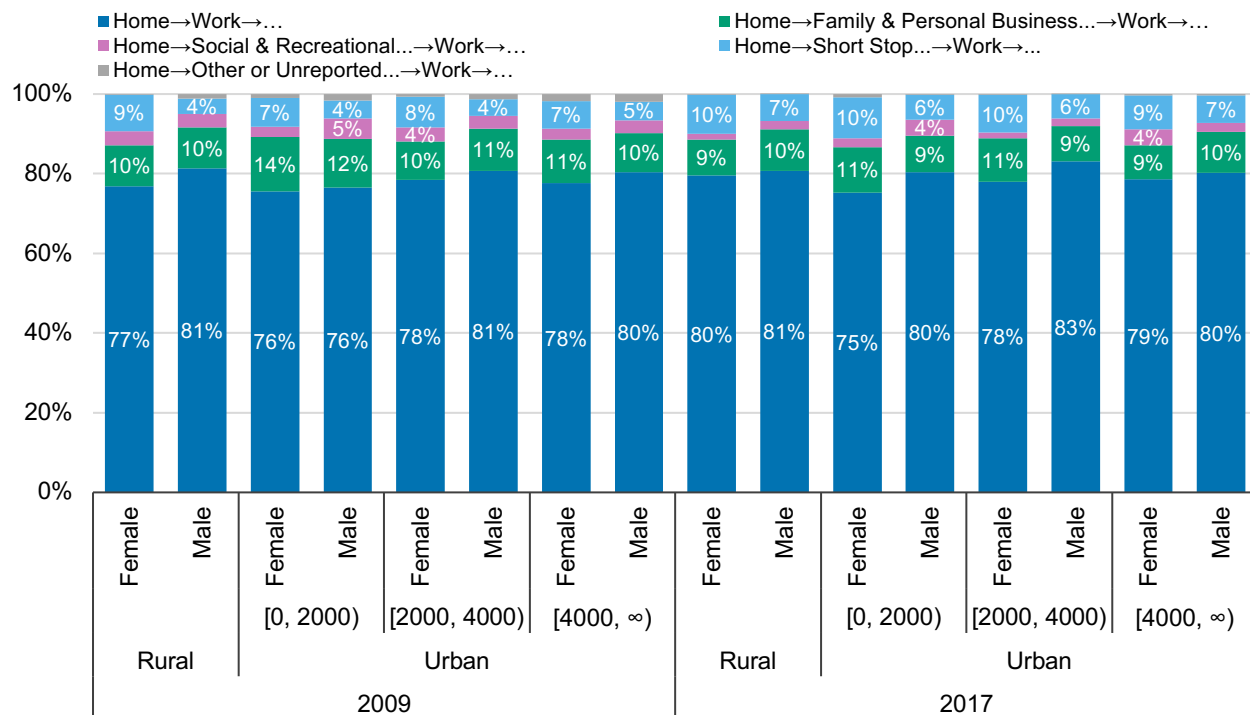


Figure 4-4. Pre-Work Activity Patterns by Sex and Population Density

4.2.2 Post-Work Trip Chaining

In both 2009 and 2017, women and men showed similar likelihoods of returning directly home after work (Figure 4-5). Differences between sexes were mostly not statistically significant within the same population density group. For example, in 2009 across all regions, men and women had nearly identical rates of direct Home→Work→Home patterns; about 71–77%, with no statistically significant gaps. The variation by population density became more pronounced in 2017, particularly among women. In high-density urban areas (4,000+ people per square mile), only 59% of women returned directly home, significantly lower than the 75% in mid-density areas and 70% in rural areas. This decline is statistically significant, suggesting that women in denser urban environments were more likely to engage in post-work activities than their counterparts elsewhere. For men, the difference was also present (70% in high-density vs. 74–77% elsewhere), but not statistically significant.

The largest difference by sex emerged in family and personal business activities. In high-density urban areas in 2017, 29% of women stopped for family/personal business after work, compared to 18% of men, a statistically significant difference. This gap suggests women may shoulder a greater share of household-related errands in dense urban contexts. In contrast, in rural areas and less dense urban settings, the male/female gap in this activity category was not statistically significant.

4.2.3 Transportation Mode Transition

Across both years, the most common continuation mode after a POV trip was nonmotorized travel (e.g., walking or biking) for both men and women, though women generally had a higher share (Figure 4-6). In 2009 rural areas, 81% of women transferred to nonmotorized modes compared to 64% of men, a statistically significant difference. This pattern of women having a statistically higher share of nonmotorized travel held in most urban density groups in both years, although the gap was narrower in higher-density areas.

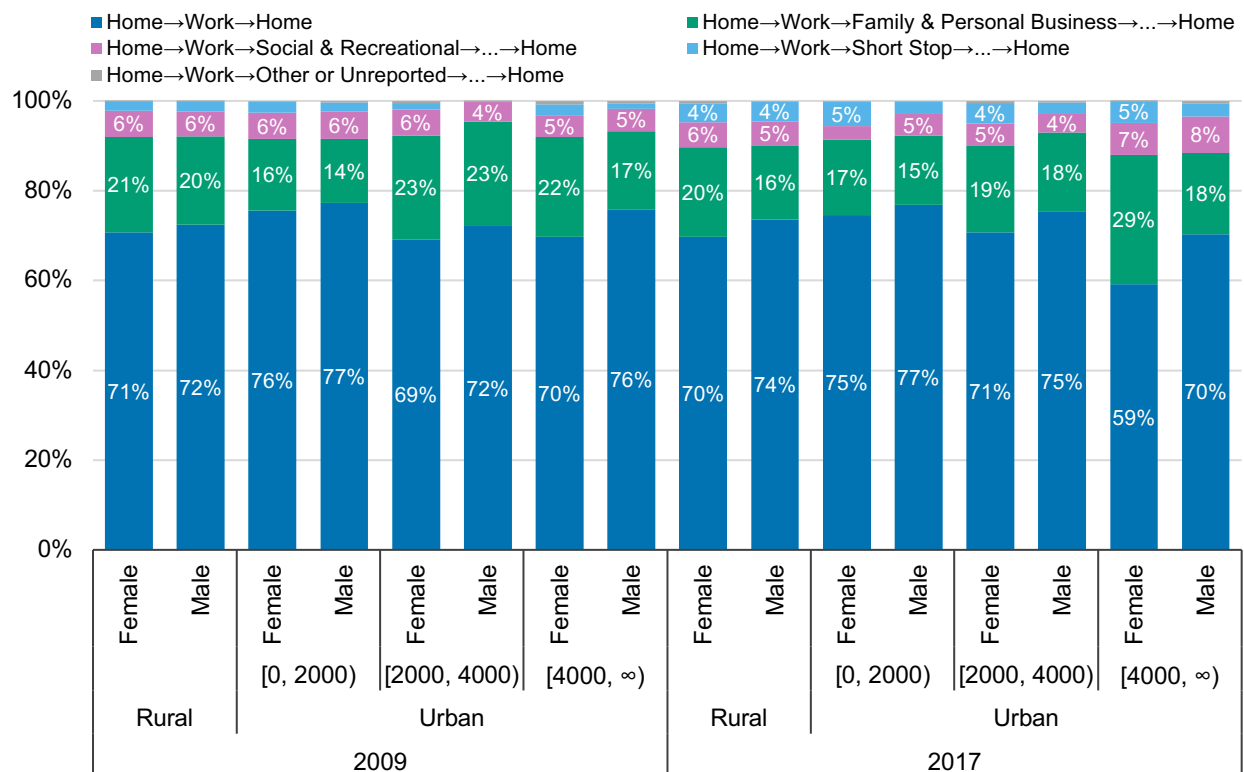


Figure 4-5. Post-Work Activity Patterns by Sex and Population Density: Home→Work Trip Chains

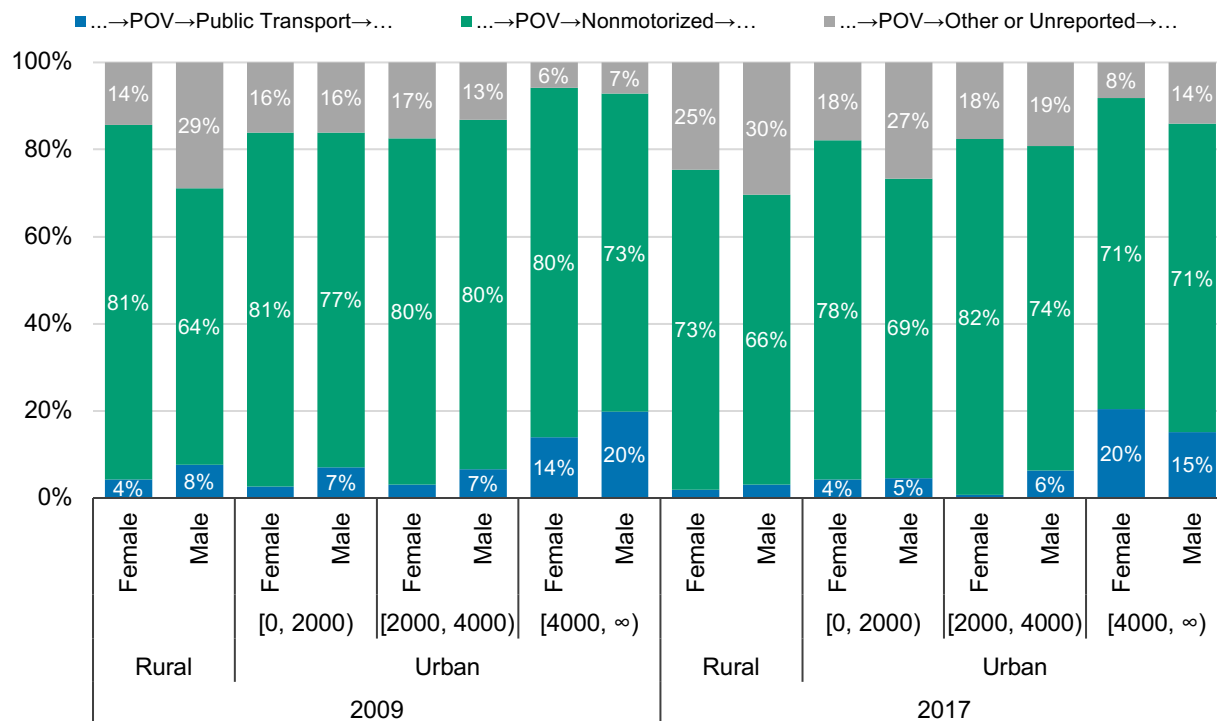


Figure 4-6. Transportation Mode Transition Following a POV Trip by Sex and Population Density

Transfers from POV to public transport were generally more common in higher-density urban areas and more frequent among men. For instance, in 2009 high-density urban areas (4,000+ people/sq mi), 20% of men used public transport after a POV trip compared to 14% of women, a statistically significant difference. By 2017, this pattern had reversed in high-density areas, with 20% of women making the shift compared to 15% of men, though the difference was not statistically significant. The largest jump for women occurred in high-density areas from 2009 to 2017 (14% to 20%).

Comparing across population densities, POV-to-public transport transfers were expectedly higher in high-density areas, while POV-to-nonmotorized shares were generally high across all densities for women but declined with increasing density for men. For example, in 2009 men's nonmotorized share dropped from 64% in rural areas to 73% in high-density areas, while women's share stayed around 80% regardless of density.

4.3 TRIP CHAINING BY LIFECYCLE

4.3.1 Pre-Work Trip Chaining

Across all regions and both years, the majority of workers followed a direct Home→Work pattern. This share was generally higher for workers with children than for those without children, though the differences were mostly not statistically significant (Figure 4-7). For instance, in 2017, in mid-density urban areas (2,000–4,000 people per square mile), 82% of workers with children went directly to work, compared to 79% without children, not a statistically significant difference. Short stops (likely for errands) were relatively more common among workers with children, especially in 2017. In high-density areas (4,000+ people/sq mi), 13% of workers with children made short stops before work, statistically significantly higher than the 2% among those without children. This may suggest time-sensitive errands related to childcare or school drop-offs.

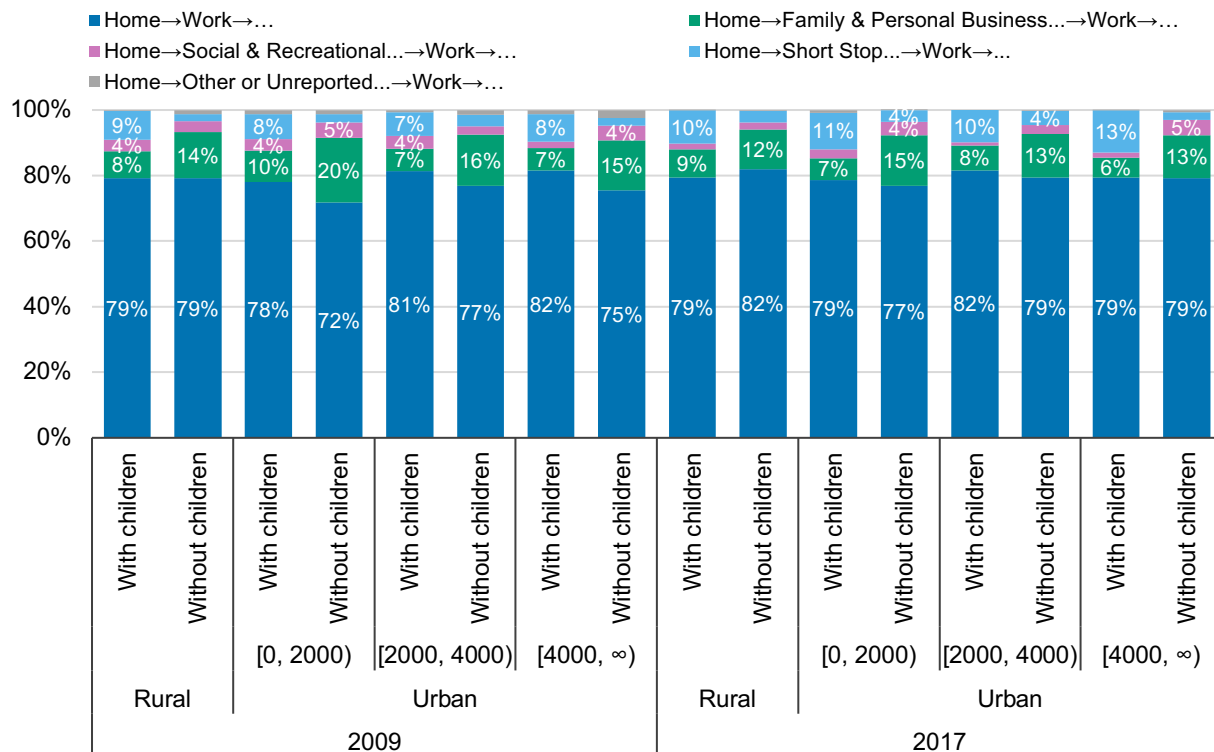


Figure 4-7. Pre-Work Activity Patterns by Lifecycle and Population Density

4.3.2 Post-Work Trip Chaining

Across all groups, the most common behavior is going directly home after work. In both 2009 and 2017, individuals with children were statistically more likely to follow the Home→Work→Home pattern than those without children (Figure 4-8). This was especially evident in rural and mid- to high-density urban areas. For example, in 2009, 75% of rural workers with children went directly home compared to 65% without children; a statistically significant difference that remained consistent in 2017. In contrast, workers without children were more likely to stop for family and personal business activities after work across nearly all regions and both years. These differences were statistically significant in most cases, particularly in urban areas with population densities between 2000 and 4000 people per square mile. There was a slight decrease in the share of workers going directly home in high-density urban areas between 2009 and 2017 for both groups, but these changes were not statistically significant.

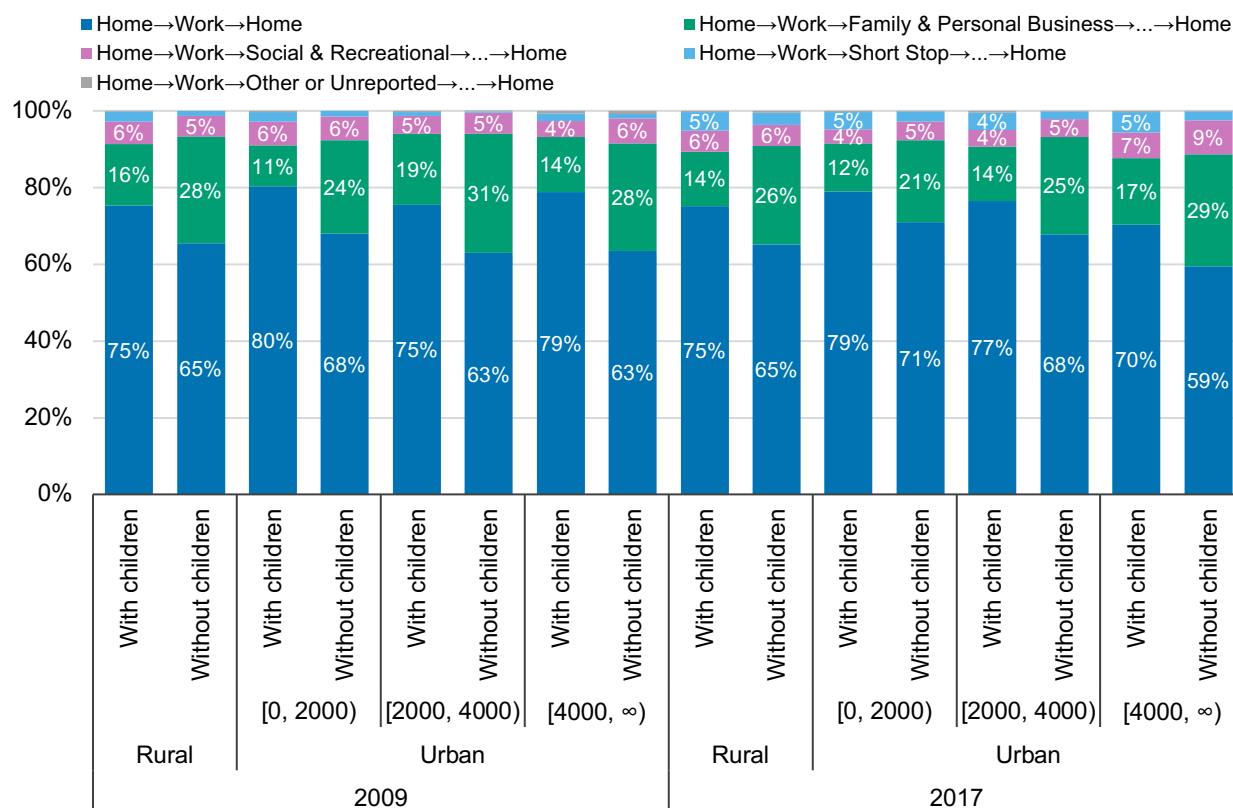


Figure 4-8. Post-Work Activity Patterns by Lifecycle and Population Density: Home→Work Trip Chains

4.3.3 Transportation Mode Transition

In 2009, individuals without children were more likely to switch from public transport to private vehicles, especially in rural and low-density urban areas (Figure 4-9). For example, in rural areas, 69% of individuals without children transitioned to a POV, statistically significantly higher than the 53% among those with children. Meanwhile, transfers to nonmotorized modes were generally more common among those without children, particularly in urban areas. In high-density urban areas (4,000+ people per square mile), 84% of childless individuals used nonmotorized modes after public transport in 2009, compared to 67% of those with children, statistically significantly higher. This suggests greater walkability and reliance on active modes among younger or single populations in denser neighborhoods.

By 2017, a strong shift toward nonmotorized modes occurred across nearly all groups, especially for those without children. In mid-density urban areas, 75% of childless individuals used nonmotorized modes after transit, significantly higher than the 43% in 2009. In high-density areas, both groups saw increases, with 83% of with-child and 79% of childless individuals continuing their trips on foot or by bike; differences that are not statistically significant but reflect a general trend. In contrast, POV use declined notably between 2009 and 2017 for most groups. For example, among with-child individuals in rural areas, the share dropping from 53% to 21% is statistically significant.

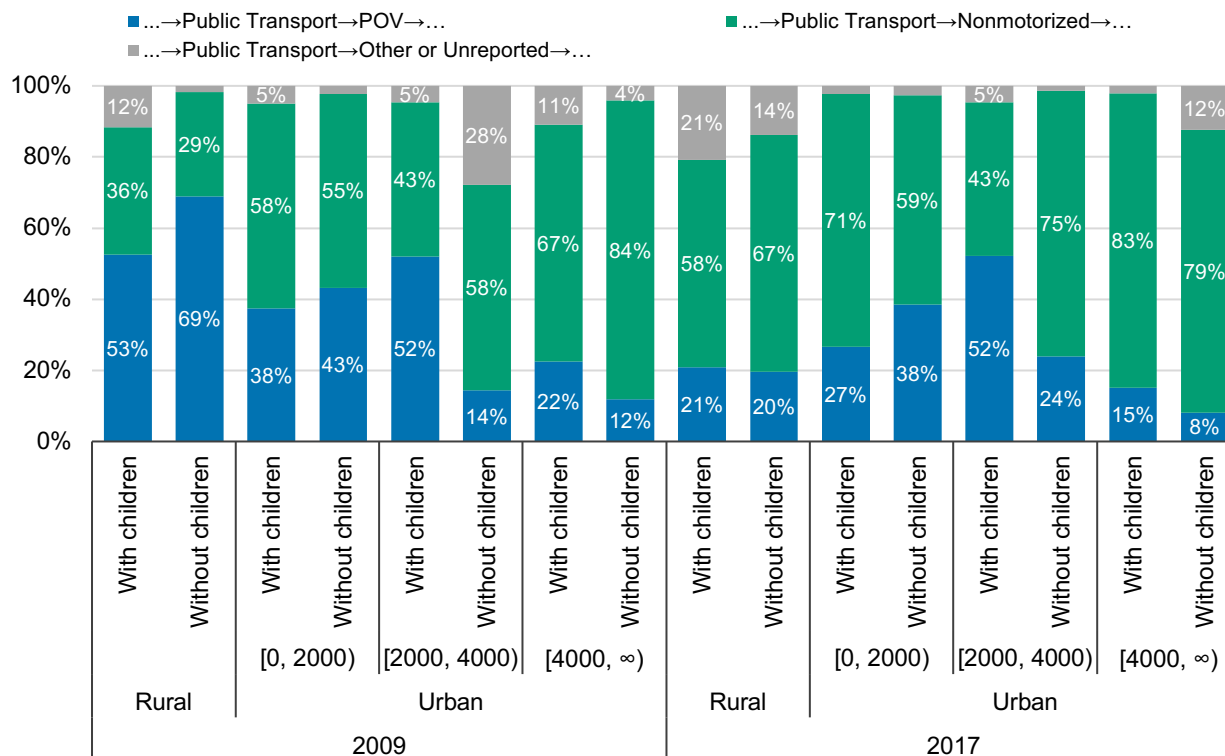


Figure 4-9. Transportation Mode Transition Following a Public Transport Trip by Lifecycle and Population Density

4.4 TRIP CHAINING BY ALICE

4.4.1 Pre-Work Trip Chaining

In both 2009 and 2017, most workers went directly from home to work, with rates typically above 75% for all groups (Figure 4-10). Differences between ALICE and non-ALICE were not statistically significant in most density categories. An exception appears in 2009 low-density urban areas (<2,000 people/sq mi), where ALICE workers had a lower direct-to-work share (75%) than non-ALICE (76%), though the gap was small and not statistically significant. By 2017, direct-to-work rates had converged further, with no statistically significant gaps in any density group. In both years, rural and low-density urban areas had slightly higher short stop rates and slightly lower direct-to-work shares than mid- and high-density areas. Family and personal business stops were somewhat more common in rural areas for ALICE workers in 2009 but evened out by 2017.

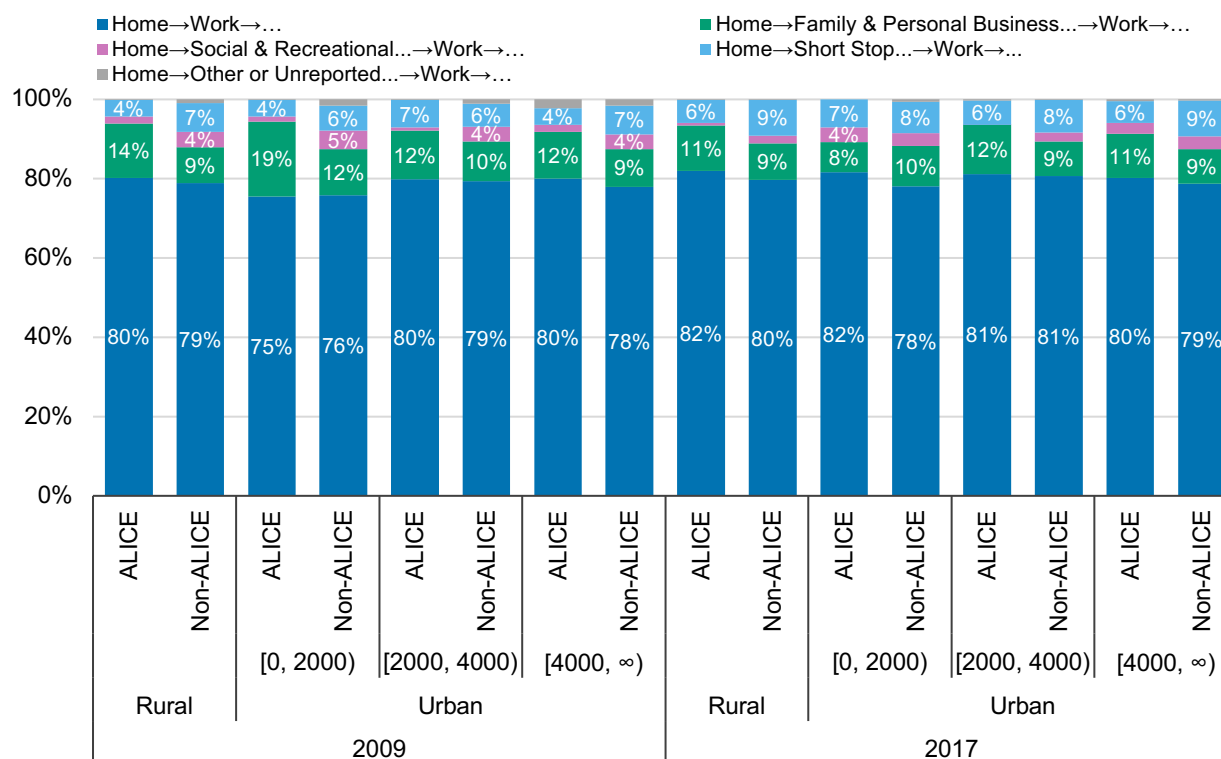


Figure 4-10. Pre-Work Activity Patterns by ALICE and Population Density

4.4.2 Post-Work Trip Chaining

In both 2009 and 2017, the majority of workers went directly home after work, and in most density groups the differences between ALICE and non-ALICE were not statistically significant (Figure 4-11). An exception appears in 2009 high-density urban areas (4,000+ people/sq mi), where 77% of ALICE workers went straight home compared to 68% of non-ALICE workers, a statistically significant gap. In 2017 mid-density urban areas (2,000–4,000 people/sq mi), the pattern reversed: 76% of non-ALICE workers went directly home compared with 68% of ALICE workers, also statistically significant. Family and personal business activities after work were more common among non-ALICE workers in high-density areas. In 2017 high-density urban areas, 26% of non-ALICE workers made such stops, statistically significantly higher than the 19% among ALICE workers. In contrast, in mid-density urban areas, ALICE workers had a higher rate (22%) than non-ALICE (17%), but this difference was not statistically significant. Rural and low-density areas showed minimal differences between groups.

Comparing across population densities, a consistent pattern emerges. In rural areas, both ALICE and non-ALICE workers were more likely to go straight home and less likely to engage in other activities. In high-density urban areas, non-ALICE workers were more likely to engage in family/personal business and less likely to go straight home. In mid-density areas, ALICE workers tended to have lower direct-to-home shares and higher short stop rates than non-ALICE, suggesting greater integration of errands into their commute.

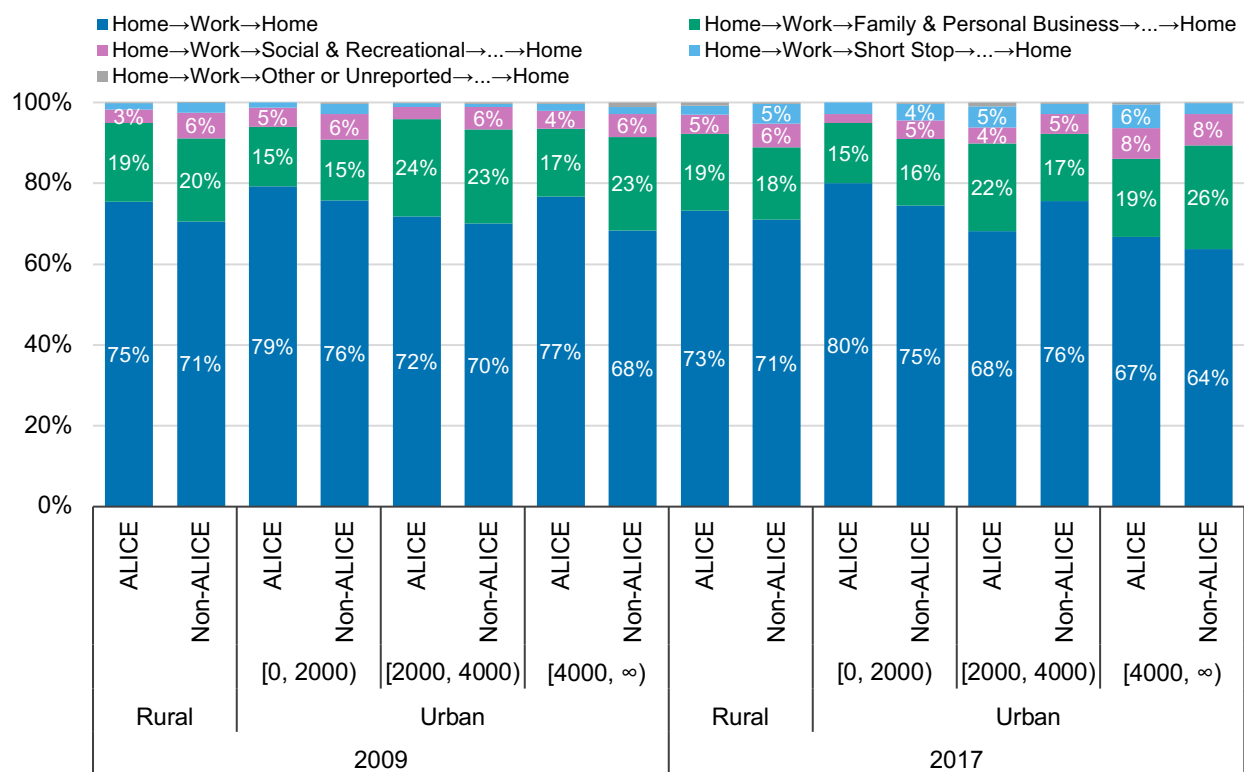


Figure 4-11. Post-Work Activity Patterns by ALICE and Population Density: Home→Work Trip Chains

4.4.3 Transportation Mode Transition

Figure 4-12 shows that in 2009 rural areas, non-ALICE workers were statistically more likely to use a POV after transit (65% vs. 55% for ALICE) and less likely to walk/bike (30% vs. 44%). This gap suggests that rural ALICE workers were more reliant on nonmotorized travel once off transit, while non-ALICE workers had greater car access. In low-density urban areas (<2,000 people/sq mi), differences were minimal and not statistically significant, with both groups showing a majority nonmotorized continuation.

A striking 2009 gap appears in mid-density areas (2,000–4,000 people/sq mi): ALICE workers were statistically less likely to use a POV (15% vs. 35%) and more likely to use “other or unreported” modes (49% vs. 4%). This unusual pattern may reflect unique mobility constraints or a concentration in specific communities with less typical mode mixes. In high-density urban areas (4,000+ people/sq mi) in 2009, both groups had similar and very high nonmotorized shares (~75%), with no statistically significant difference.

By 2017, rural differences largely disappeared; POV and non-motorized shares were almost identical between ALICE and non-ALICE. However, in low-density urban areas, ALICE workers became statistically more likely to use POV after transit (53% vs. 28%) and less likely to walk/bike (45% vs. 69%), reversing the 2009 pattern. In mid-density areas, both groups were predominantly nonmotorized (~61–64%) with no significant gap. High-density areas again showed no meaningful ALICE/non-ALICE difference, with nonmotorized continuation dominant (79–83%).

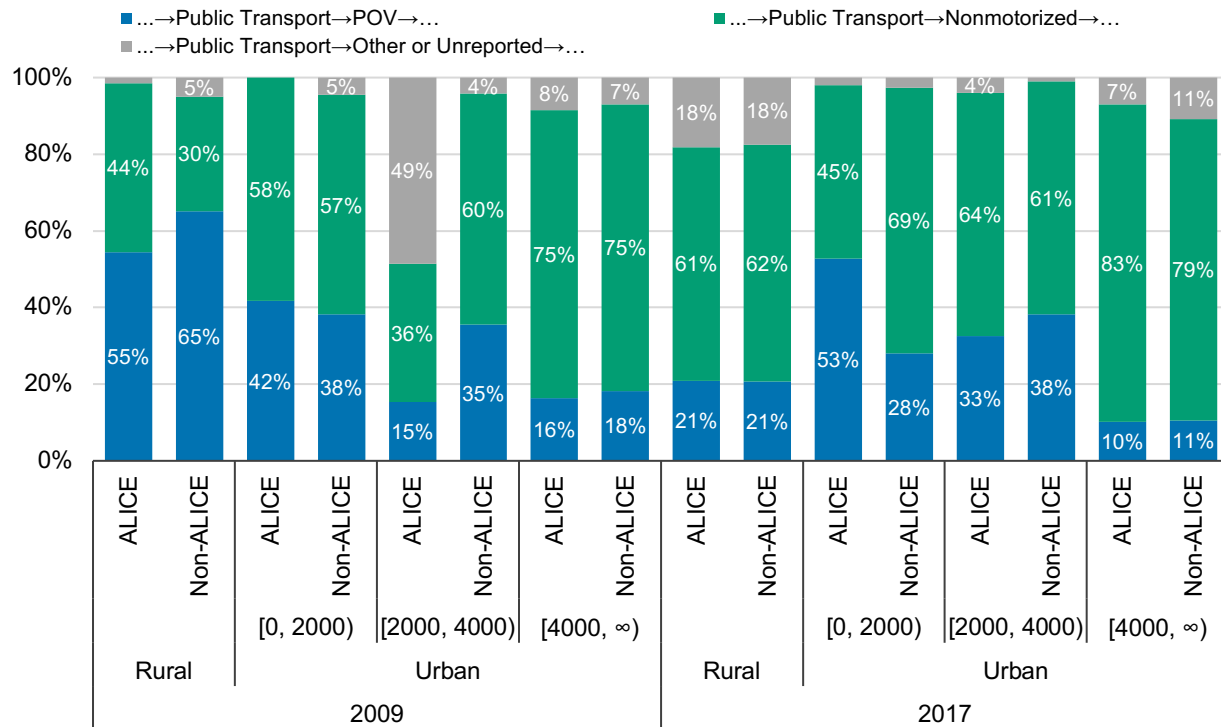


Figure 4-12. Transportation Mode Transition Following a Public Transport Trip by ALICE and Population Density

5. TRIP CHAINING BEHAVIOR BETWEEN MIDDLE ATLANTIC CENSUS DIVISION AND THE REST OF THE UNITED STATES

This chapter focuses on trip chaining behavior using the 2022 NHTS data. Compared with 2009 and 2017, a smaller share of respondents in 2022 reported work activities on their survey day (Table 5-1), only 35.3%, whereas over 44% of respondents in the earlier two NHTS years reported making at least one work trip. This lower share may also reflect a different composition of activities or trip purposes in 2022 compared with 2009 and 2017.

Table 5-1. Summary of Individual's Work Activities

NHTS year	Share of respondents that had at least one work trip on the survey day
2009	46.6%
2017	44.5%
2022	35.3%

5.1 TRIP CHAINING BY SEX

5.1.1 Pre-Work Trip Chaining

In 2009, women in the Middle Atlantic reported 78% direct Home→Work trips compared to 82% among men, a statistically significant gap (Figure 5-1). By 2017, the Middle Atlantic showed a narrowing trend. Women reported 79% direct trips and men 78%, no longer statistically different. Both sexes showed about 8% making short stops. In 2022, in the Middle Atlantic, men had 94% direct Home→Work trips compared to 89% for women. Both values are statistically significantly higher than in 2017 and 2009, but men's share is also statistically higher than women's in 2022. Family and personal business trips nearly disappeared, dropping to 2–3% from the 9–10% reported in earlier years.

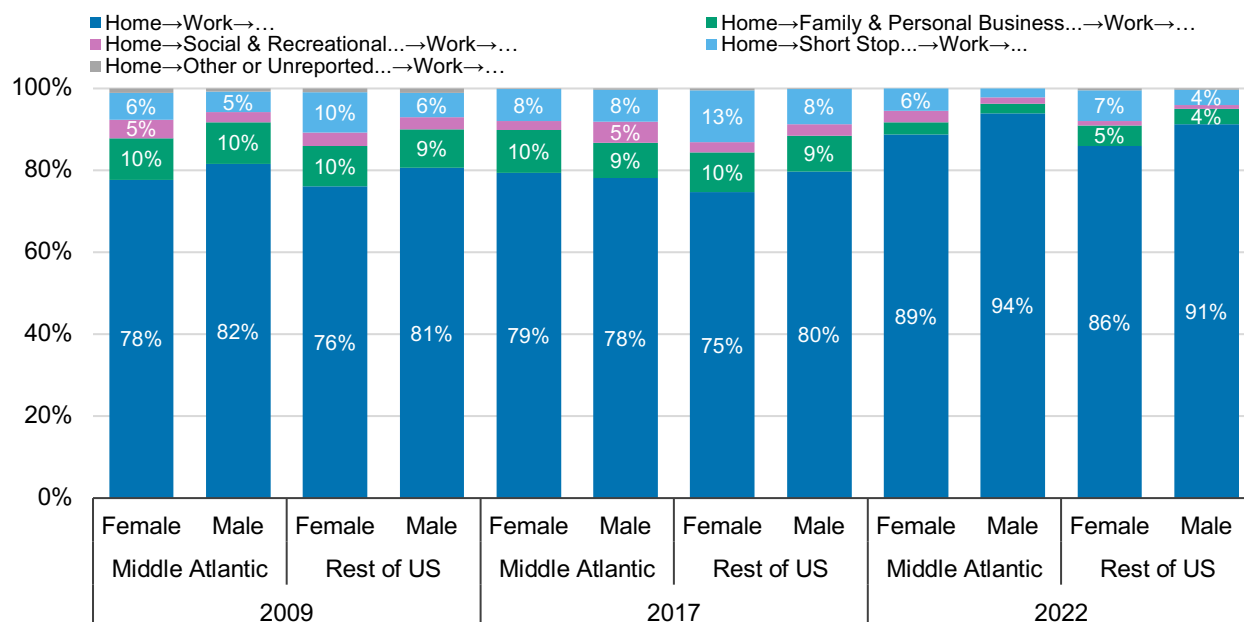


Figure 5-1. Pre-Work Activity Patterns by Sex in the Middle Atlantic Census Division and Rest of the United States

5.1.2 Post-Work Trip Chaining

The 2022 data reveals the strongest departure from earlier years (Figure 5-2). In the Middle Atlantic, 93% of women returned directly home after work, compared with 88% of men. Both shares are statistically significantly higher than in 2017 and 2009. Family and personal business trips nearly disappeared, dropping to just 4% among women and 6% among men, both statistically significantly lower than the 20–24% shares in 2017. Social and recreational trips also fell to 2–4%. In the rest of the United States, the pattern is similar: 86% of women and 89% of men went directly home in 2022, statistically significantly higher than in 2017, while family and personal business trips fell to 7–8% from earlier shares above 18%.

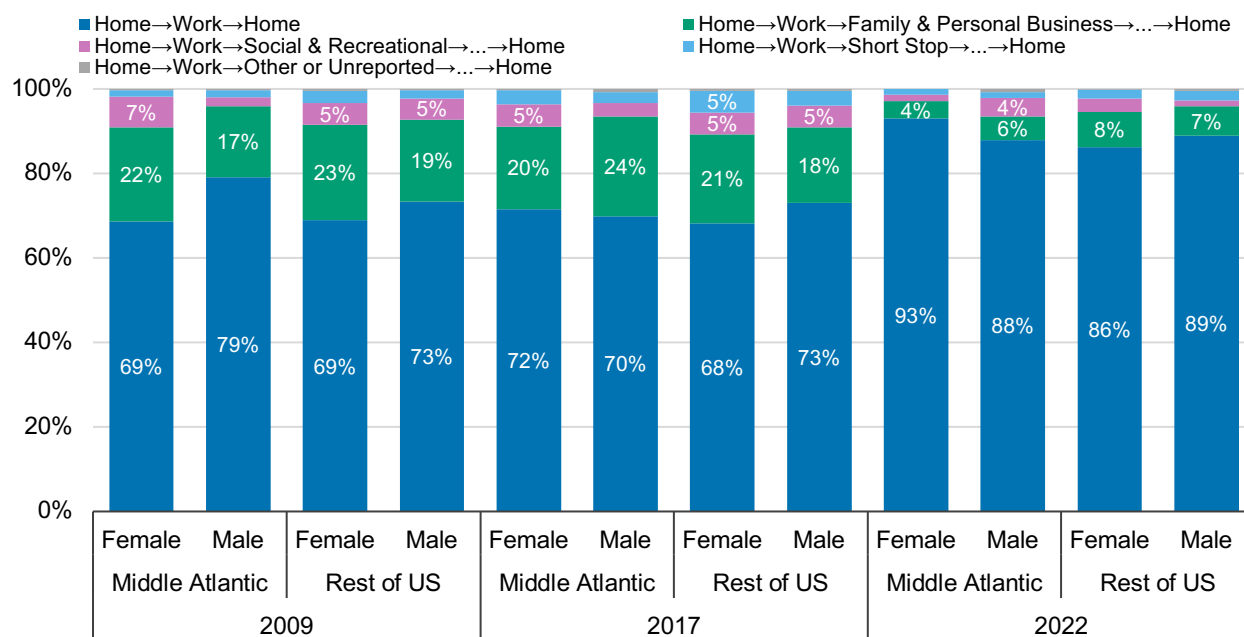


Figure 5-2. Post-Work Activity Patterns by Sex in the Middle Atlantic Census Division and Rest of the United States: Home→Work Trip Chains

5.1.3 Transportation Mode Transition

In 2009, both men and women in the Middle Atlantic were most likely to transition from POV to nonmotorized modes (80–81%). By 2017, women in the Middle Atlantic reported a slightly higher share of nonmotorized transitions, 87%, compared with 80% in 2009, though the difference is not statistically significant. Men’s nonmotorized share slightly fell to 72% from 81% in 2009. In the Middle Atlantic Census Division in 2022, women’s nonmotorized transitions decreased slightly to 78%. Men, however, reported only 54% nonmotorized, statistically significantly lower than their own earlier levels. Men also showed a statistically significant increase in public transport transitions, reaching 13% compared with their earlier shares of 2–3%

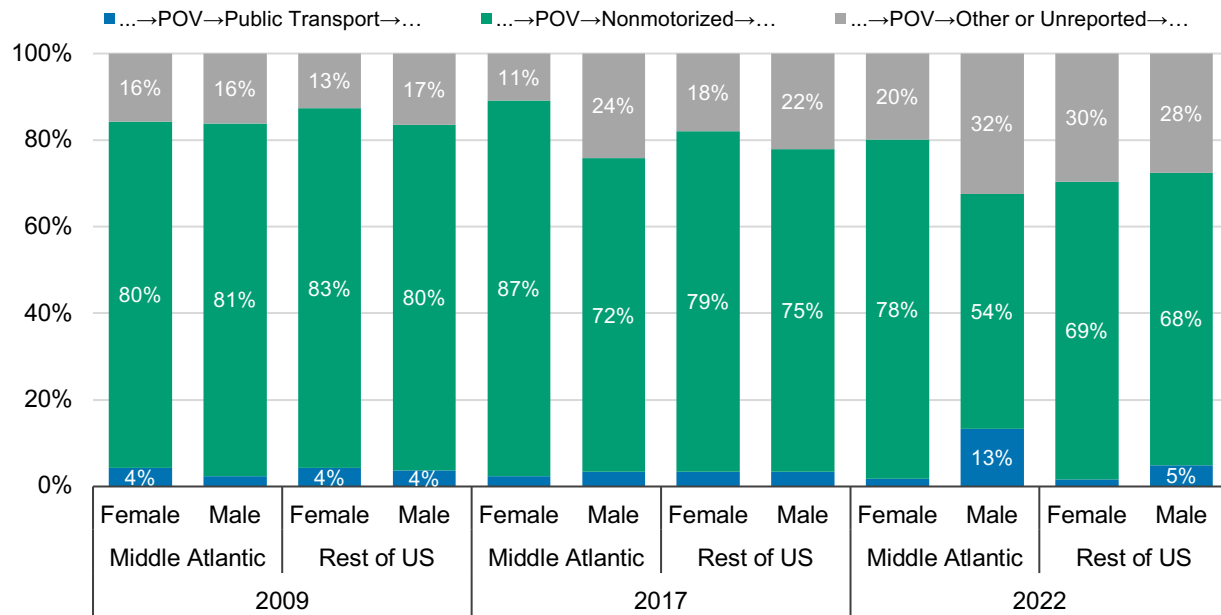


Figure 5-3. Transportation Mode Transition Following a POV Trip by Sex in the Middle Atlantic Census Division and Rest of the United States

5.2 TRIP CHAINING BY LIFECYCLE

5.2.1 Pre-Work Trip Chaining

In 2009 and 2017, the Middle Atlantic Census Division saw about 79–80% of workers with children and 78–79% without children going directly to work from home (Figure 5-4). In 2022, the share of direct Home→Work trips rose sharply in the Middle Atlantic, reaching 93% for workers with children and 89% for those without. These increases are statistically significant compared with earlier years. For instance, 93% in 2022 for with-children households is significantly higher than 79% in 2017. Similarly, the 89% direct-to-work share among those without children in 2022 is statistically higher than both 78% in 2017 and 79% in 2009.

The rest of the United States reflects a similar but slightly less pronounced trend. In 2009 and 2017, about 76–80% of workers went directly from home to work. By 2022, direct Home→Work trips rose significantly to 87% among workers with children and 91% among those without. These increases are statistically significant compared with both 2009 and 2017.

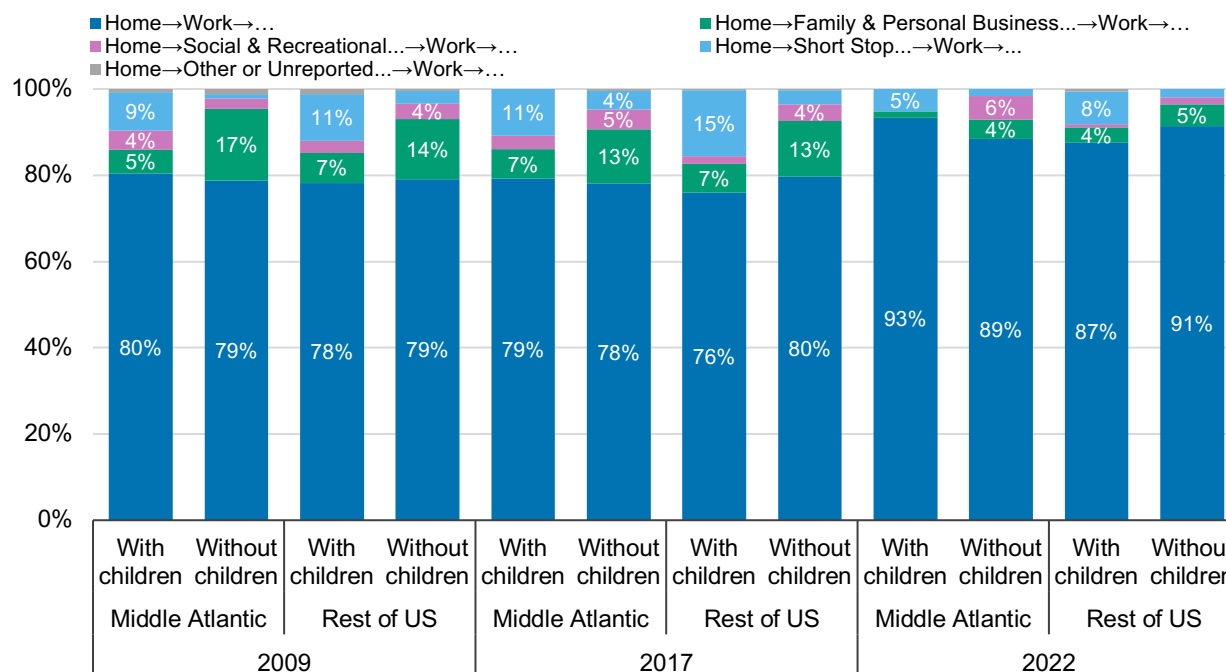


Figure 5-4. Pre-Work Activity Patterns by Lifecycle in the Middle Atlantic Census Division and Rest of the United States

5.2.2 Post-Work Trip Chaining

In 2009 and 2017, the majority of workers followed a simple Home→Work→Home chain, but a notable portion also incorporated family and personal business, especially among those without children (Figure 5-5). In 2022, however, the trend changed substantially. The proportion of Home→Work→Home chains increased sharply, dominating activity patterns for both groups. In the Middle Atlantic, 93% of workers with children and 87% without children returned directly home, with family and personal business trips dropping to 5% in both groups. These differences are statistically significant compared to earlier years: for example, 93% in 2022 for with-children households is significantly higher than 77% in 2017, while the 5% making family and personal business trips is statistically significantly lower than the 15% in 2017. Among households without children, the 87% share going straight home in 2022 is statistically higher than both 62% in 2017 and 68% in 2009, while the 5% making family and personal business trips is significantly lower than 30% in 2017. The rest of the United States followed a similar pattern, though the changes were somewhat less extreme.

5.2.3 Non-Work Days Trip Chaining

The 2022 results for the Middle Atlantic Census Division showed a difference in the first activity after leaving home compared with 2009 and 2017 (Figure 5-6). Among with-children households, recreation increased to 42%, statistically significantly higher than in 2017, while family and personal business declined slightly to 47%. Among those without children, family and personal business reached 78%, statistically significantly higher than in both 2009 and 2017, while recreation dropped to 21%, statistically significantly lower than in 2009. The gap between the two lifecycle groups therefore widened: with-children households continued to balance between family obligations and recreation, while without-children households overwhelmingly focused on family and personal business activities on non-work days. In the rest of the United States, the pattern is more consistent across time.

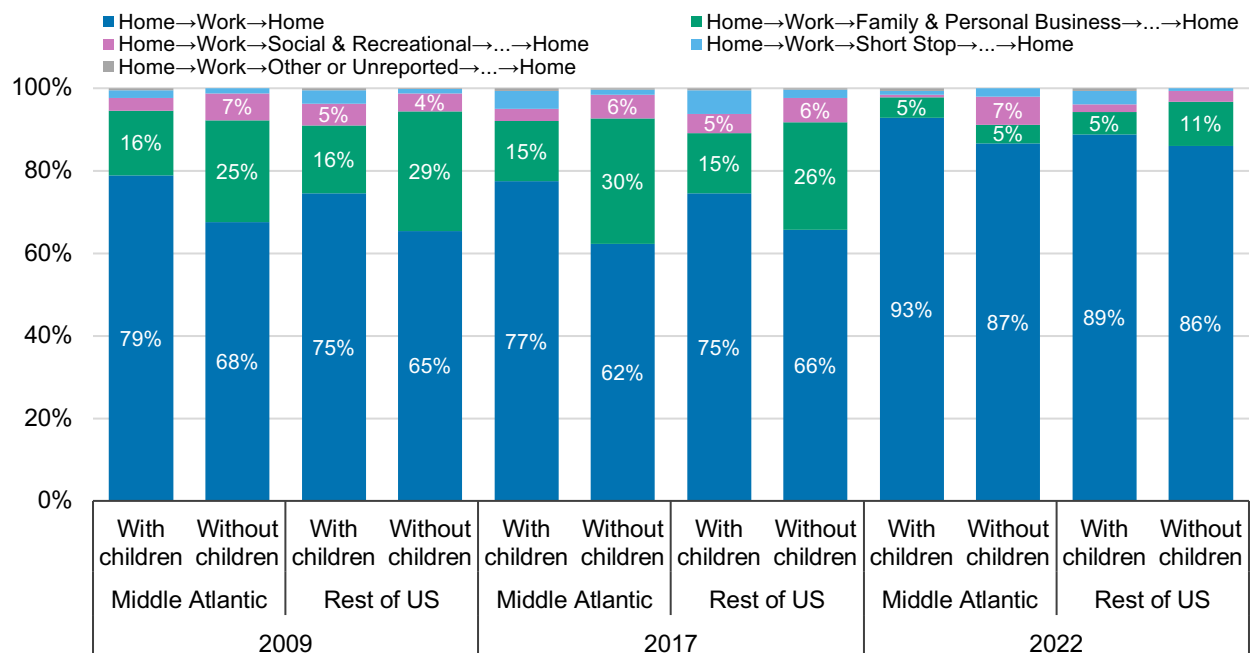


Figure 5-5. Post-Work Activity Patterns by Lifecycle in the Middle Atlantic Census Division and Rest of the United States: Home→Work Trip Chains

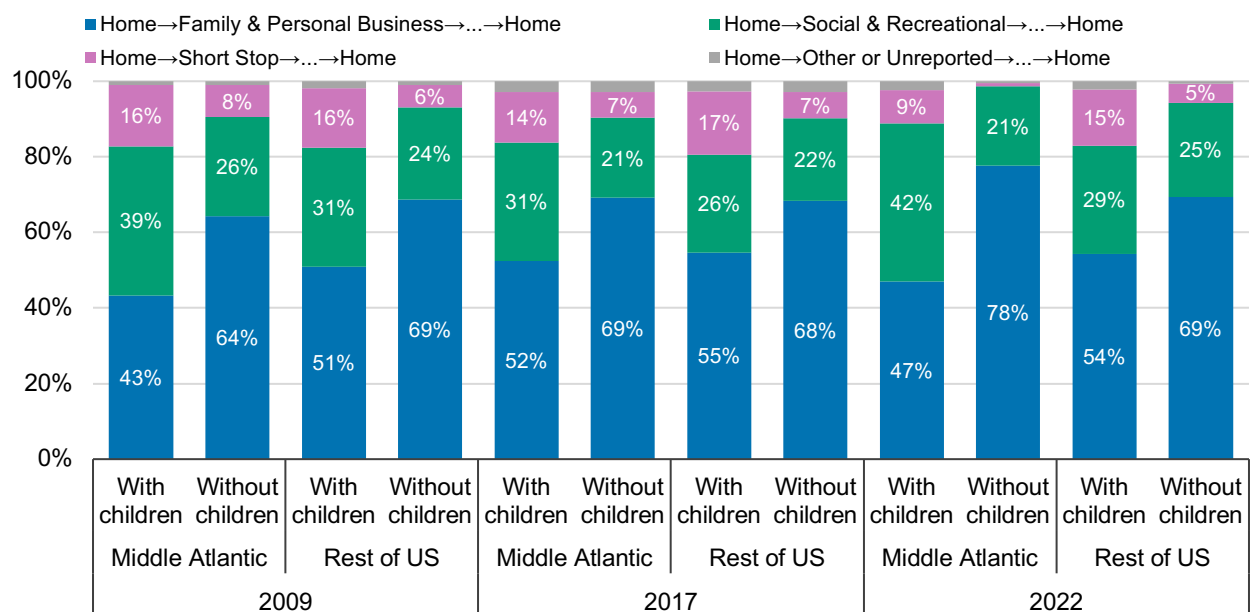


Figure 5-6. Activity Patterns on Non-Work Days by Lifecycle in the Middle Atlantic Census Division and Rest of the United States

5.2.4 Transportation Mode Transition

In earlier years, most workers transferred from POV to nonmotorized travel, with relatively small shares moving to public transport or other modes. For example, in 2009 in the Middle Atlantic, 75% of with-

children households and 89% of without-children households continued by nonmotorized mode. By 2017, the distribution remained largely stable. In 2022, among Middle Atlantic households with children, nonmotorized transitions dropped to 61%, statistically significantly lower than the 72% in 2017. At the same time, other or unreported modes rose to 34%, a statistically significant increase compared with 26% in 2017. In the rest of the United States, for with-children households, nonmotorized transitions fell from 69% in 2017 to 58% in 2022, a statistically significant decline, while other or unreported modes jumped from 28% to 39%, also statistically significant.

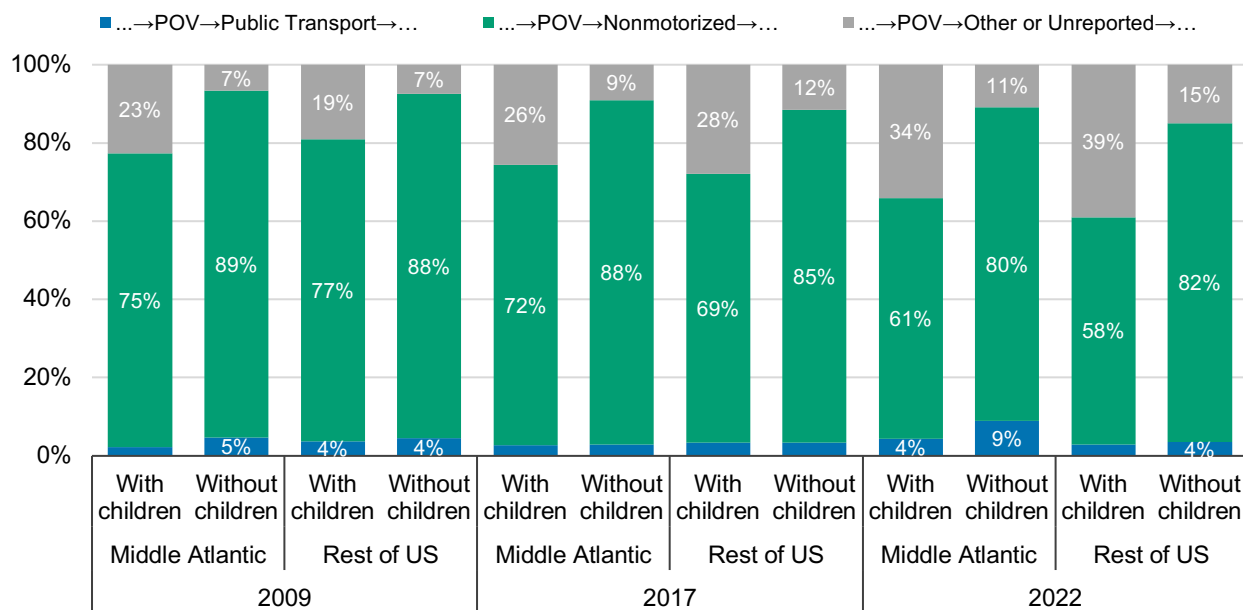


Figure 5-7. Transportation Mode Transition Following a POV by Lifecycle in the Middle Atlantic Census Division and Rest of the United States

5.3 TRIP CHAINING BY EDUCATION LEVEL

5.3.1 Pre-Work Trip Chaining

In the Middle Atlantic Census Division in 2022, 90% of both the college-educated and their counterparts went directly to work, statistically significantly higher than in earlier years for both groups (Figure 5-8). Family and personal business dropped to very low levels (3–4%), and short stops remained minor (3–5%). None of the differences between education groups in 2022 are statistically significant. In the rest of the United States, the same pattern holds: 87% of the college-educated and 90% of their counterparts went directly to work, both statistically significantly higher than in 2009 and 2017. The very small differences in family and personal business and short stops are not statistically significant, however.

5.3.2 Post-Work Trip Chaining

Individuals in the Middle Atlantic with a college degree or higher were less likely to return directly home after work (67%) compared with those with a high school diploma or less (77%) in 2009 (Figure 5-9). This difference is statistically significant. By 2017, these education-based gaps persisted. By 2022, however, the trend changed dramatically. Direct returns home became the dominant pattern for all groups. In the Middle Atlantic, 85% of college-educated workers and 94% of workers with high school degree or less went straight home. Both shares represent statistically significant increases compared with 2009 and

2017. While the workers with high school degree or less remained statistically significantly more likely to go directly home than the college-educated group, the gap between them narrowed substantially.

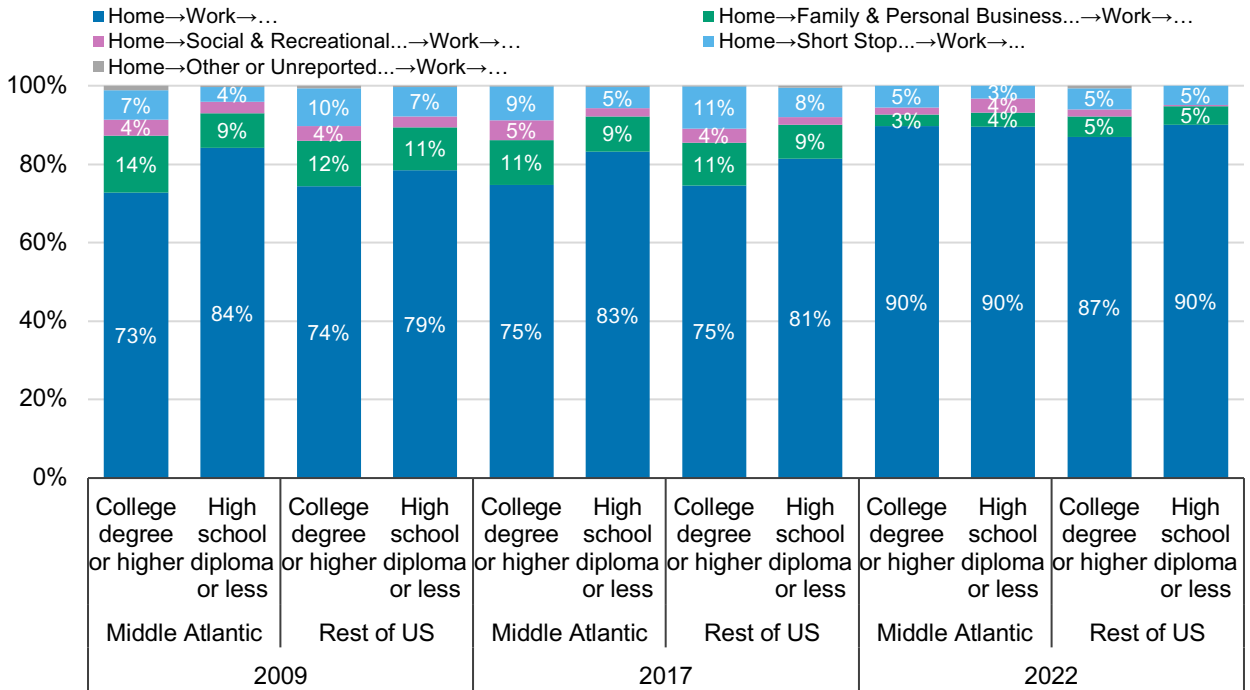


Figure 5-8. Pre-Work Activity Patterns by Education Level in the Middle Atlantic Census Division and Rest of the United States

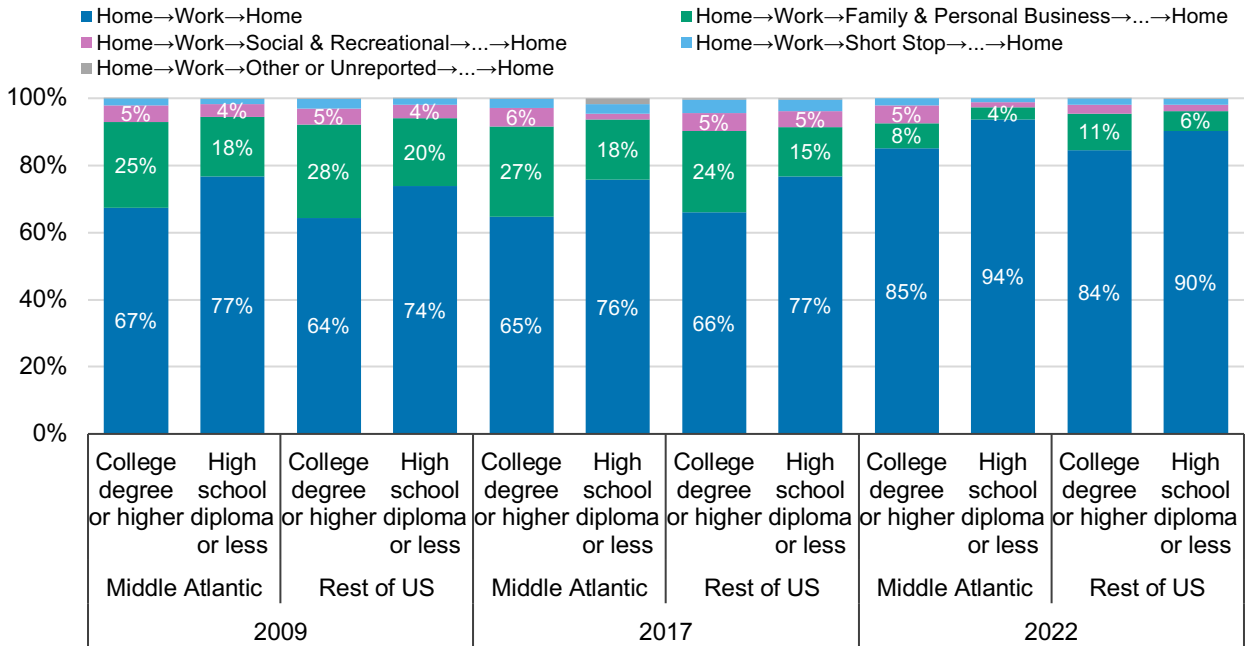


Figure 5-9. Post-Work Activity Patterns by Education Level in the Middle Atlantic Census Division and Rest of the United States: Home→Work Trip Chains

5.3.3 Non-Work Days Trip Chaining

In 2009, in the Middle Atlantic, individuals with a college degree or higher devoted 58% of their non-work day travel to family and personal business, compared with 64% among those with a high school diploma or less (Figure 5-10). Although those with a high school diploma or less appeared more family-focused, this difference is not statistically significant. By 2017, patterns remained stable. By 2022, the Middle Atlantic saw both groups increase their reliance on family and personal business, reaching 64% among the college-educated and 69% among the less-educated. The differences between the two groups are not statistically significant.

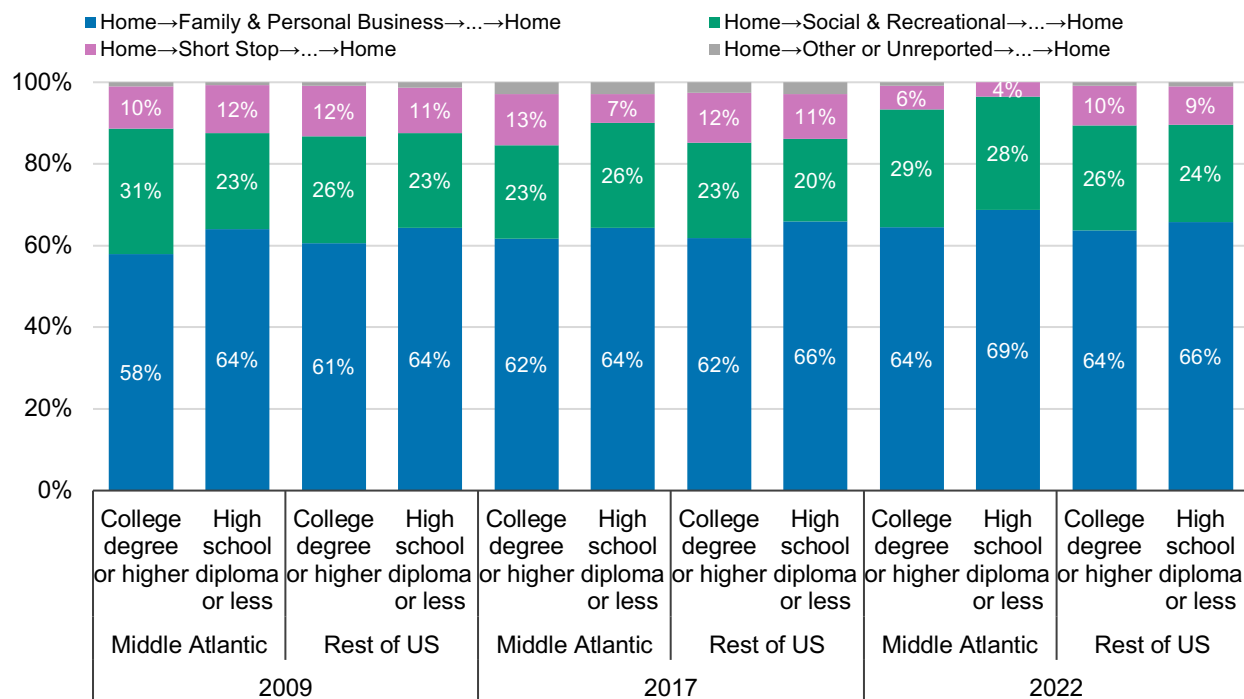


Figure 5-10. Activity Patterns on Non-Work Days by Education Level in the Middle Atlantic Census Division and Rest of the United States

5.4 TRIP CHAINING BY VEHICLE OWNERSHIP

5.4.1 Pre-Work Trip Chaining

In 2009, not zero vehicle households in the Middle Atlantic Census Division reported 79% of trips going directly from home to work, while zero vehicle households showed an even higher 93% (Figure 5-11). This difference is not statistically significant, however. By 2017, the Middle Atlantic Census Division displayed some divergence. Not zero vehicle households remained stable at 78% going directly to work, nearly identical to 2009, with no statistically significant change. Zero vehicle households, however, dropped to 85% direct-to-work, while family and personal business rose to 13%, though these changes were not statistically significant. By 2022, in the Middle Atlantic Census Division, not zero vehicle households rose sharply to 93% going directly to work, a statistically significant increase compared with 78% in 2017. For zero vehicle households, however, direct-to-work fell to 82% from 85% in 2017. This decline is not statistically significant. Instead, zero vehicle households in 2022 showed higher shares of short stops (10%) and social and recreational activities (6%) before work, though these increases are also not statistically significant.

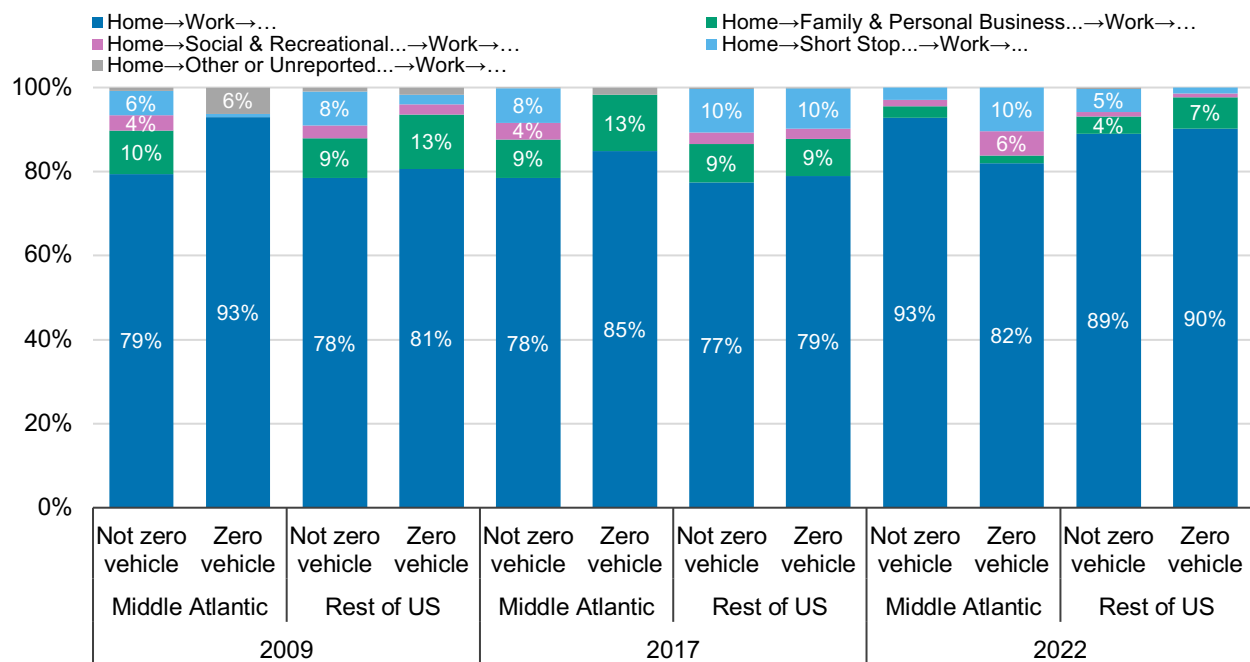


Figure 5-11. Pre-Work Activity Patterns by Vehicle Ownership in the Middle Atlantic Census Division and Rest of the United States

5.4.2 Post-Work Trip Chaining

In 2009, post-work travel behavior looked quite similar between households with and without vehicles in both the Middle Atlantic and the rest of the United States (Figure 5-12). In the Middle Atlantic, 74% of not zero vehicle households and 76% of zero vehicle households went directly home after work. By 2017, zero vehicle households showed a dramatic departure: only 46% went directly home, while nearly half (47%) stopped for family and personal business. Compared with 2009, this represented a statistically significant drop in direct returns and a rise in family and personal business trips. By 2022, in the Middle Atlantic, households with vehicles recorded 90% direct returns home and zero vehicle households 92%, with family and personal business dropping to 1–5%. These increases in direct returns are statistically significant compared with 2017, especially for zero vehicle households.

5.4.3 Non-Work Days Trip Chaining

In 2022, in the Middle Atlantic Census Division, households with vehicles reported 61% family and personal business activities and 33% recreation activities, a pattern similar to 2009 (Figure 5-13). Zero vehicle households continued to devote far more time to family and personal business activities on non-work days, at 74%, compared with only 20% recreation. These differences between zero vehicle and vehicle owning households are statistically significant. In the rest of the United States, the differences between the two groups were smaller, with 62% of vehicle owning households and 68% of zero vehicle households in family and personal business.

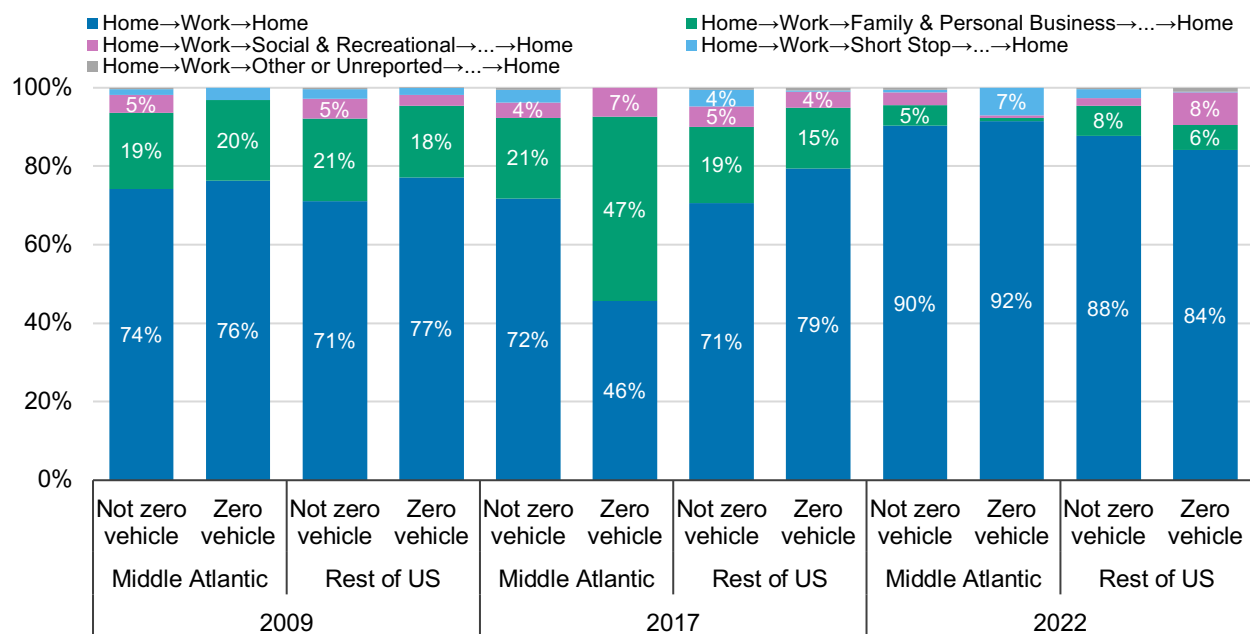


Figure 5-12. Post-Work Activity Patterns by Vehicle Ownership in the Middle Atlantic Census Division and Rest of the United States: Home→Work Trip Chains

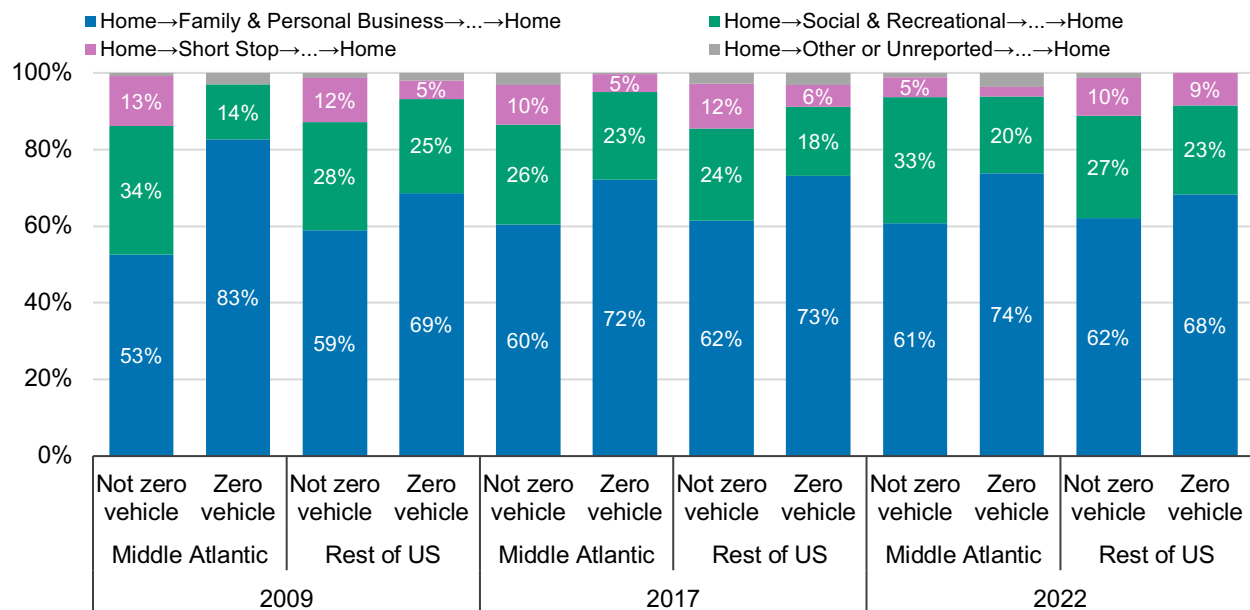


Figure 5-13. Activity Patterns on Non-Work Days by Vehicle Ownership in the Middle Atlantic Census Division and Rest of the United States

CONCLUSIONS

This analysis presents a comprehensive empirical examination of trip chaining behaviors utilizing NHTS datasets, focusing on NYS, NYC, and comparative assessments between the Middle Atlantic Census Division and the rest of the United States. The findings demonstrate that demographic characteristics, including occupational category, age, and educational attainment, along with environmental factors, systematically influence pre-work and post-work trip chaining patterns, non-work day activity sequences, and mode transition decisions. Across all temporal periods examined, direct home→work→home travel emerged as the predominant pattern, particularly pronounced among workers in occupations characterized by limited schedule flexibility, including manufacturing, construction, maintenance, and farming, as well as among households with dependent children. These results suggest that occupational constraints and household responsibilities serve as primary determinants of simplified trip chaining behaviors.

Workers residing in high-density urban environments, particularly within NYC, demonstrated significantly elevated rates of post-work activity engagement. This pattern aligns with established accessibility and urban form theoretical frameworks, which demonstrate that increased density and enhanced service proximity facilitate trip chaining behaviors (Cheng et al. 2016). In such urban contexts, opportunities for commercial, retail, and recreational activities exhibit greater spatial concentration, thereby reducing both travel impedance and the opportunity costs associated with intermediate stops. These findings illustrate how built environment characteristics influence not only aggregate trip generation rates but also the temporal sequencing and spatial organization of daily activities. Additionally, zero-vehicle households in NYC exhibited higher propensities for post-work activity integration, demonstrating efficient activity bundling strategies while minimizing home-based trip origins. This observation corroborates earlier research on multipurpose urban travel patterns, which identified that substantial proportions of household travel occur in conjunction with work-related commuting (Hanson 1980). Conversely, zero-vehicle households in non-NYC areas of NYS showed reduced likelihood of extending travel beyond basic commuting functions. This finding reinforces spatial accessibility constraints documented in automobile ownership and social exclusion literature, where limited public transit availability and reduced walkable destination density outside major metropolitan areas constrains post-work activity participation (Gray et al. 2006).

Gender-based differences in post-work travel patterns were pronounced within NYC. Female workers exhibited significantly higher rates of family- and personal-business-related intermediate stops following work activities, while male workers demonstrated modest increases in leisure-oriented travel between 2009 and 2017. These empirical findings align with established literature documenting that women engage in complex, multipurpose trip chains characterized by constrained temporal and modal resources, whereas men demonstrate greater propensity for single-purpose travel utilizing superior transportation modes and incurring higher associated costs (Turner and Grieco 2000). Such behavioral patterns reflect the phenomenon of "time poverty," wherein competing household and professional obligations limit individuals' capacity to optimize daily activity-travel patterns or extend their spatial range of activities.

Analysis of 2022 NHTS data reveals a pronounced nationwide shift toward simplified commuting patterns, characterized by increased rates of direct home-to-work and work-to-home travel without intermediate trip segments. While these findings suggest convergence in commuting behaviors across demographic groups, interpretation requires methodological caution due to the reduced 2022 NHTS sample size and the diminished proportion of respondents reporting work-related trips during their designated survey period. These sampling limitations may introduce selection bias that could influence the observed behavioral patterns.

Future research should expand this analytical framework through integration of comprehensive behavioral, spatial, and temporal datasets with enhanced granularity. Examining the relationship between

trip chaining behaviors and emerging mobility paradigms, including telecommuting adoption, e-commerce penetration, micromobility integration, and innovative transit service deployment, could elucidate how contemporary travel patterns respond to technological and societal transformations. Enhanced spatial analysis at finer geographic scales, coupled with causal inference methodologies, would facilitate clearer identification of mechanisms through which built environment characteristics and accessibility measures influence trip consolidation versus direct travel decisions. Furthermore, integrating travel behavior analysis with public health and environmental impact frameworks would yield critical insights for transportation planners and policymakers developing sustainable mobility systems that accommodate diverse population needs while simultaneously reducing traffic congestion. Such interdisciplinary approaches would advance both theoretical understanding of mobility behavior and practical applications for evidence-based transportation policy development.

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APPENDIX A. GLOSSARY OF NHTS TERMS

APPENDIX A. GLOSSARY OF NHTS TERMS

This glossary provides the most commonly used terms in the National Household Travel Survey (NHTS) and the terms used in this report, along with definitions. These definitions are provided to assist the user in interpreting the NHTS data.

Adult	For NHTS, this is defined as a person 18 years or older.
Block Group	A subdivision of a census tract that contains 600–3,000 people. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Census Tract	A small subdivision of a county, generally having a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. The geographic size of the tract may vary considerably, depending on population density. Tracts were designed to be homogeneous regarding population characteristics, economic status, and living conditions when they were first delineated. Because the first tracts were delineated for the 1890 Census, current tracts may be far from homogeneous. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Destination	For travel day trips, the destination is the point at which there is a break in travel, except if the break is only to change vehicles or means of transport.
Elderly	A person age 65 years or older.
Elderly Household	Households with one or more persons age 65 years or older.
Employed	A person is considered employed if that person worked for pay, either full time or part time, during the week before the interview. This includes persons who work at home and/or have more than one job.
Household	A group of persons whose usual place of residence is a specific housing unit; these persons may or may not be related to each other. The total of all US households represents the total civilian noninstitutionalized population. A household does not include group quarters (i.e., 10 or more persons living together, none of whom are related).
Household Income	Household income is the money earned by all family members in a household, including those temporarily absent. Annual income consisted of the income earned 12 months preceding the interview. Household income includes monies from all sources, such as wages and salary, commissions, tips, cash bonuses, income from a business or farm, pensions, dividends, interest, unemployment or worker's compensation, social security, veterans' payments, rent received from owned property (minus the operating costs), public assistance payments, regular gifts of money from friends or relatives not living in the household, alimony, child support, and other kinds of periodic money income other than earnings. Household income excludes in-kind income such as room and board, insurance payments, lump-sum inheritances, occasional gifts of money from persons not

	living in the same household, withdrawal of savings from banks, tax refunds, and the proceeds of the sale of one's house, car, or other personal property.
Household Members	Household members include all people, whether present or temporarily absent, whose usual place of residence is in the sample unit. Household members also include people staying in the sample unit who have no other usual place of residence elsewhere.
Household Vehicle	A household vehicle is a motorized vehicle that is owned, leased, rented, or company-owned and available to be used regularly by household members during the 2 week travel period. Household vehicles include vehicles used solely for business purposes or business-owned vehicles if they are driven home and can be used for the home-to-work trip (e.g., taxicabs and police cars). Household vehicles include all vehicles that were owned or available for use by members of the household during the travel period, even though a vehicle may have been sold before the interview. Vehicles excluded from household vehicles are (1) those that were not working and were not expected to be working within 60 days and (2) those that were purchased or received after the designated travel day.
Journey-to-Work Trips (Commute Trips)	Includes travel to and from a place where one reports for work. Does not include any other work-related travel. Does not include trips for persons who work at home.
Means of Transportation	A mode of travel used for going from one place (origin) to another (destination). A means of transportation includes private and public transit modes, as well as walking. The following transportation modes, grouped by major mode, are included in the NHTS data. Active modes—include modes by which a person must actively move from one place to the next. • Walk: This category includes walking and jogging. • Bicycle: This category includes bicycles of all speeds and sizes that do not have a motor. Private vehicle—a stipulation for being a private vehicle is that the vehicle is privately operated, including rental cars. • Car: Includes cars and station wagons. Leased and rented cars are included if privately operated and not used for picking up passengers in return for fare. • Sport utility vehicle: Includes vehicles that are a hybrid of design elements from a van, pickup truck, and station wagon. Examples include a Ford Explorer, Jeep Cherokee, or Nissan Pathfinder. • Van: Includes vans or minivans designed to carry 5–13 passengers or to haul cargo. • Pickup truck: Includes vehicles with an enclosed cab that usually accommodates two to three passengers and has an open cargo area in the rear. Late-model pickups often have a back seat that allows for total seating of four to six passengers. Pickup trucks usually have a wheelbase the same size as that of a full-size station wagon. This category also includes pickups with campers.

- **Motorcycle/moped:** This category includes large, medium, and small motorcycles and mopeds.
- **RV (motor home, ATV, snowmobile):** An RV or motor home includes a self-powered recreational vehicle that is operated as a unit without being towed by another vehicle (e.g., a Winnebago motor home). This category also includes ATVs and snowmobiles.

Rental car—Includes Zipcar and Car2Go, in addition to commercially rented cars for private use.

- **Golf cart/Segway:** This includes all electric or gas-operated vehicles designed for use on a golf course, but whose use has recently extended to within smaller, often gated, communities. NOTE: Travel taken via golf cart is NOT included in vehicle trips and vehicle miles of travel.

Public transportation—as used in Federal Highway Administration publications and analysis of NHTS data, typically includes the following (indicated in italics in the following text): public or commuter bus, commuter rail, and subway/elevated rail/light rail/streetcar.

Bus—This category includes:

- *Public or commuter buses:* local public transit buses available to the public and buses used for short-distance public transport purposes (e.g., city bus or public bus), school buses
- *Private/charter/tour/shuttle buses:* private buses operating on a fixed schedule between population centers and buses that shuttle passengers from one fixed place to another (e.g., airport shuttles)
- *City-to-city buses:* buses that run from one urban center to another (e.g., Greyhound)

Train—This category includes:

- *Amtrak/commuter rail:* trains that run from one urban center to another
- *Subway/elevated rail/light rail/streetcar* (also known as rail rapid transit): a high-capacity system operated on a fixed-rail or guideway system on a private right of way, and vehicles that run on a fixed-rail system powered by electricity obtained from an overhead power distribution system

Other modes—This category includes:

- School buses
- Paratransit/dial-a-ride
- Taxi/limo: Use of a taxicab by a passenger for fare, including limousines. In 2017, this category also includes ridesharing such as Uber and Lyft.
- Airplane: includes commercial airplanes and smaller planes available for use by the public in exchange for a fare. Private and corporate planes and helicopters are also included.
- Boat/ferry/water taxi: includes travel by passenger line ferries
- Something else: includes any type of transportation not previously listed (e.g., skateboards, roller blades, sailboats, cruise ships)

Metropolitan Statistical Area (MSA)

Except in the New England states, a metropolitan statistical area is a county or group of contiguous counties that contains at least one city of 50,000 inhabitants or more, or “twin cities” with a combined population of at least 50,000. In addition, contiguous counties are included in an MSA if, according to certain criteria, they are socially and economically integrated with the central city. In the

	New England states, MSAs consist of towns and cities instead of counties. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from the Census 2010 definition).
Motorized Vehicle	Motorized vehicles include all vehicles licensed for highway driving. Snow mobiles and minibikes are specifically excluded.
New York City (NYC)	New York City is defined in this report as the five-county area: Bronx, Kings (Brooklyn), Queens, New York (Manhattan), and Richmond (Staten Island).
Origin	Origin is the starting point of a trip.
Person Trip	A person trip is a trip by one or more persons in any mode of transportation. Each person is considered as making one person trip. For example, four persons traveling together in one auto are counted as four person trips.
Population Density	Category of population density (persons per square mile) in the Census Block Group of the household's home location: 0–99, 100–499, 500–999, 1,000–1,999, 2,000–3,999, 4,000–9,999, 10,000–24,999, and 25,000–999,999.
Privately Owned Vehicle (POV)	A privately owned or privately operated vehicle. In either case, the vehicle is not available to the public for a fee (i.e., it is not a bus, subway, taxi, or other vehicle for which a fee is charged for use).
Travel Day	A travel day is a 24 h period from 4:00 a.m. to 3:59 a.m. designated as the reference period for studying trips and travel by members of a sampled household.
Travel Day Trip	A travel day trip is defined as any time the respondent went from one location to another by private motor vehicle, public transportation, bicycle, walking, or other means during the NHTS-assigned reporting travel day. However, a separate trip is not counted in two instances: 1. When the sole purpose for the trip is to get to another vehicle or mode of transportation to continue to the destination. 2. Travel within a shopping center, mall, or shopping areas of four to five blocks is to be considered as travel to one destination.
Vehicle	In the 2017 NHTS, the term <i>vehicle</i> includes autos, passenger vans, sport utility vehicles, pickups and other light trucks, RVs, motorcycles, and mopeds owned or available to the household.
Vehicle Trip	A trip by a single POV, regardless of the number of persons in the vehicle.
Work-Related Travel	Trips related to business activities except travel to the place of work; for example, a plumber drives to a wholesale dealer to purchase supplies for his business or a company executive travels from his office to another building to attend a business meeting. Business, out-of-town trips, and professional conventions are also included.
Worker	See “Employed.”

